



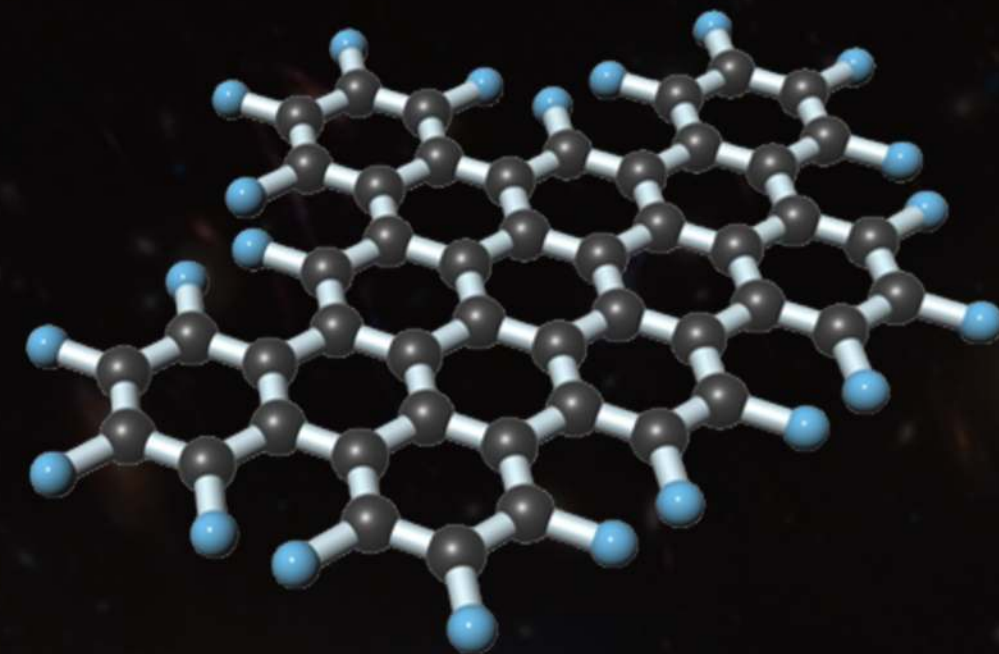
Galaxy clusters environmental effects on $\text{PAH}_{3.3}$ emission

Pietro Benotto



P
A
H

Polycyclic Aromatic Hydrocarbon

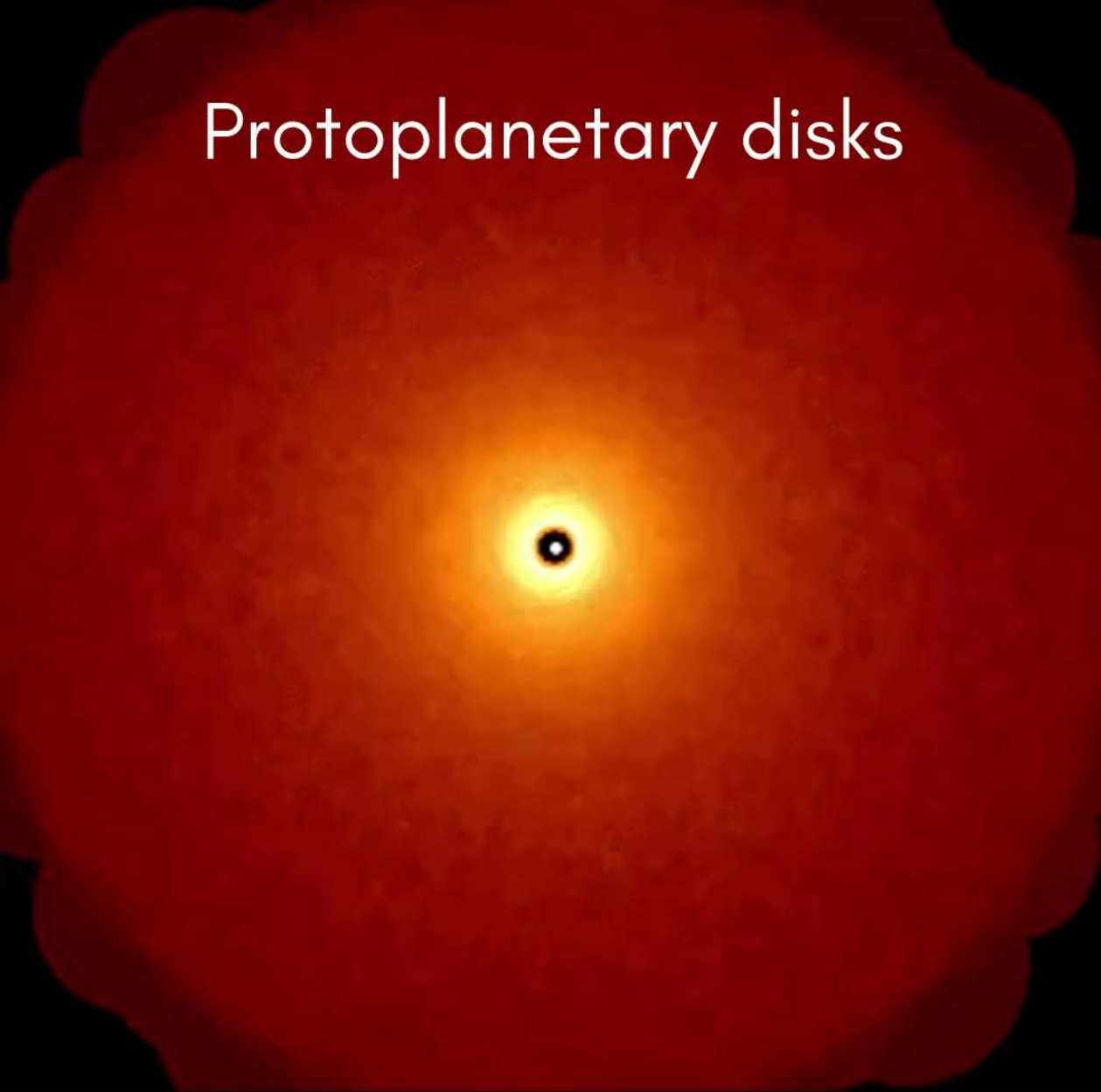


Polycyclic **A**romatic **H**ydrocarbon

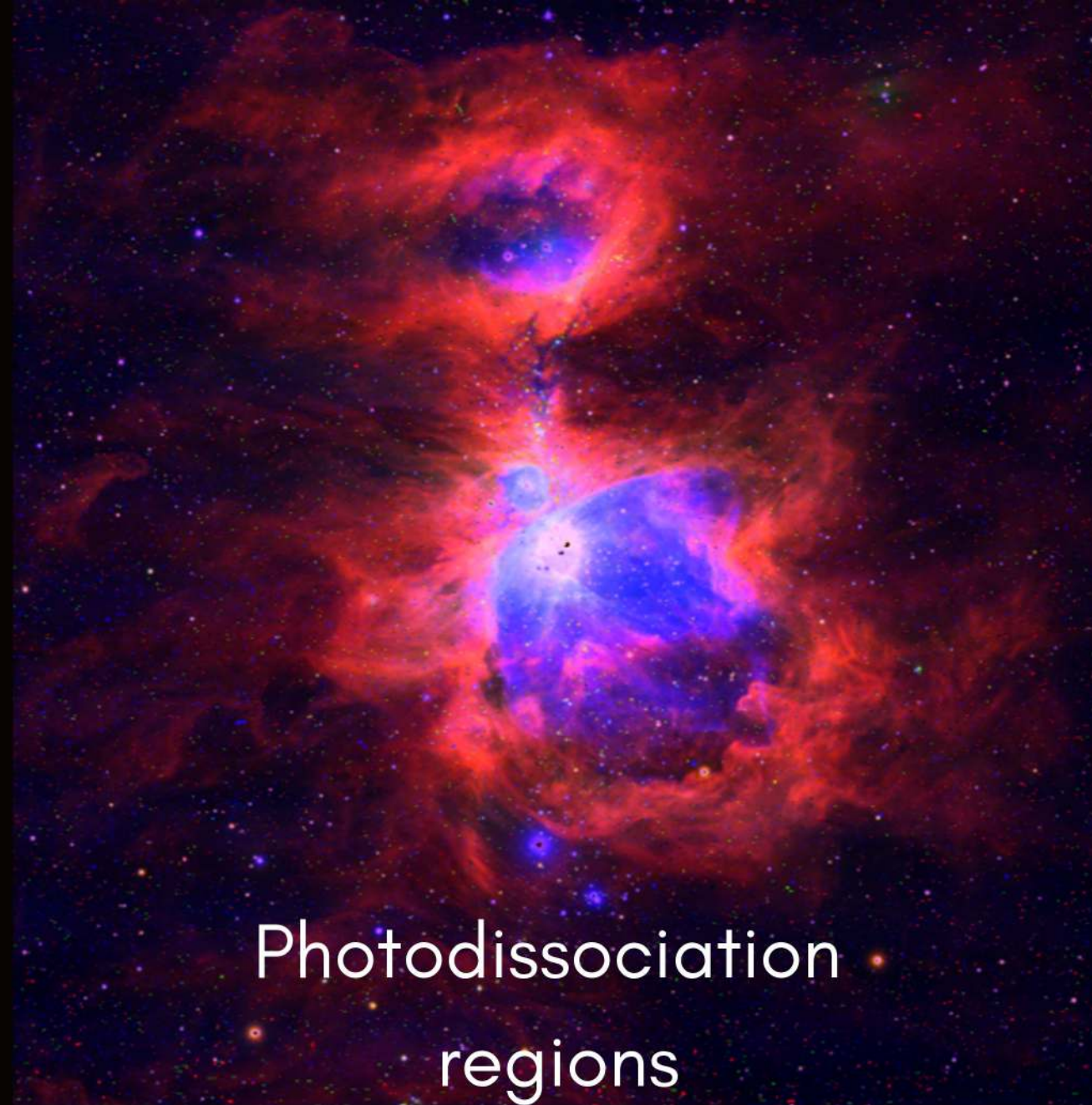
Building block:
Aromatic rings

Many
Carbon atoms

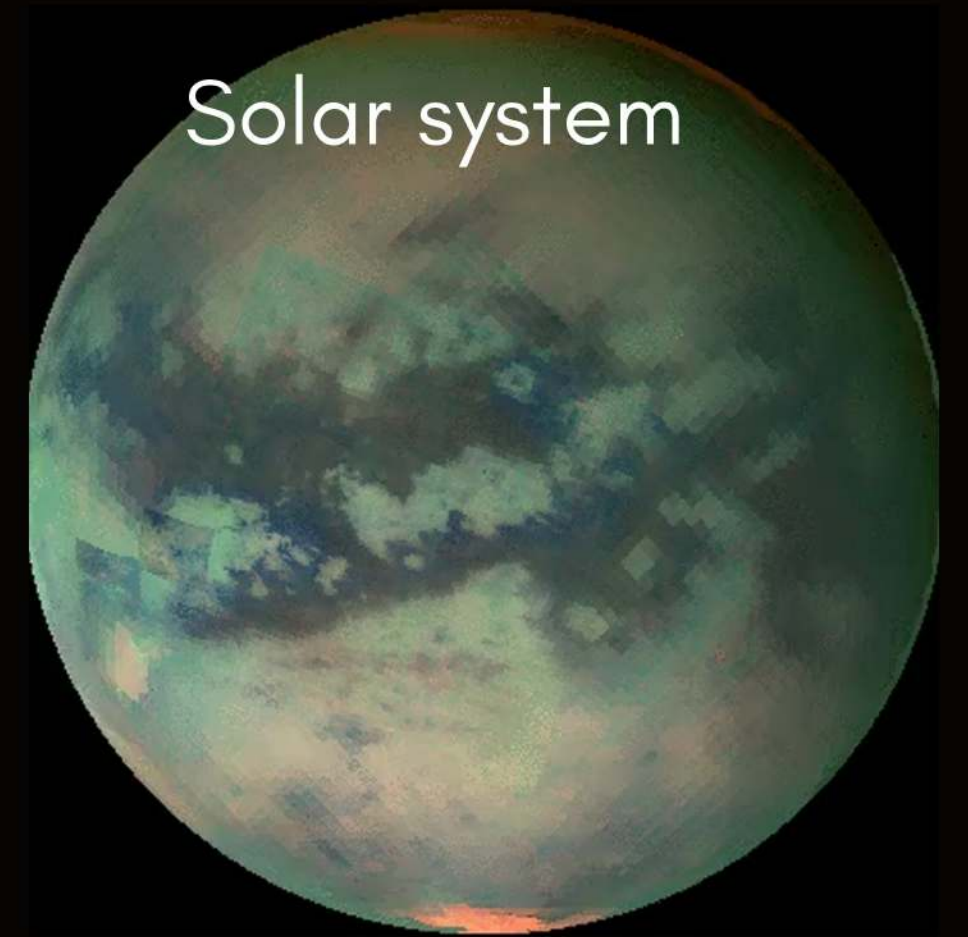
Protoplanetary disks



Photodissociation
regions



Solar system

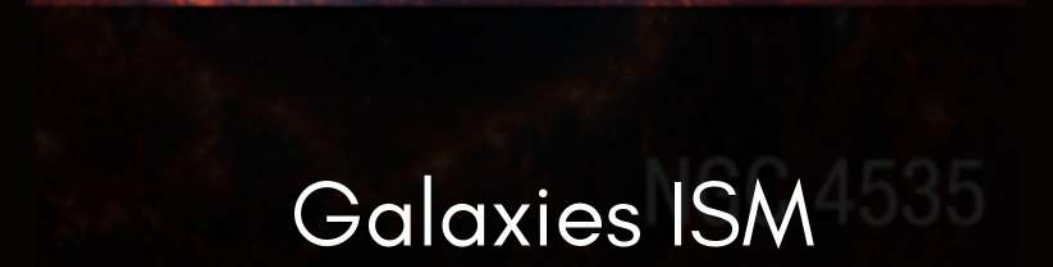


Milky way

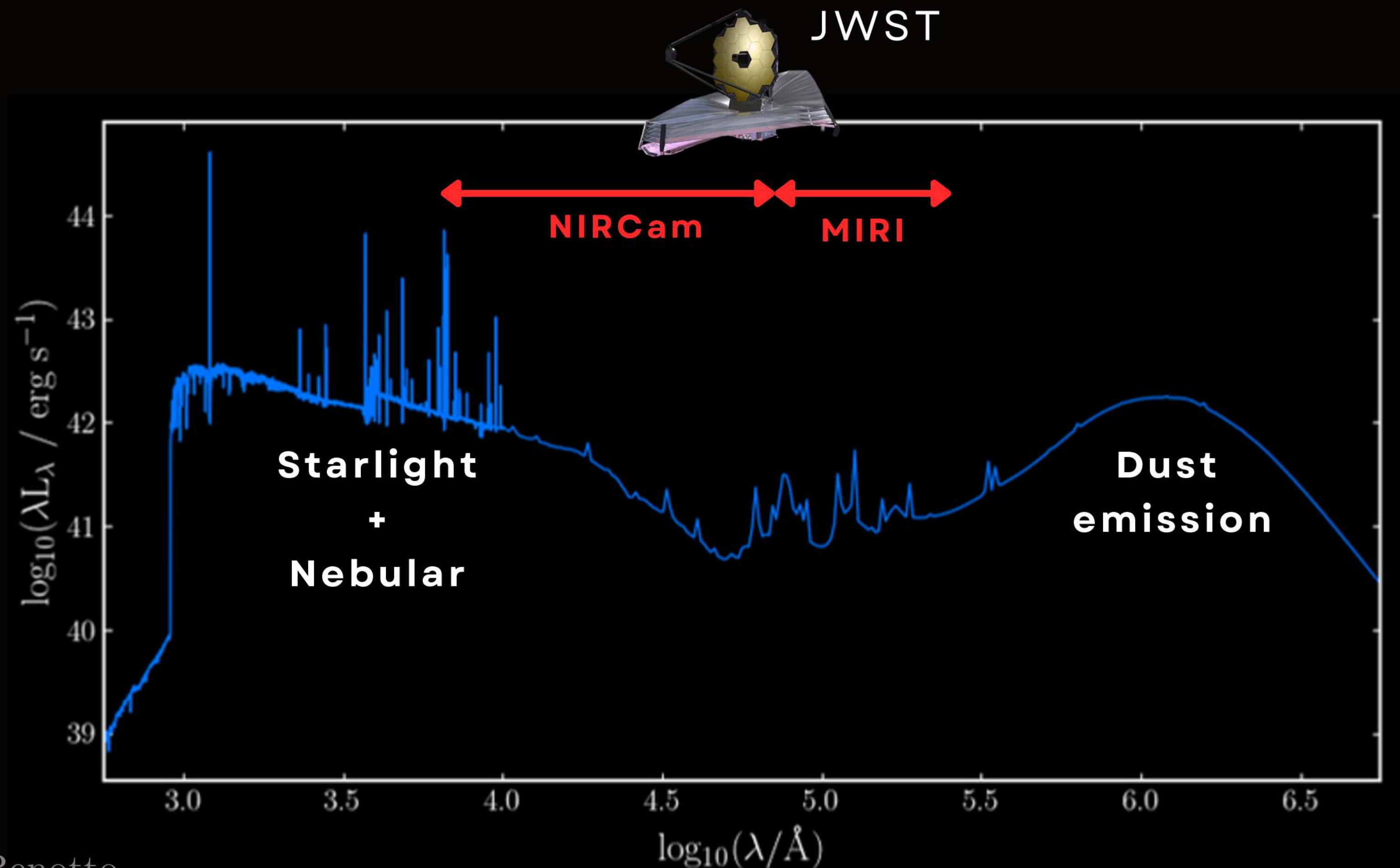
NGC 1433

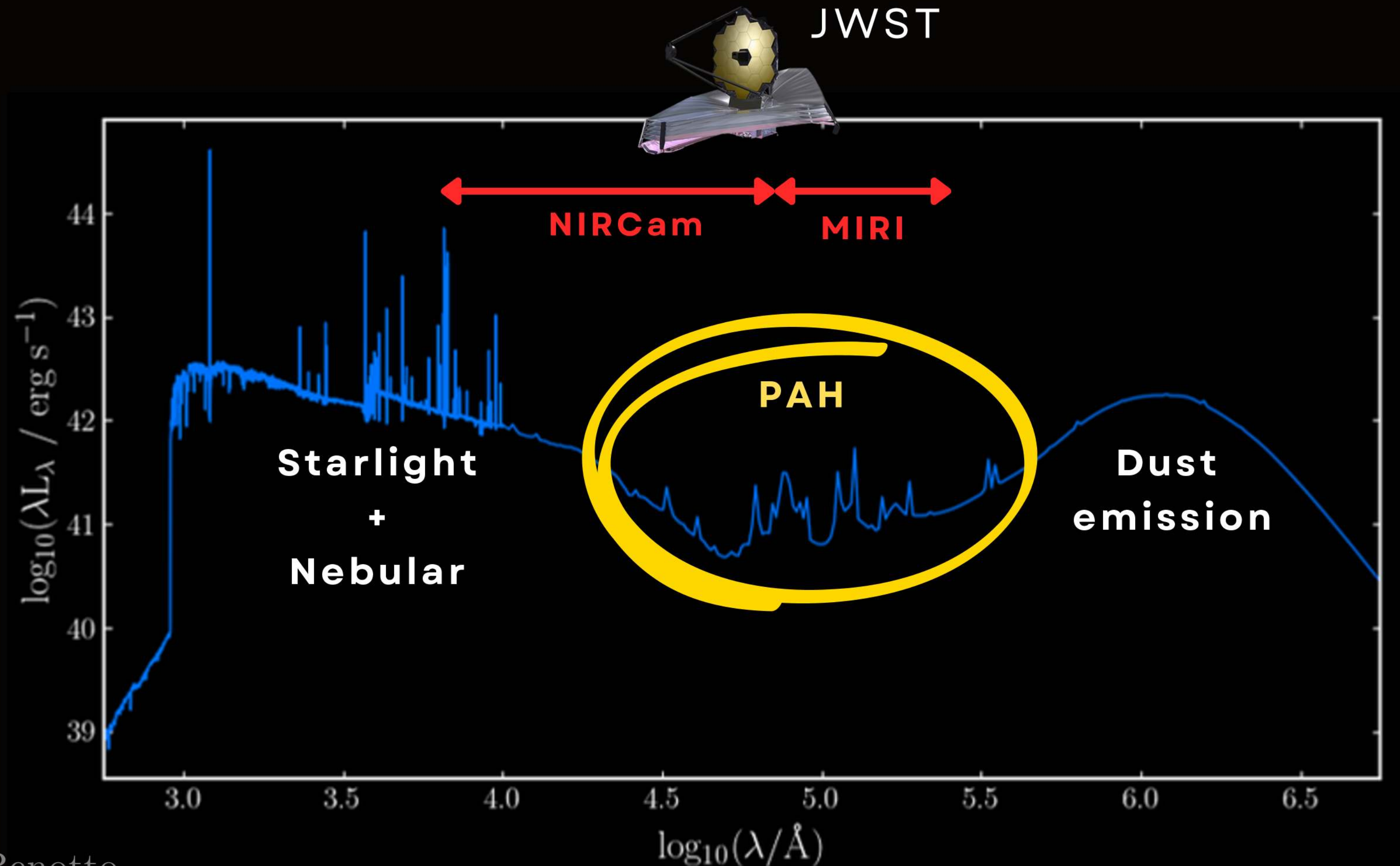


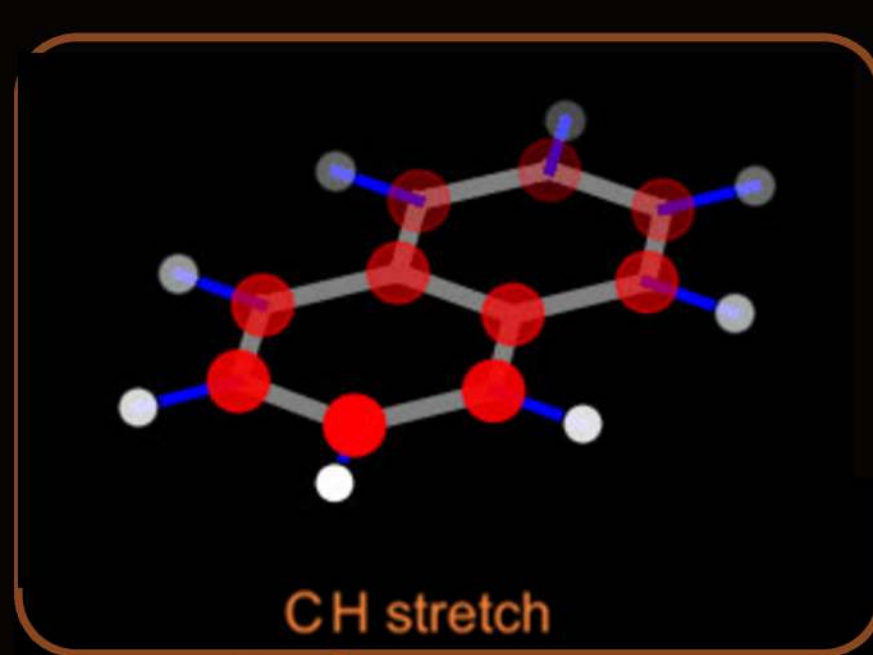
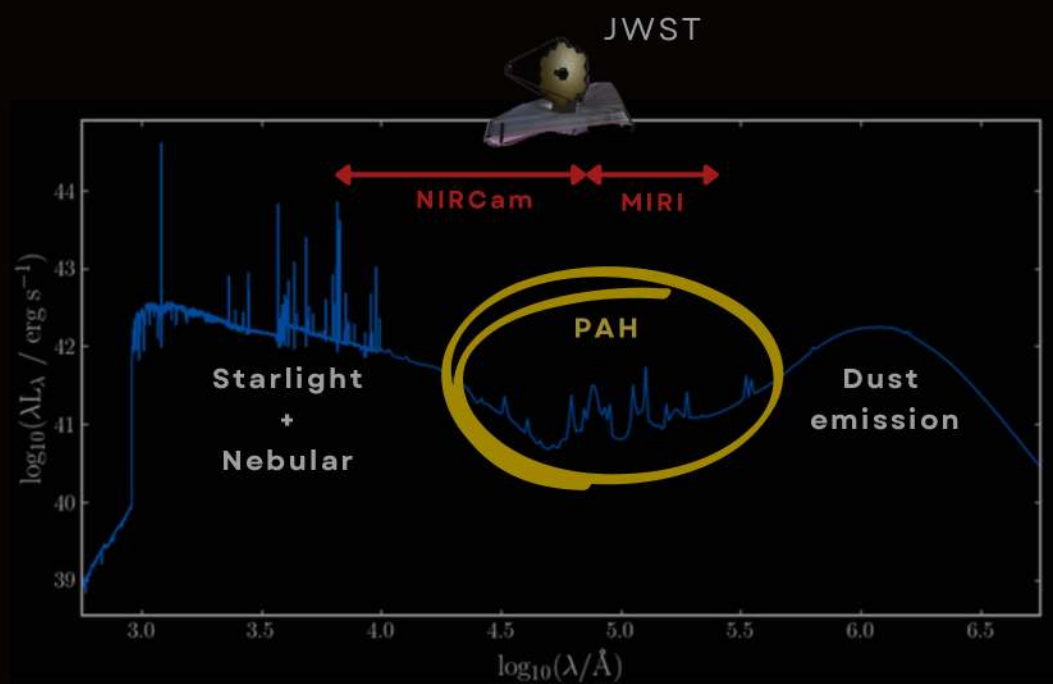
Galaxies ISM



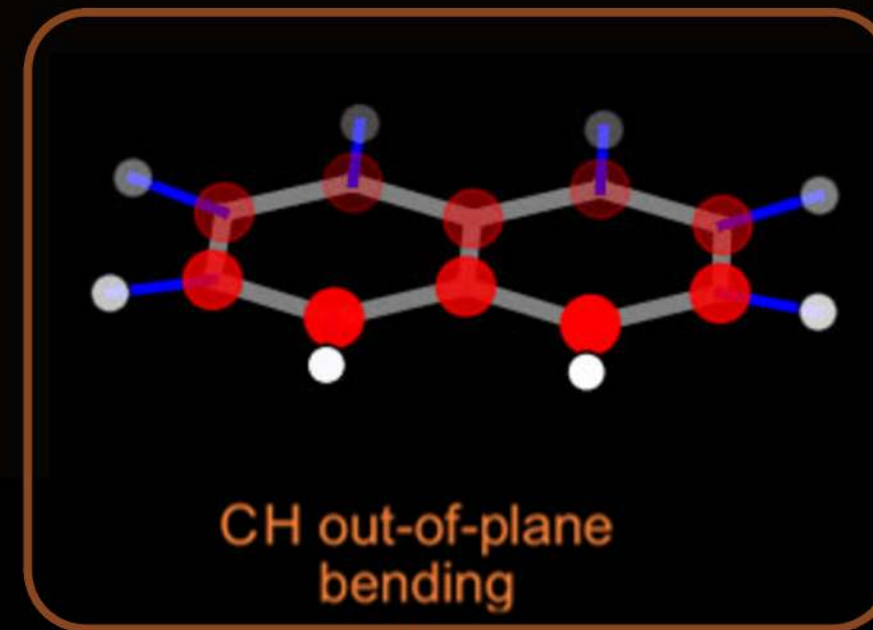
NGC 4535



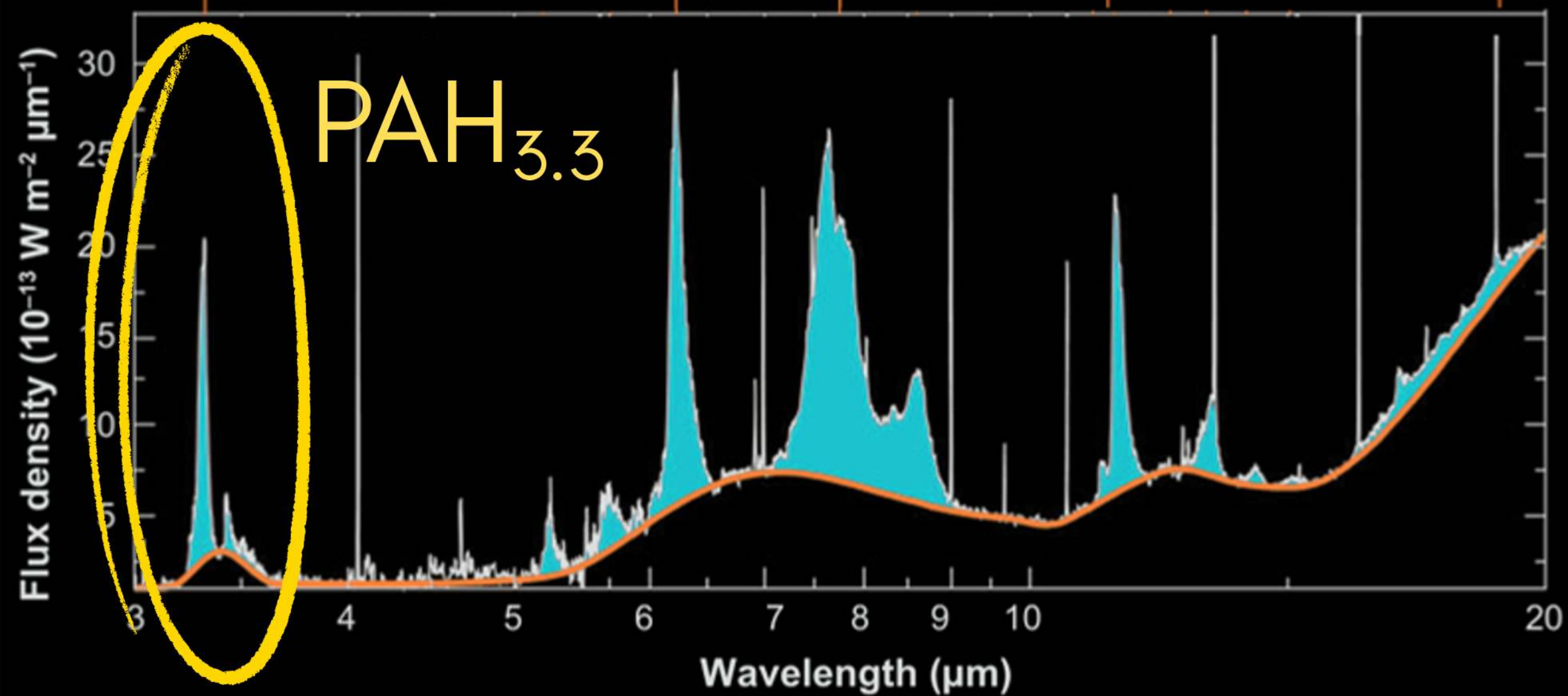


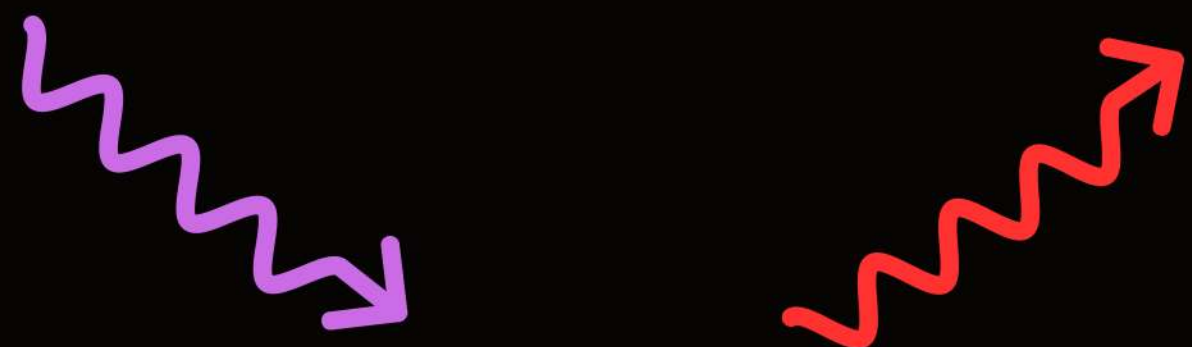


CC stretch



CC stretch

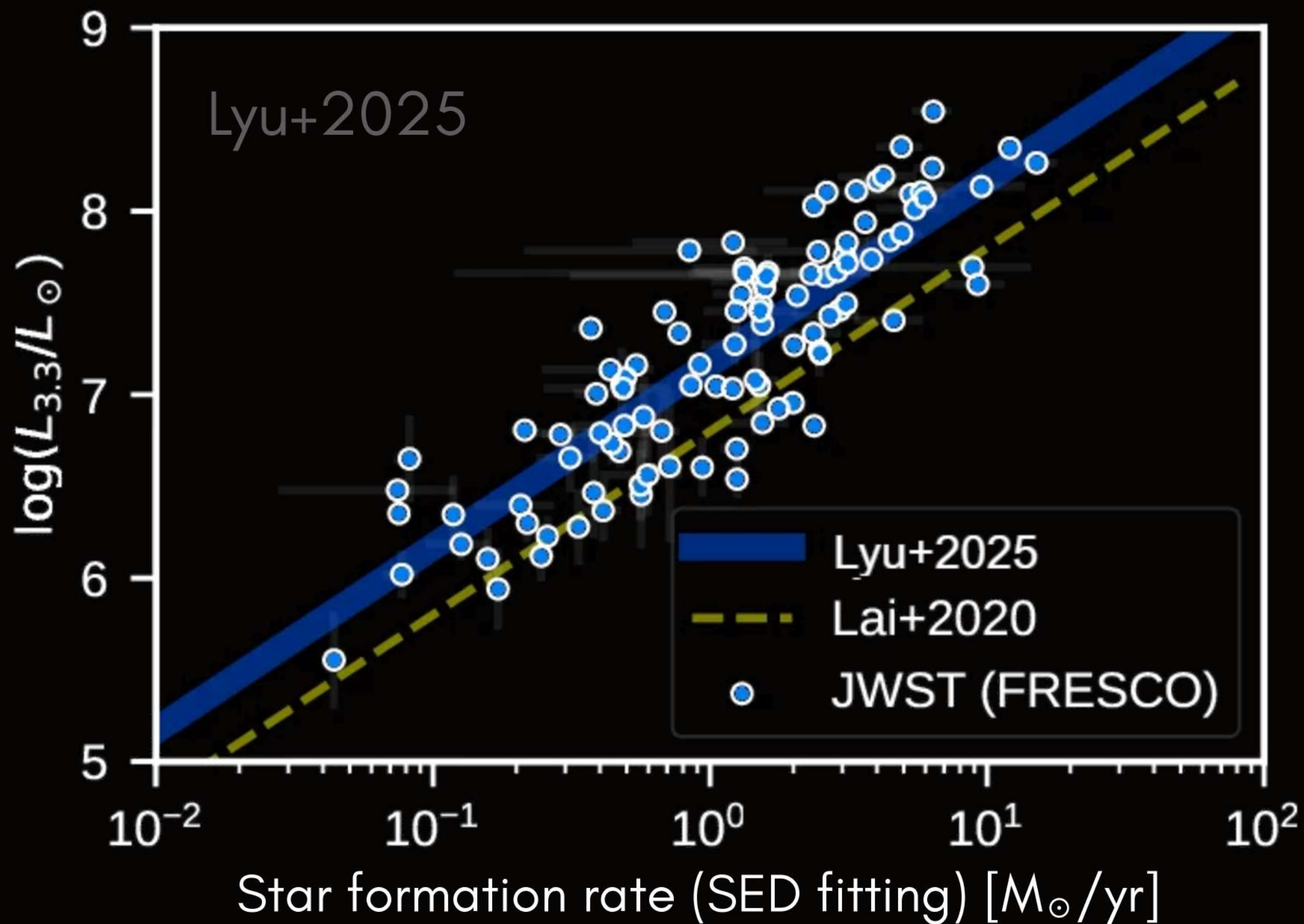
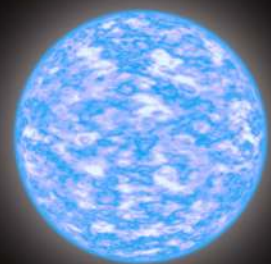




PAH molecules absorb UV photons and re-emit mid-infrared photons

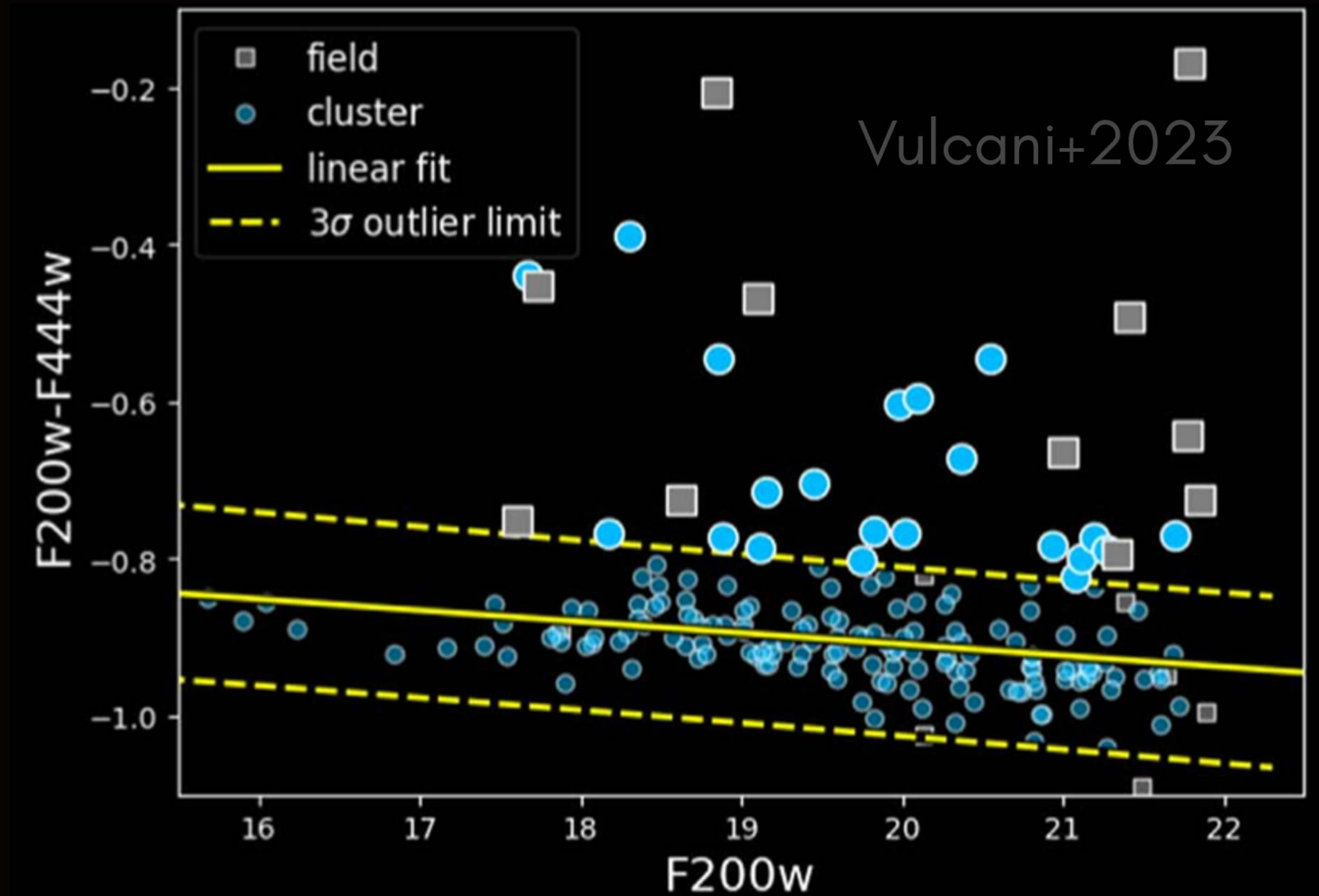


The PAH emission can be used as a **star formation tracer**



Red excess galaxies

(Vulcani+2023)



**Red excess
galaxies**

(Vulcani+2023)



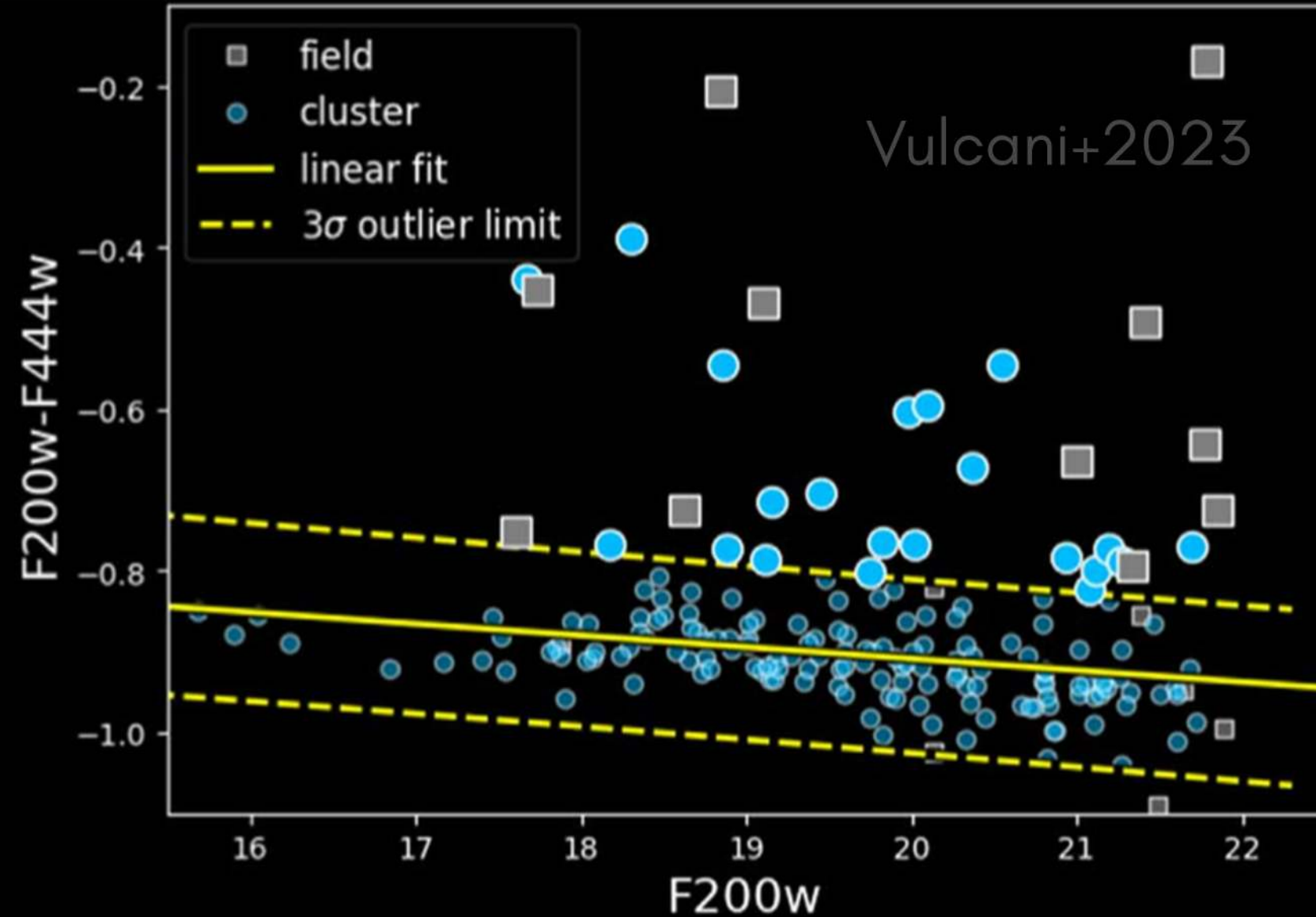
**PAH in galaxy
clusters**

(Vulcani+2025)



85% of red excess galaxies
are PAH emitters

Strong PAH emitters are
morphologically disturbed
galaxies



**Red excess
galaxies**

(Vulcani+2023)



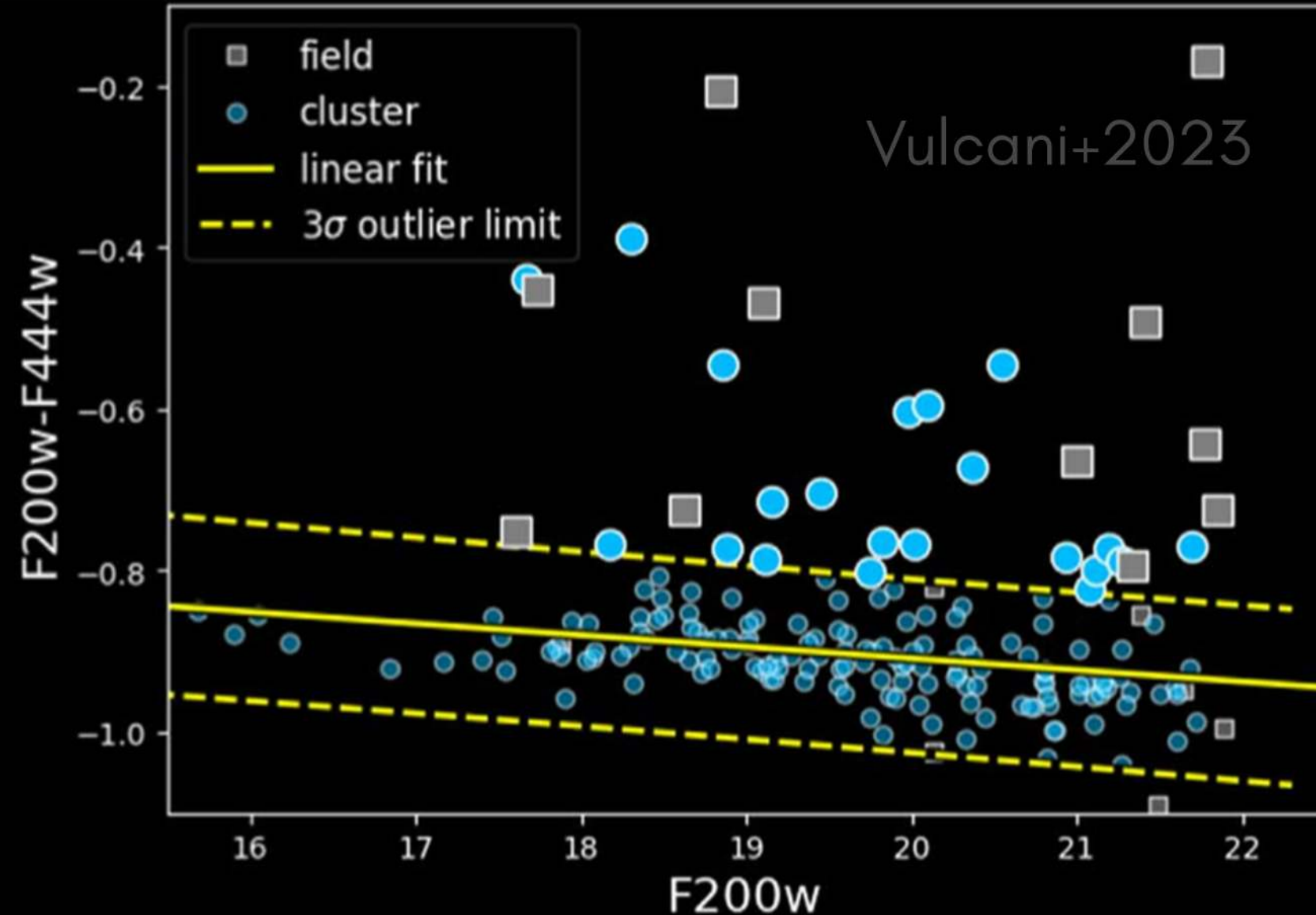
**PAH in galaxy
clusters**

(Vulcani+2025)



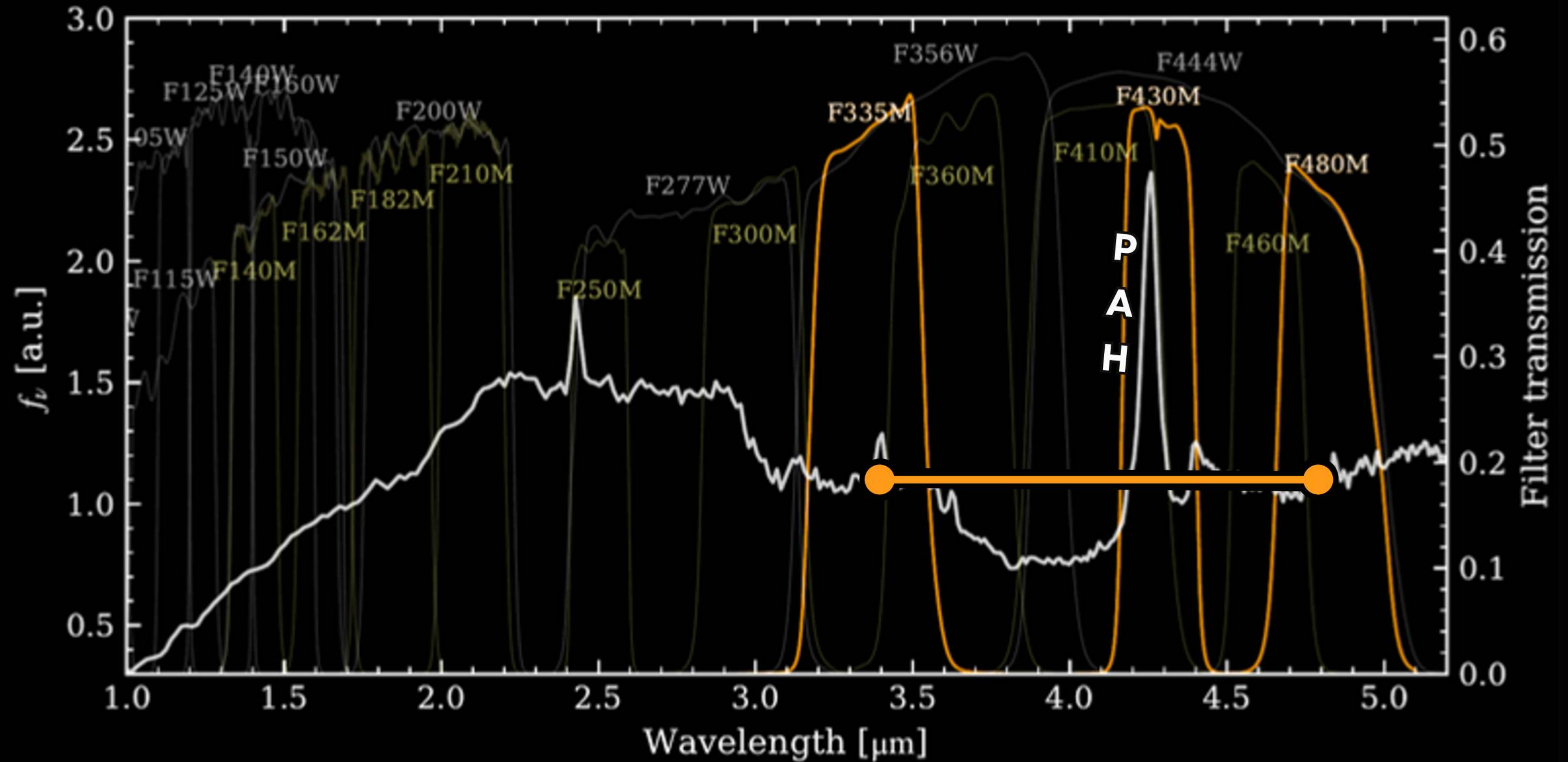
85% of red excess galaxies
are PAH emitters

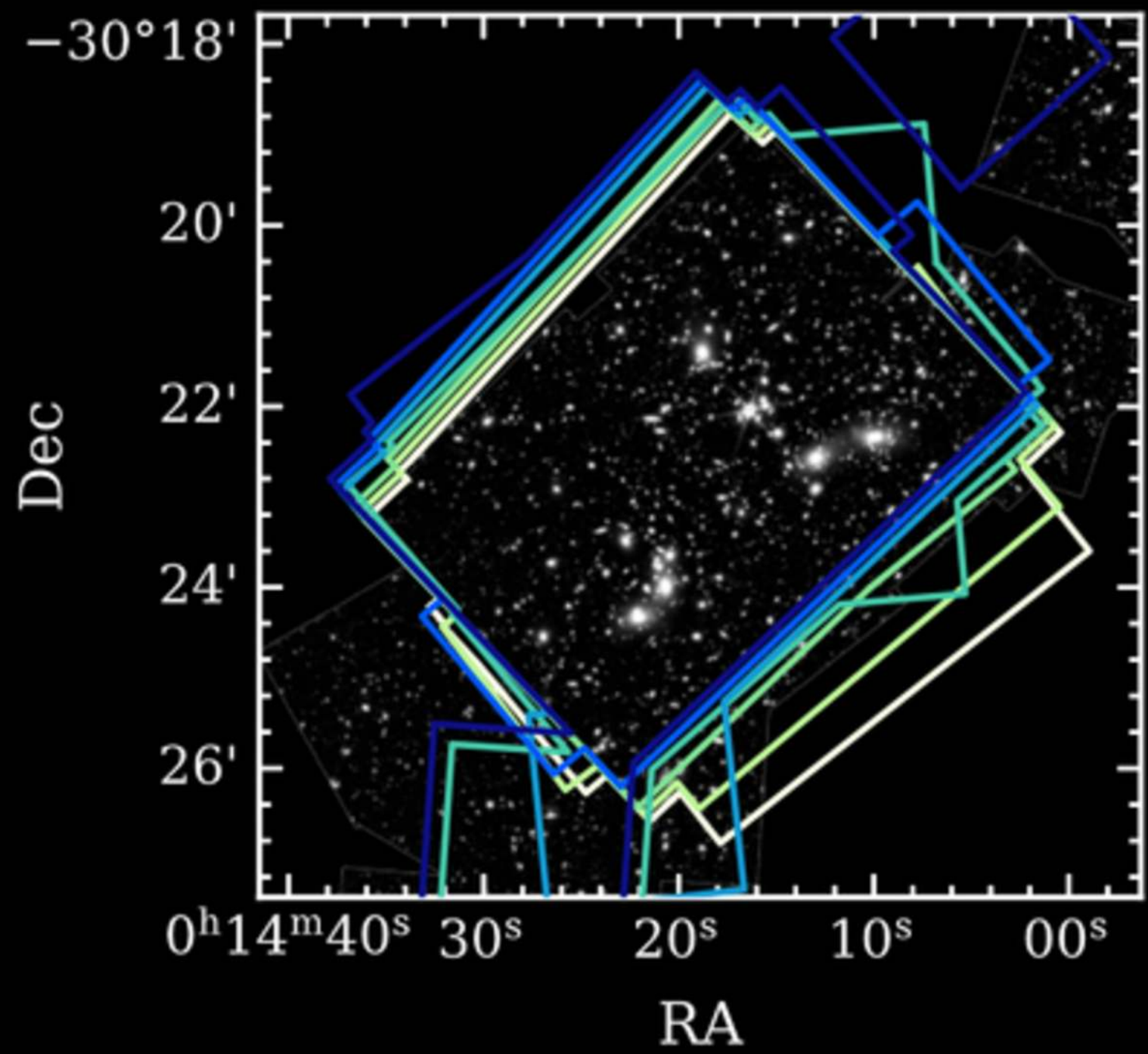
Strong PAH emitters are
morphologically disturbed
galaxies



**How does the cluster
environment affect
the PAH_{3.3} emission?**

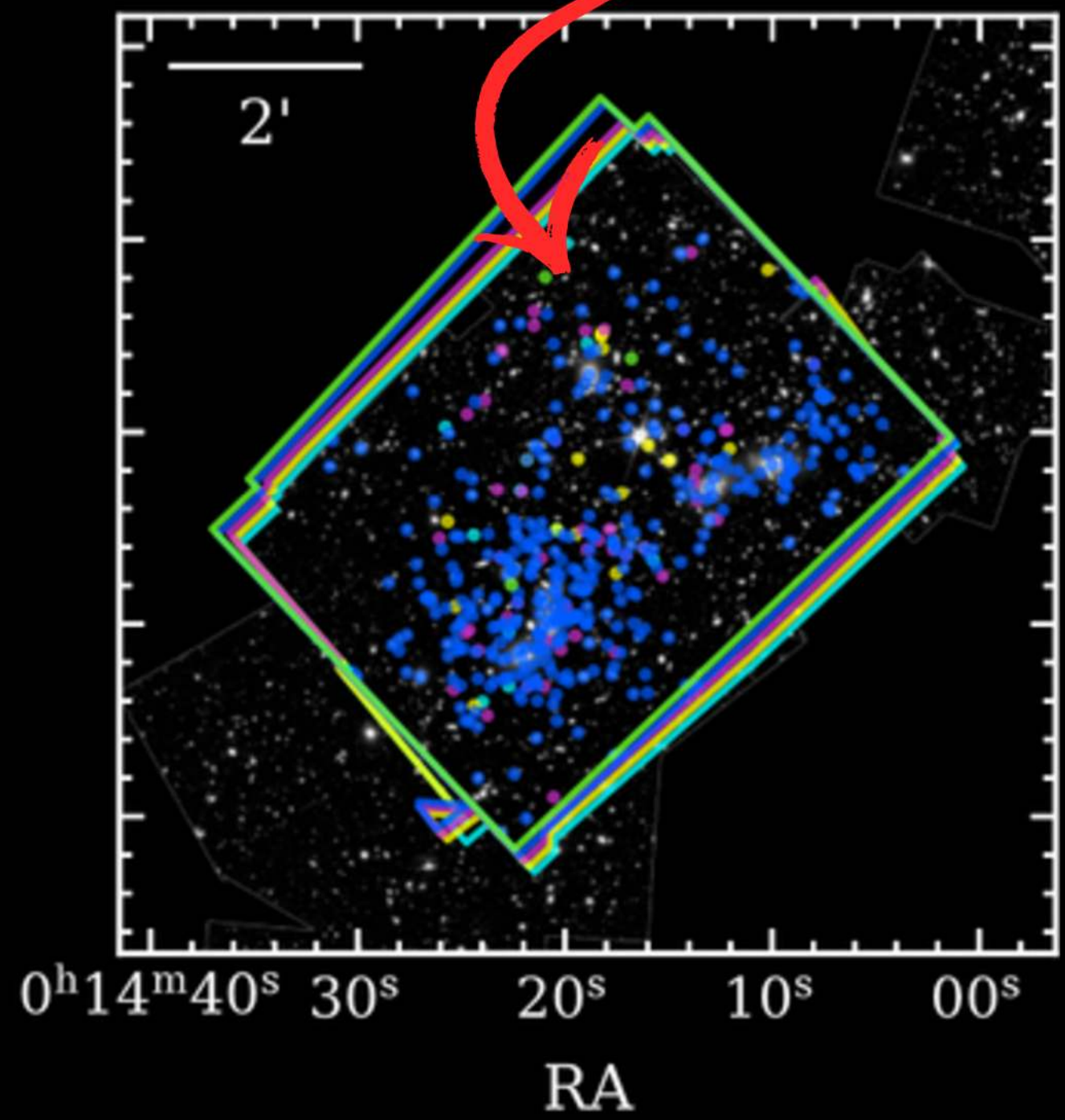
PAH flux extraction





Filter footprints

- | | |
|---|---|
| F300M | F410M |
| F335M | F430M |
| F360M | F460M |
| | F480M |

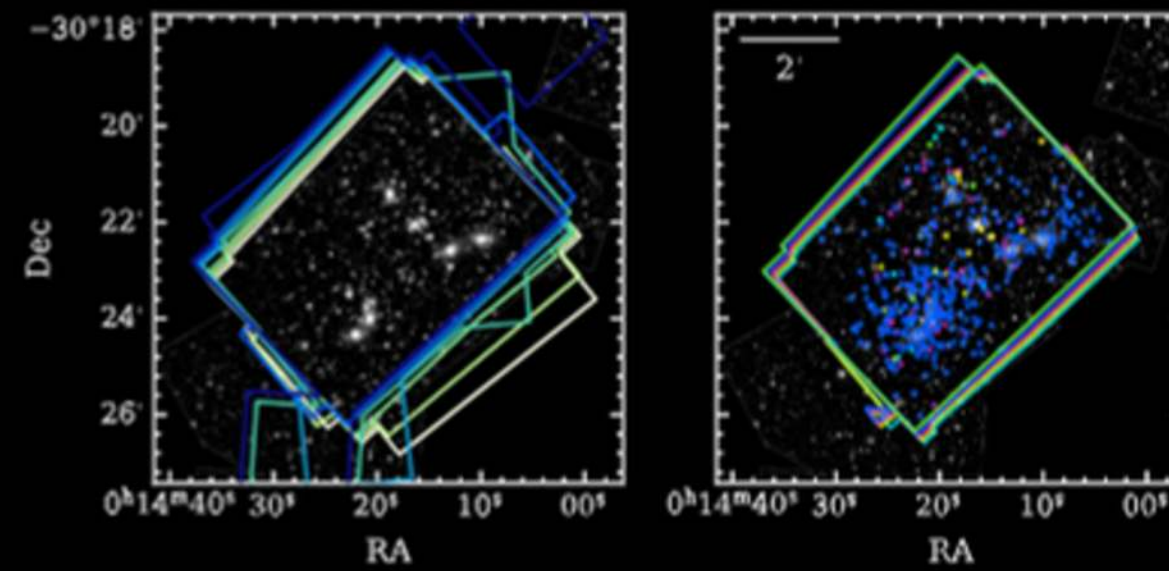
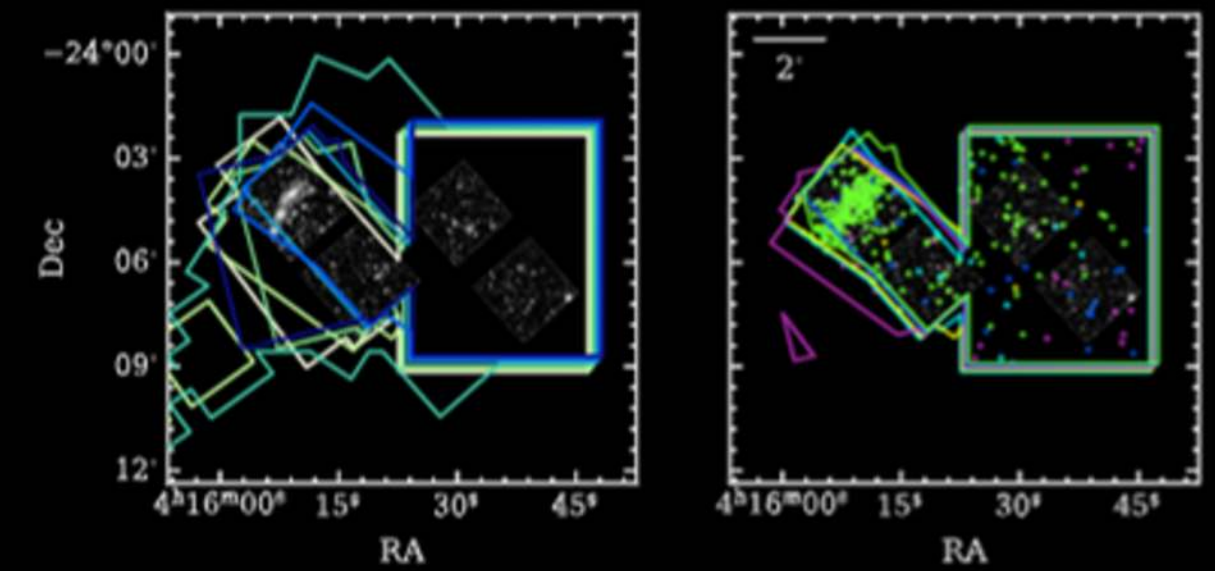
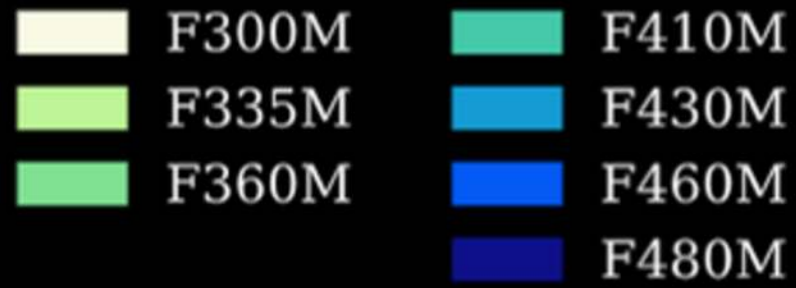
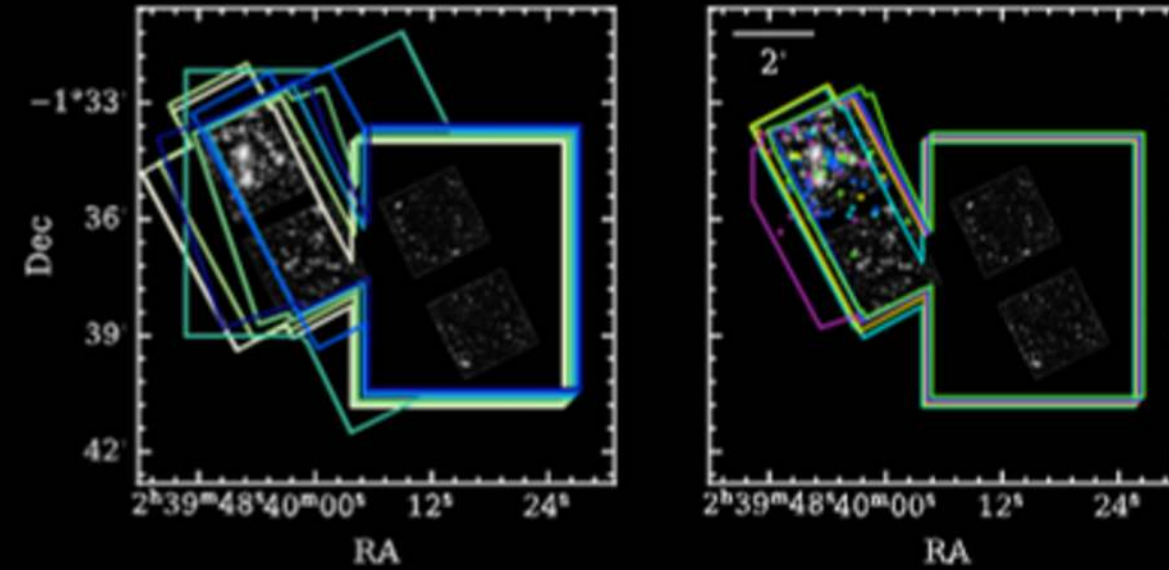
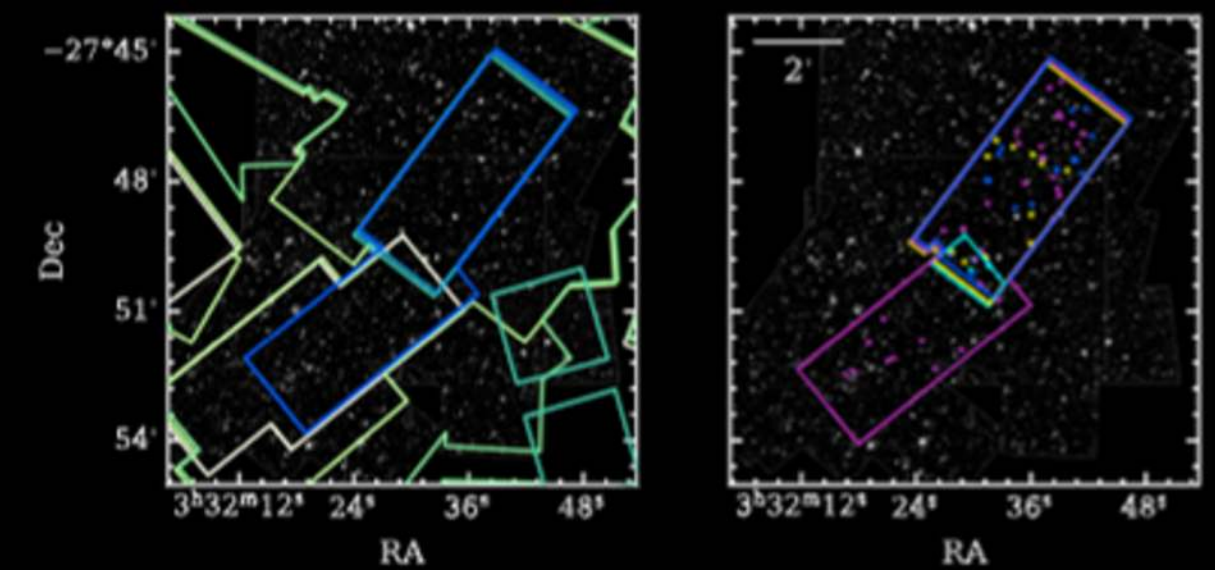
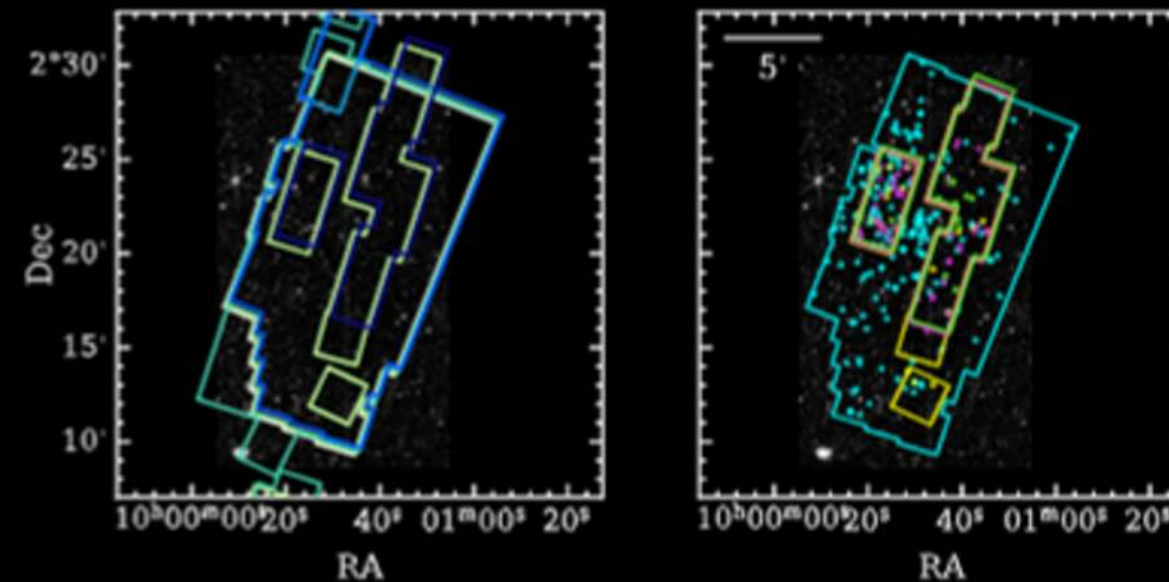
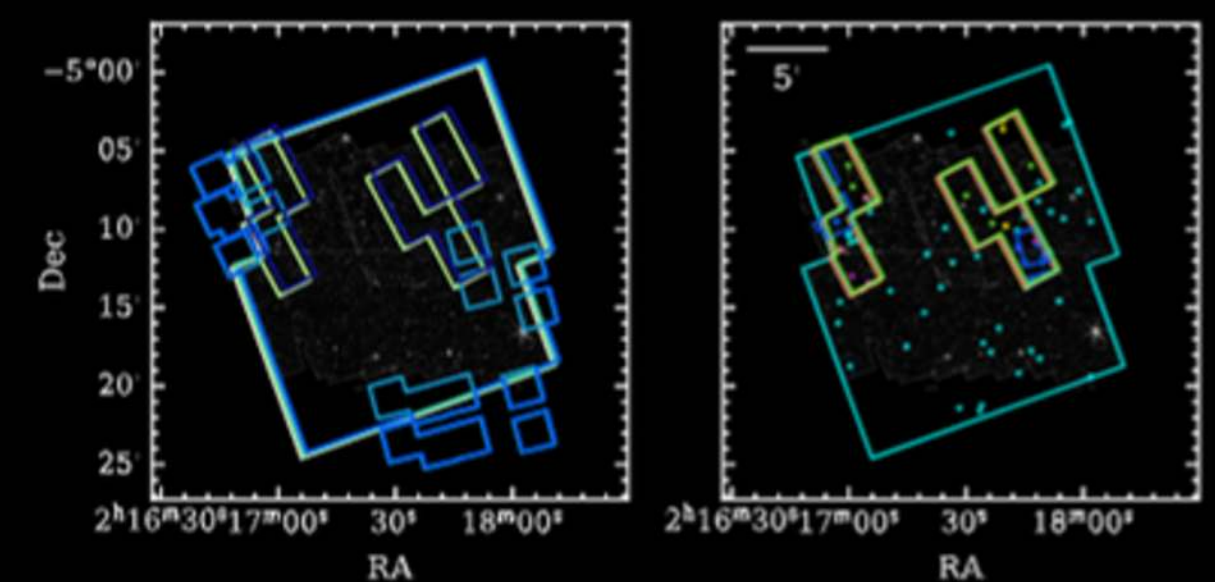


Common footprints

- | |
|---|
| F410M & F300M & F460M |
| F410M & F335M & F460M |
| F410M & F335M & F480M |
| F430M & F335M & F480M |
| F460M & F360M & F480M |

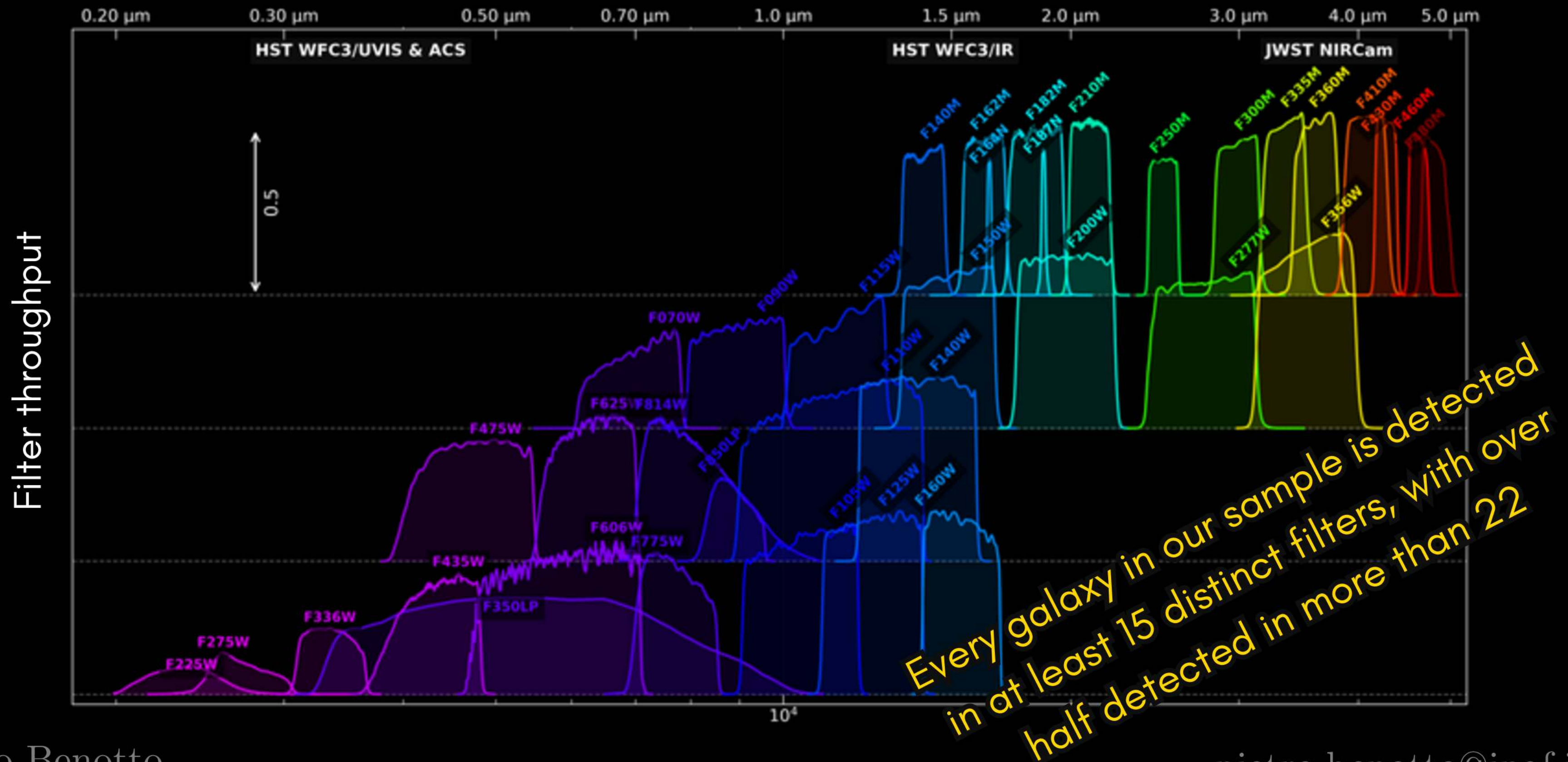
Redshift

Abell 2744

A2744**MACS0416****Filter footprints****A370****GOODS-S****Common footprints****COSMOS****UDS**

PSF matching (EPIMETHEUS) & SED fitting

Wavelength

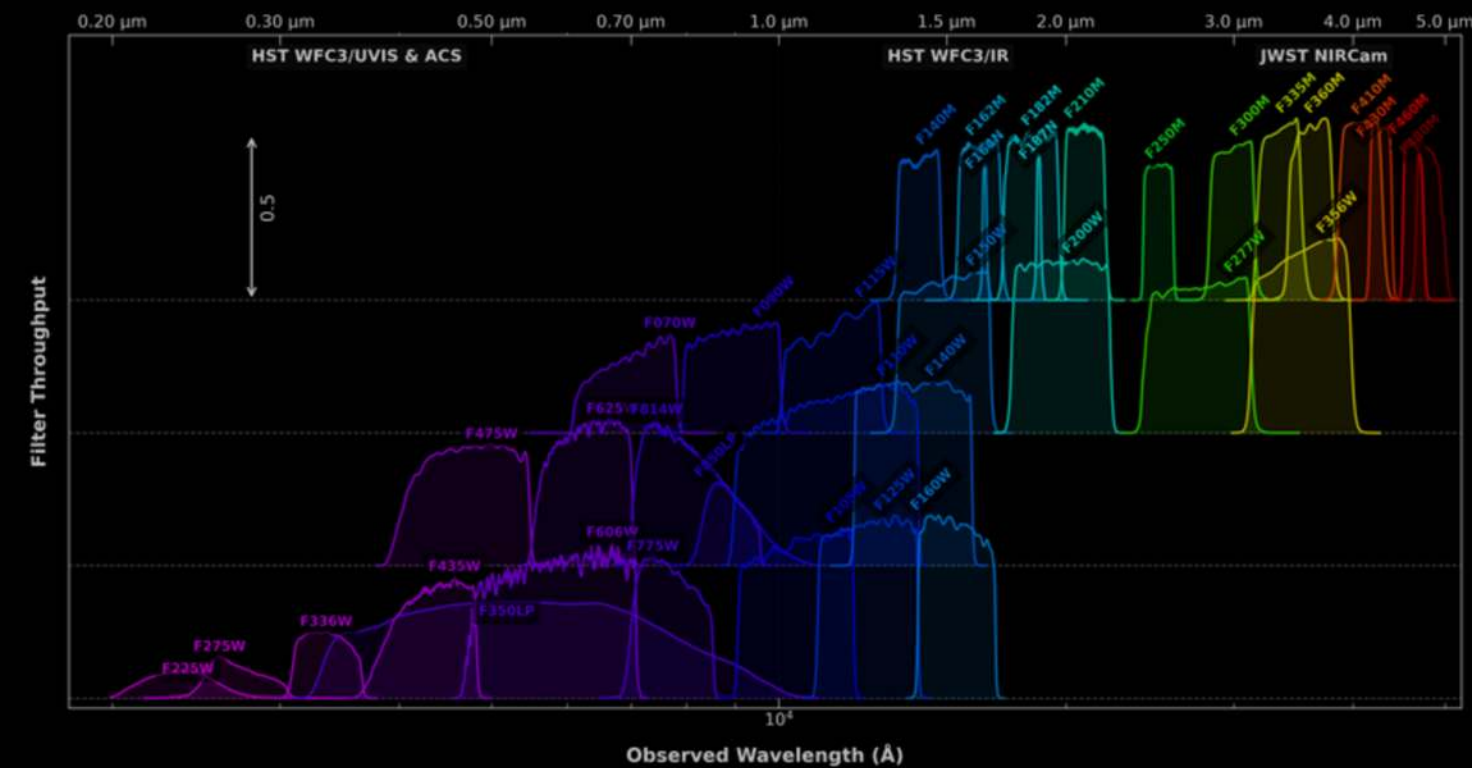


PSF matching (EPIMETHEUS) & SED fitting



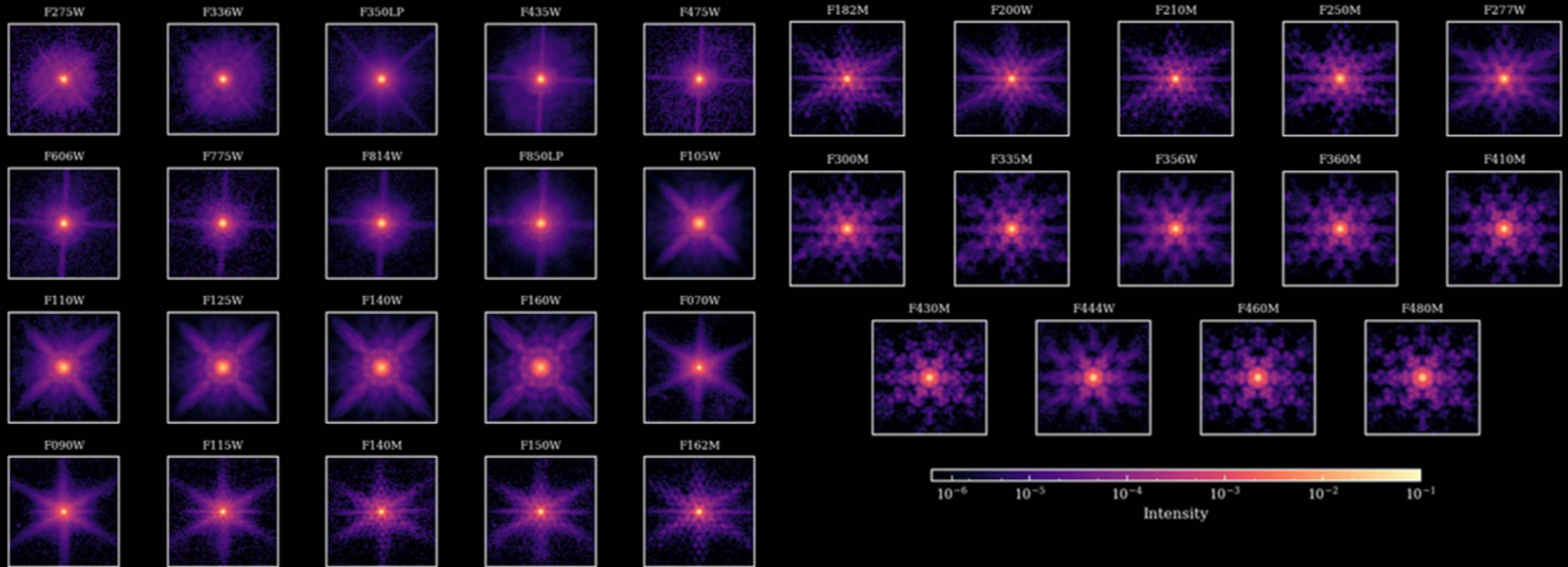
Effective
PSF
Inferred from
Mosaics using an
Estimation
Technique for
Heterogeneous
Exposures to get a
Unique PSF across the
Sensor field

- Semi-automatic star selection
- Automatic rotation computation
- Can use multiple mosaics at once
(useful in fields with few stars)



Benotto+2026b
(submitted to A&A)

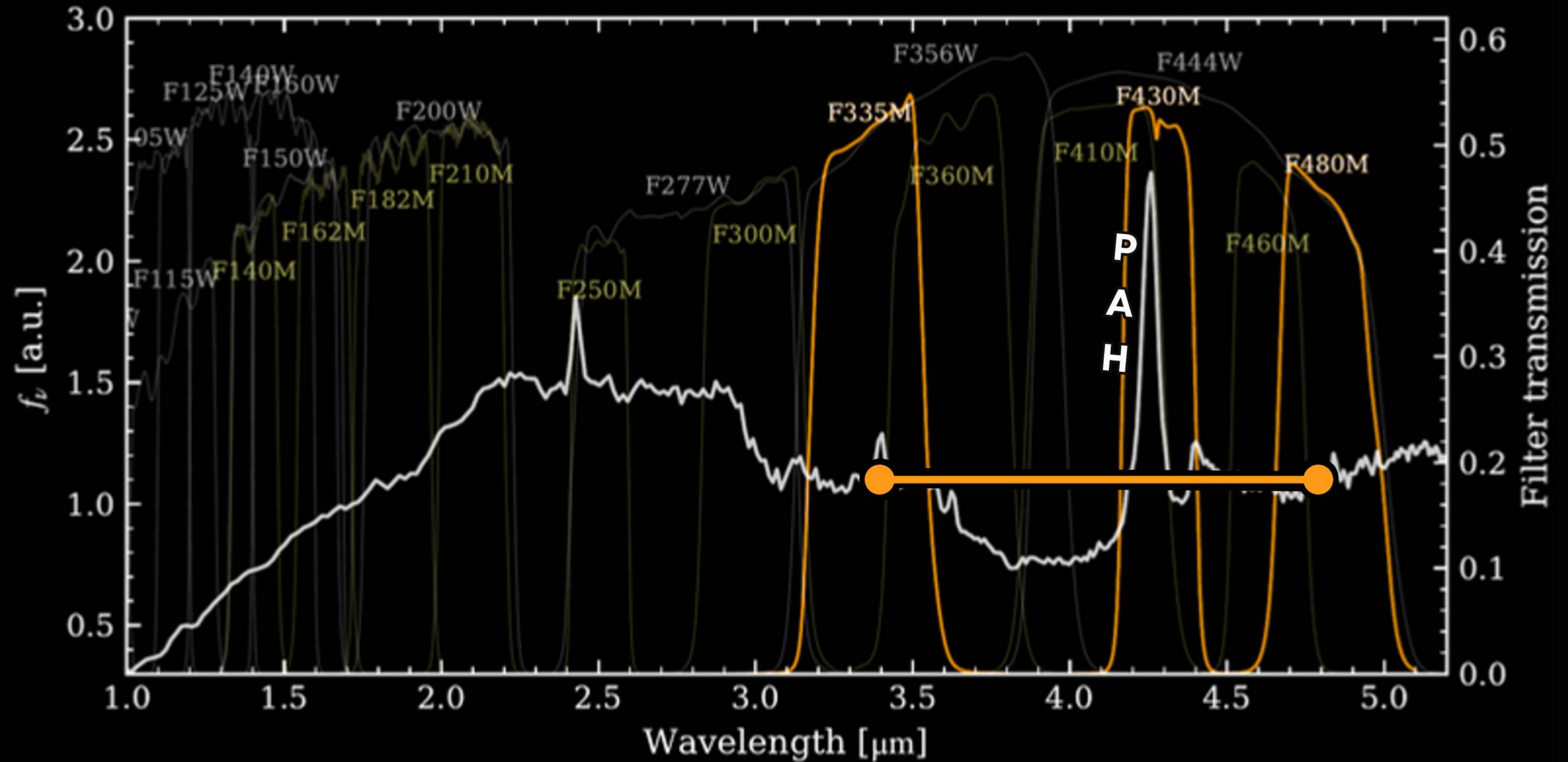
PSF matching (**EPIMETHEUS**) & SED fitting



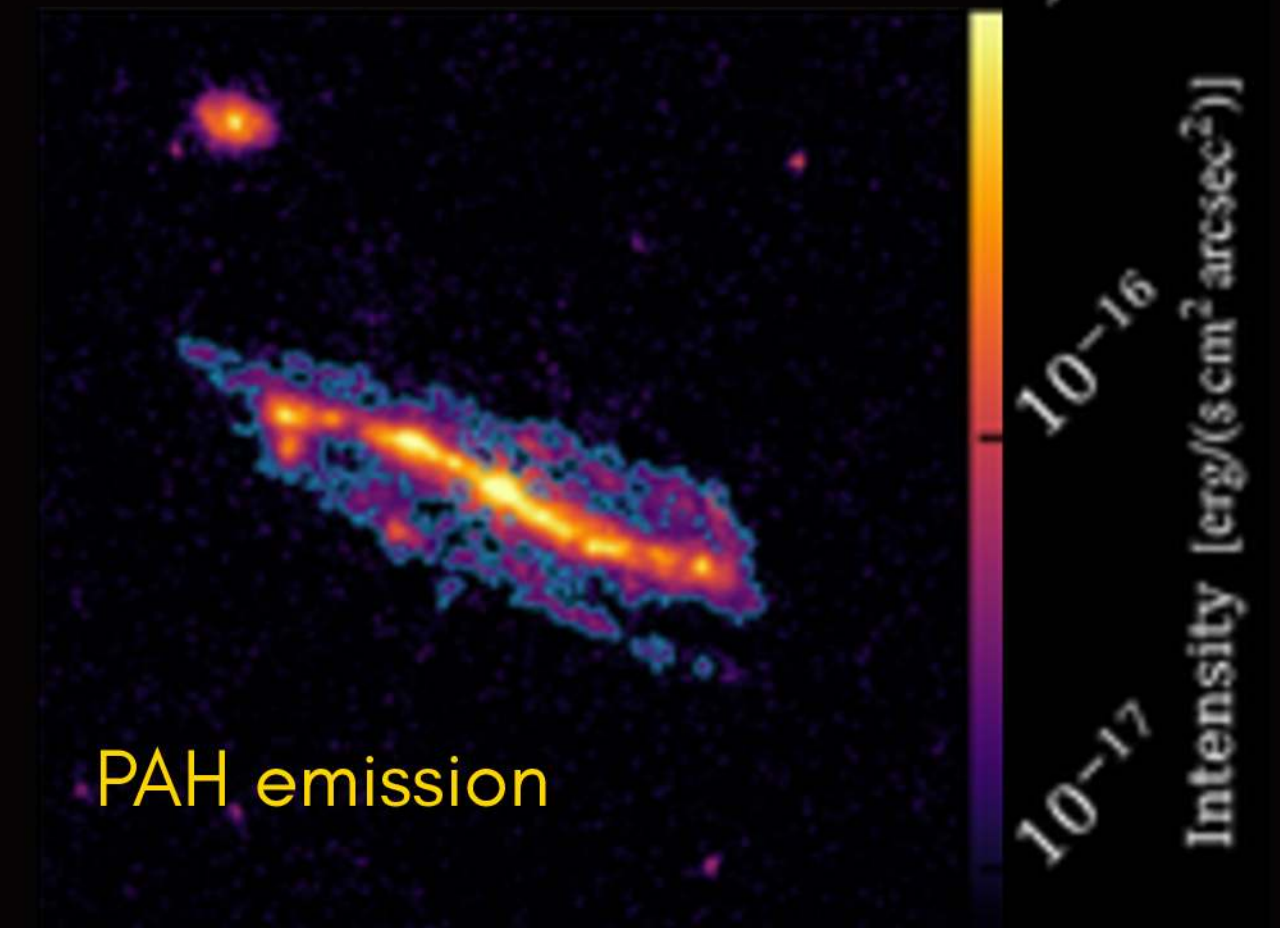
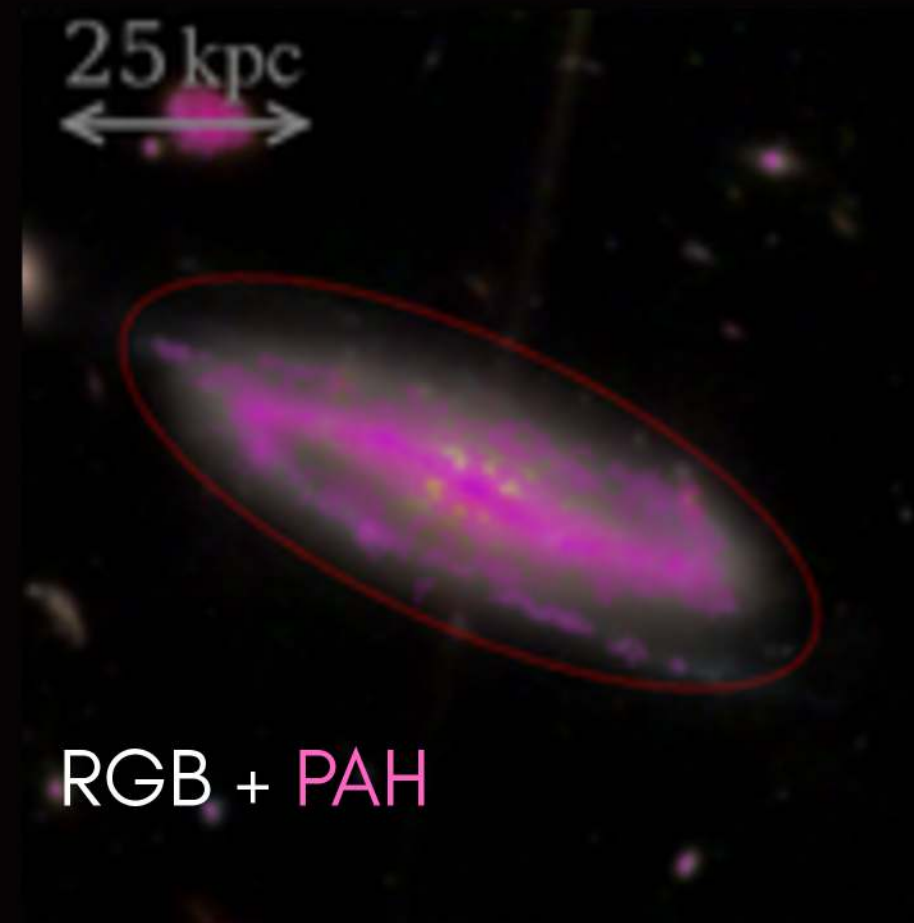
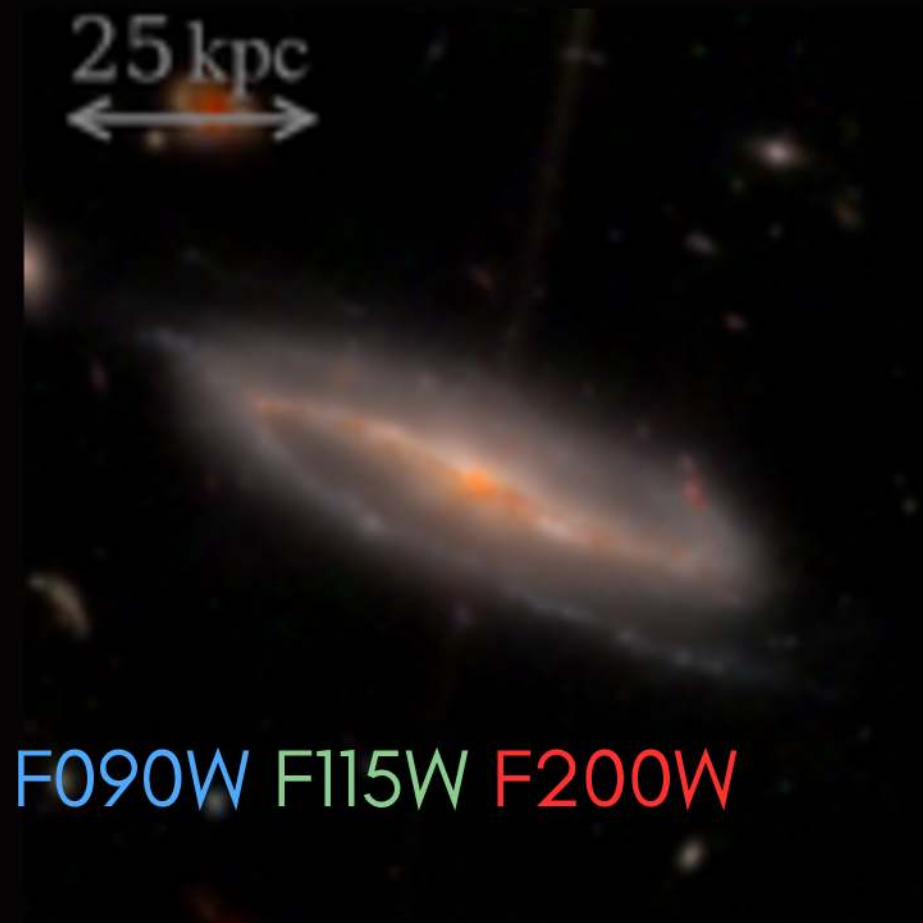
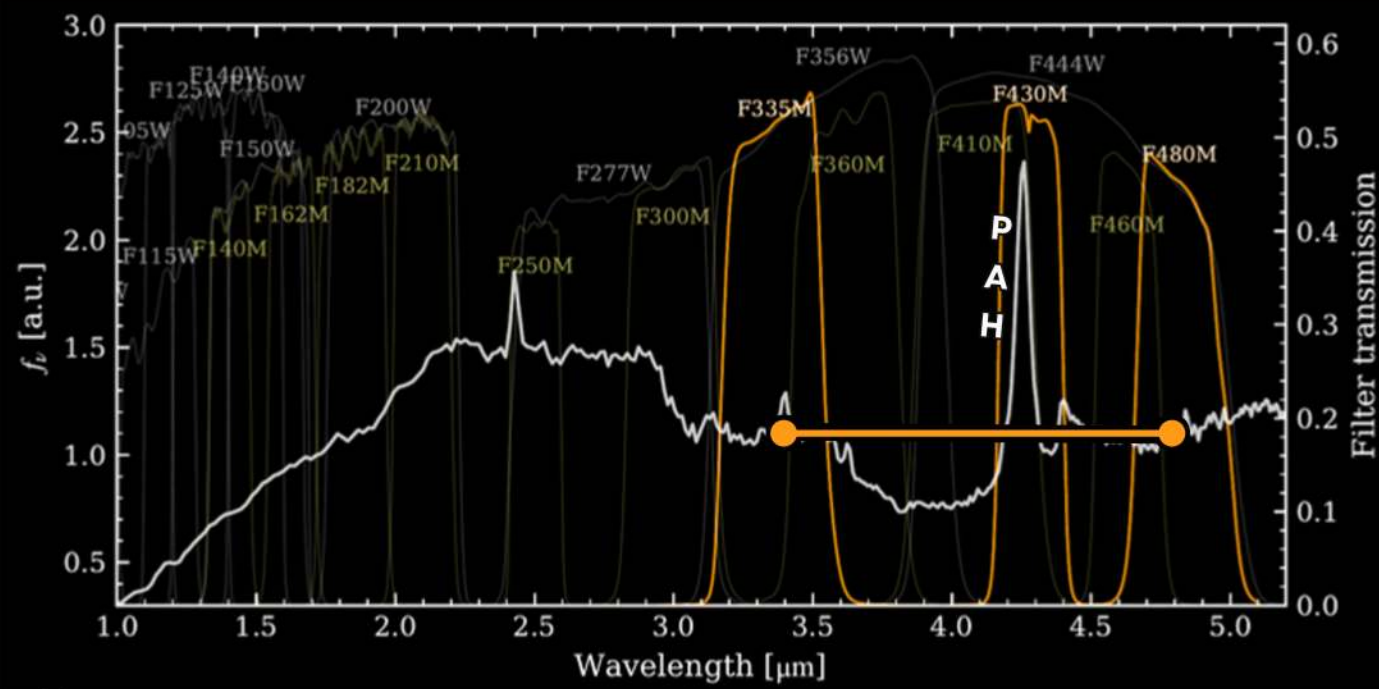
Benotto+2026b

(submitted to A&A)

PAH flux extraction



PAH flux extraction



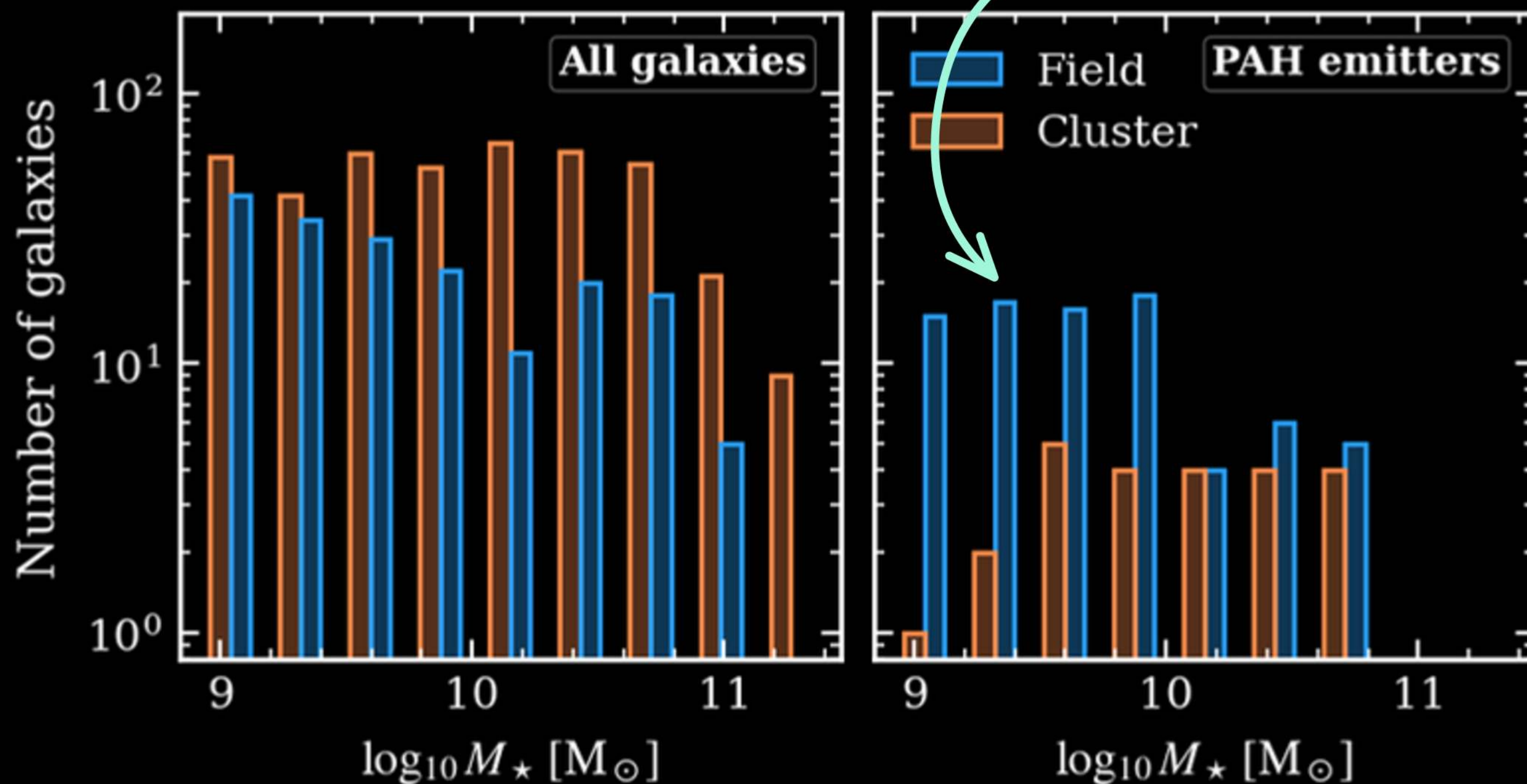
606
galaxies

425 cluster members

181 field galaxies

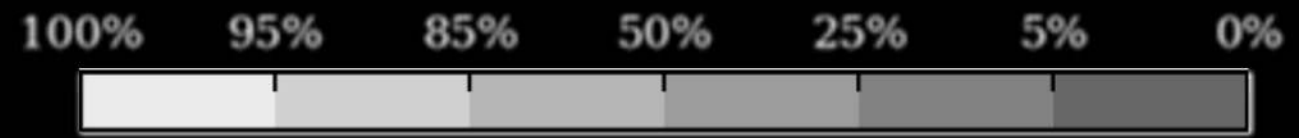
Depletion of low
mass emitters in
clusters

Sample complete
for
 $M_{\star} > 10^{8.9} M_{\odot}$

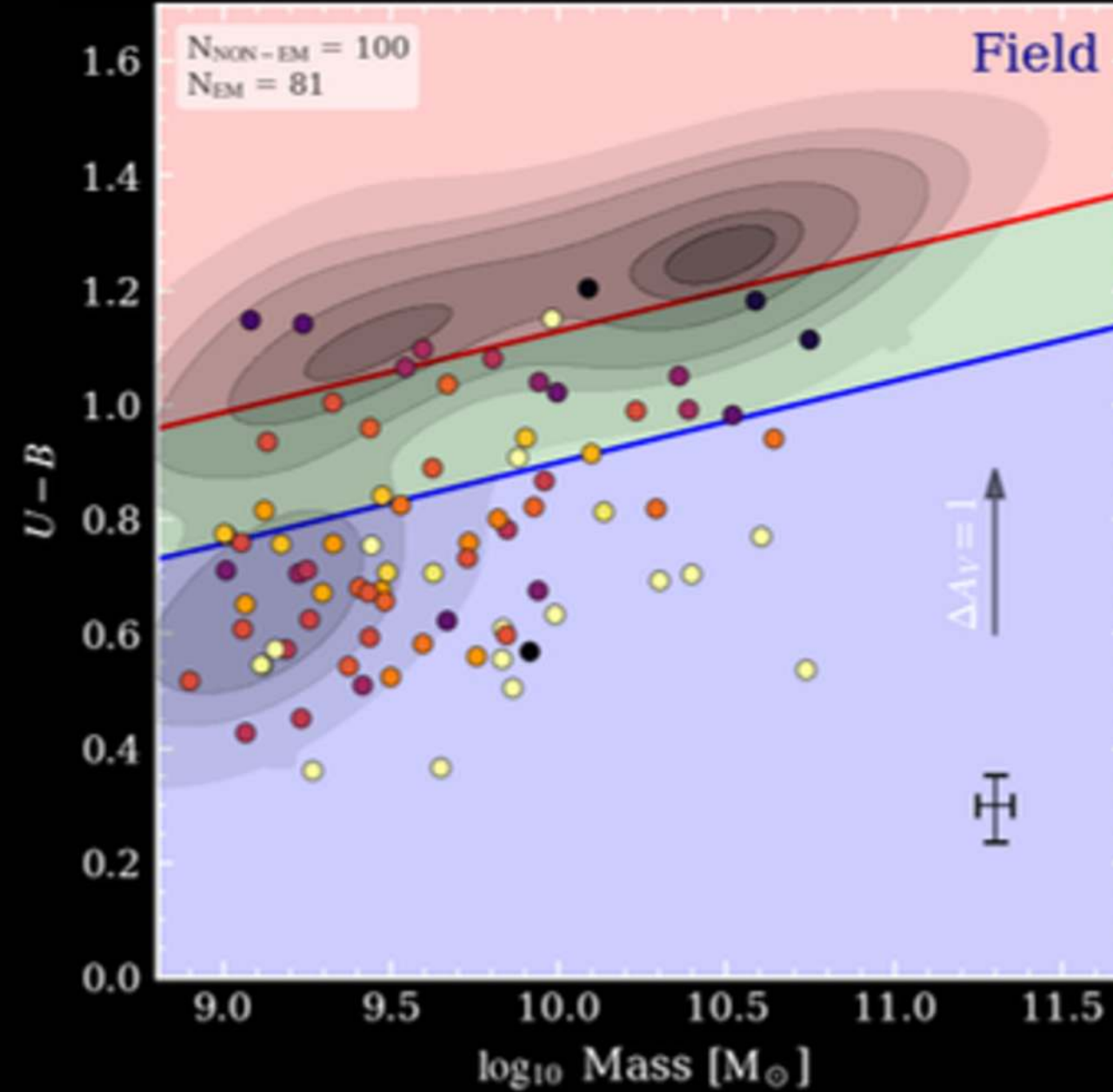
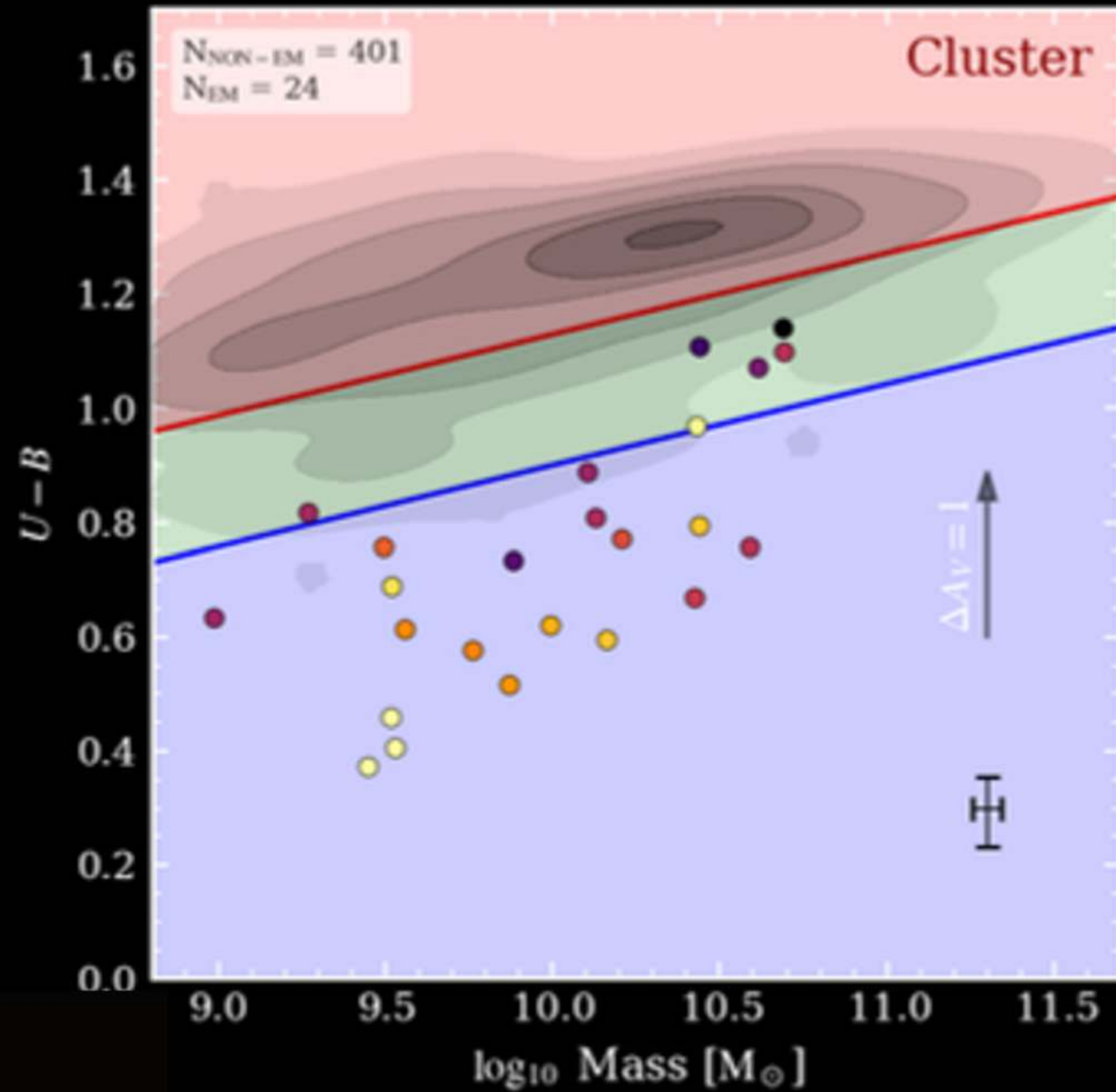
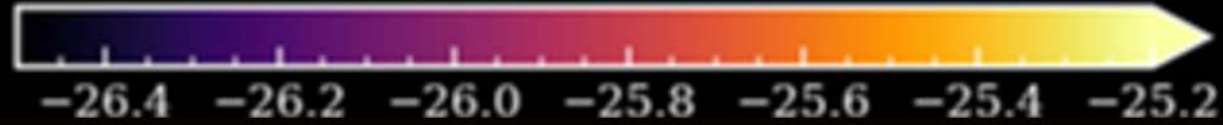


U-B vs Stellar Mass diagram

Fraction of non-emitters enclosed

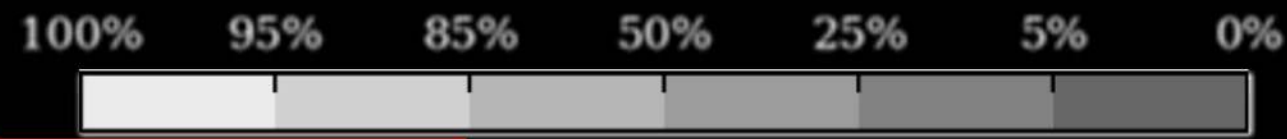


$\log s\text{PAH}[\text{erg s}^{-1} \text{cm}^{-2} \text{M}_{\odot}^{-1}]$

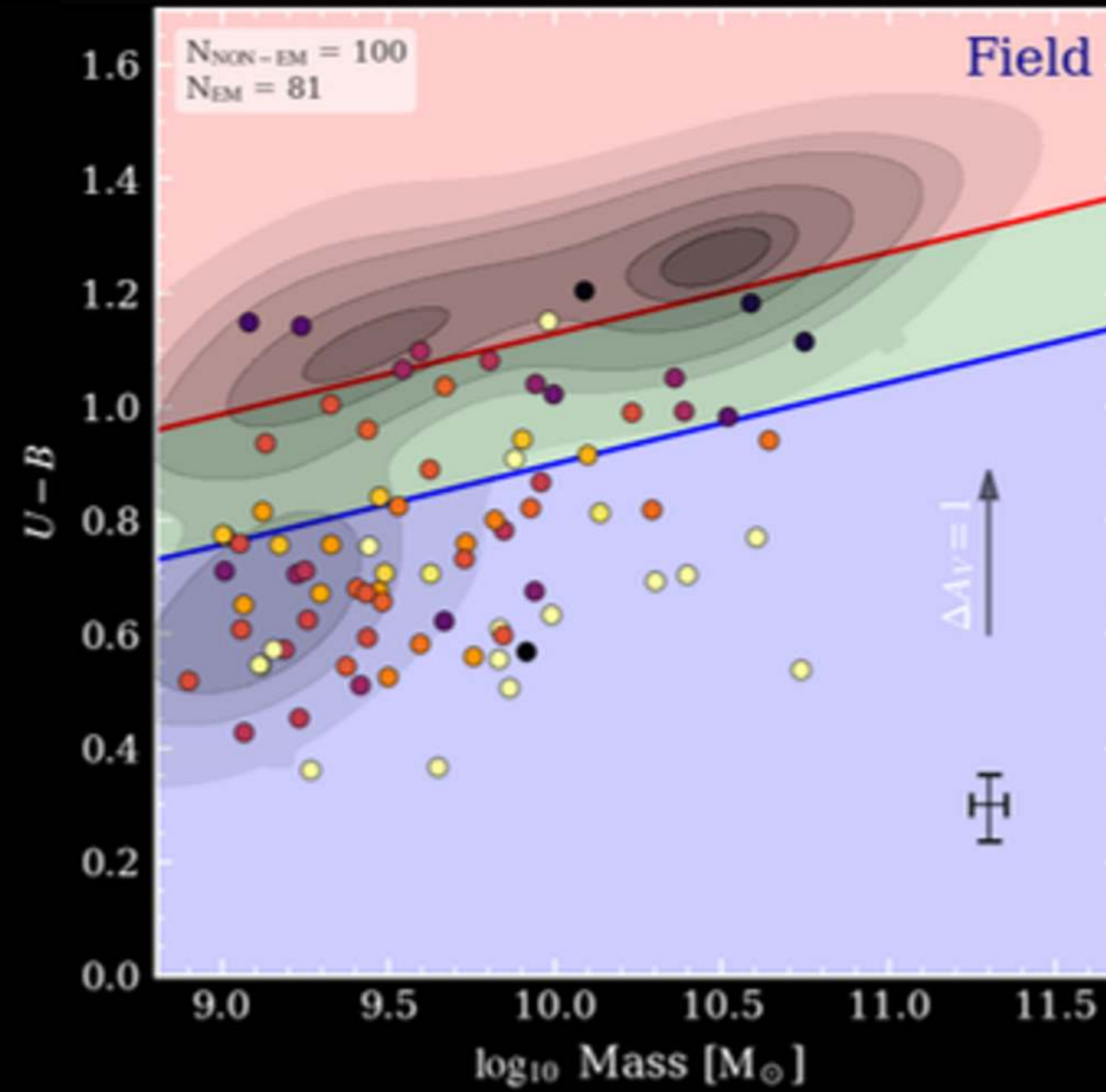
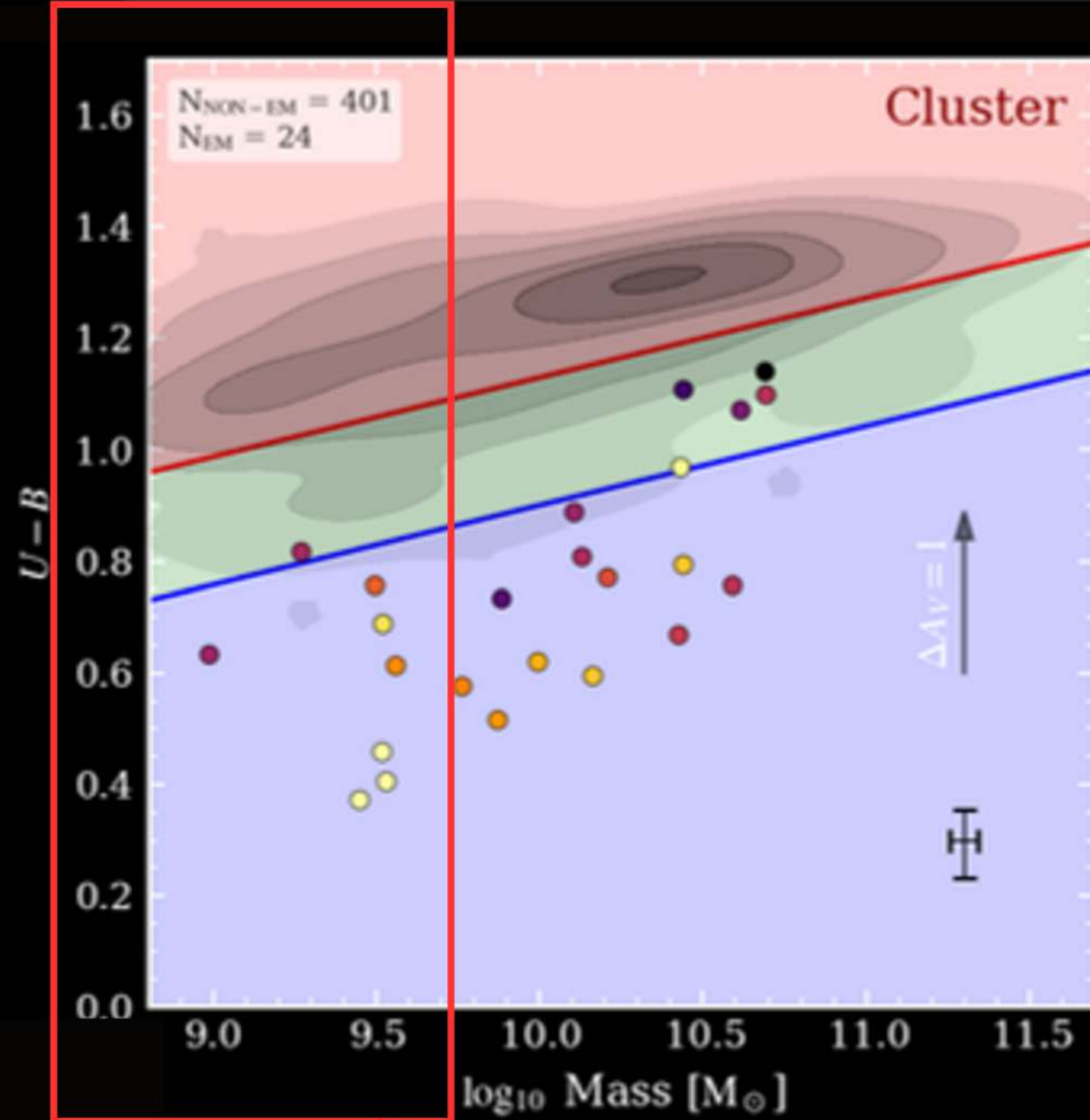
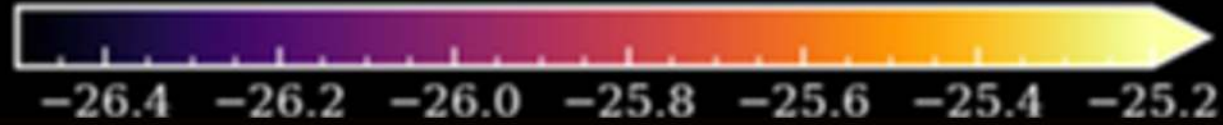


U-B vs Stellar Mass diagram

Fraction of non-emitters enclosed

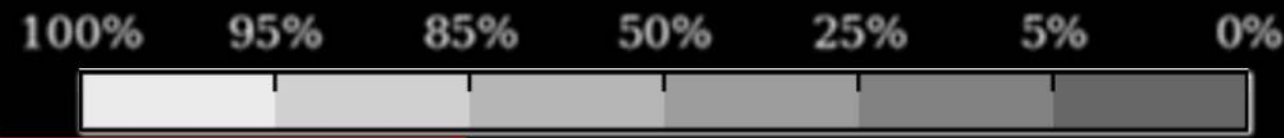


$\log s\text{PAH}[\text{erg s}^{-1} \text{cm}^{-2} \text{M}_{\odot}^{-1}]$

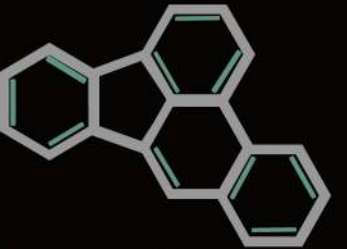
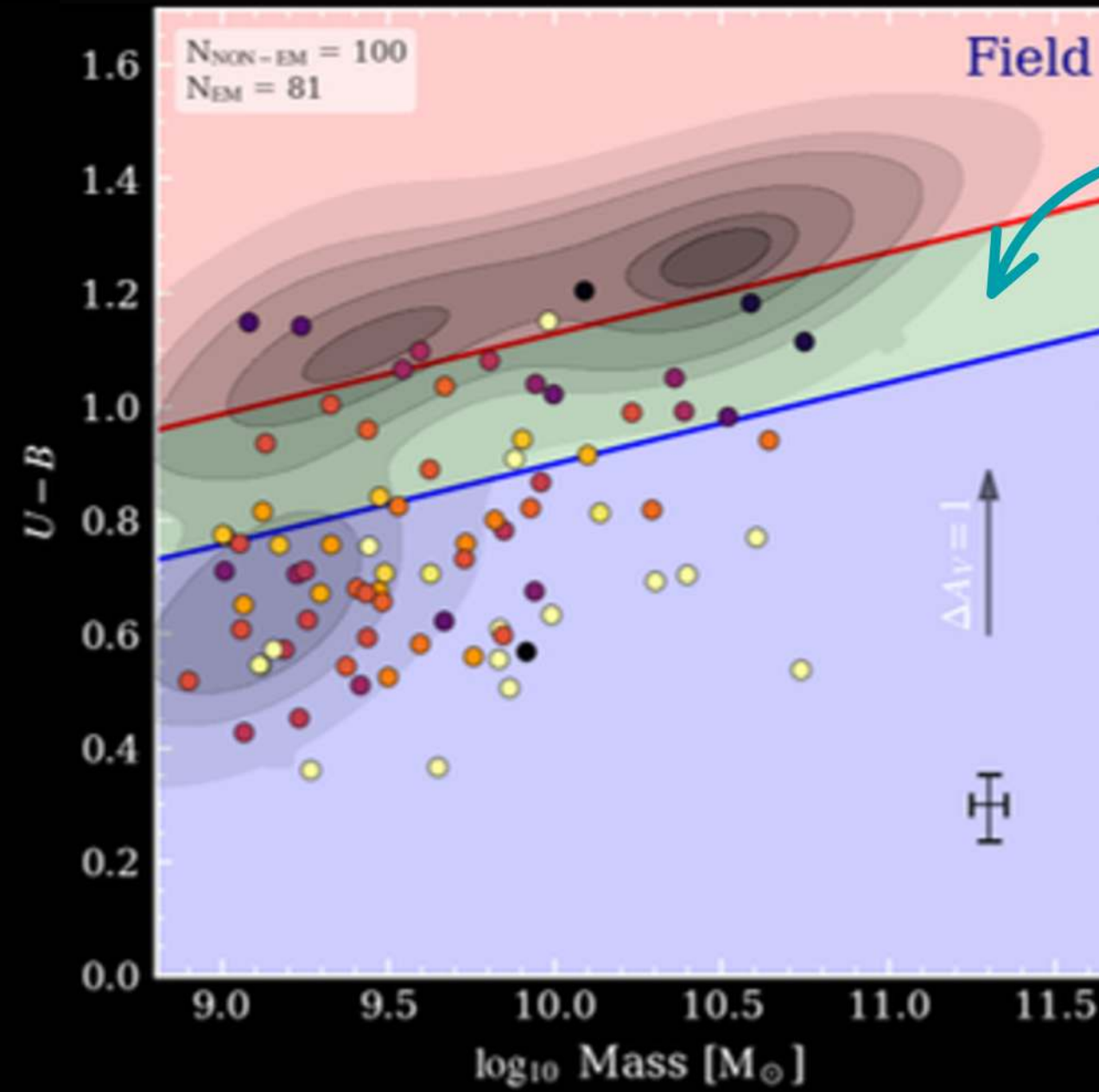
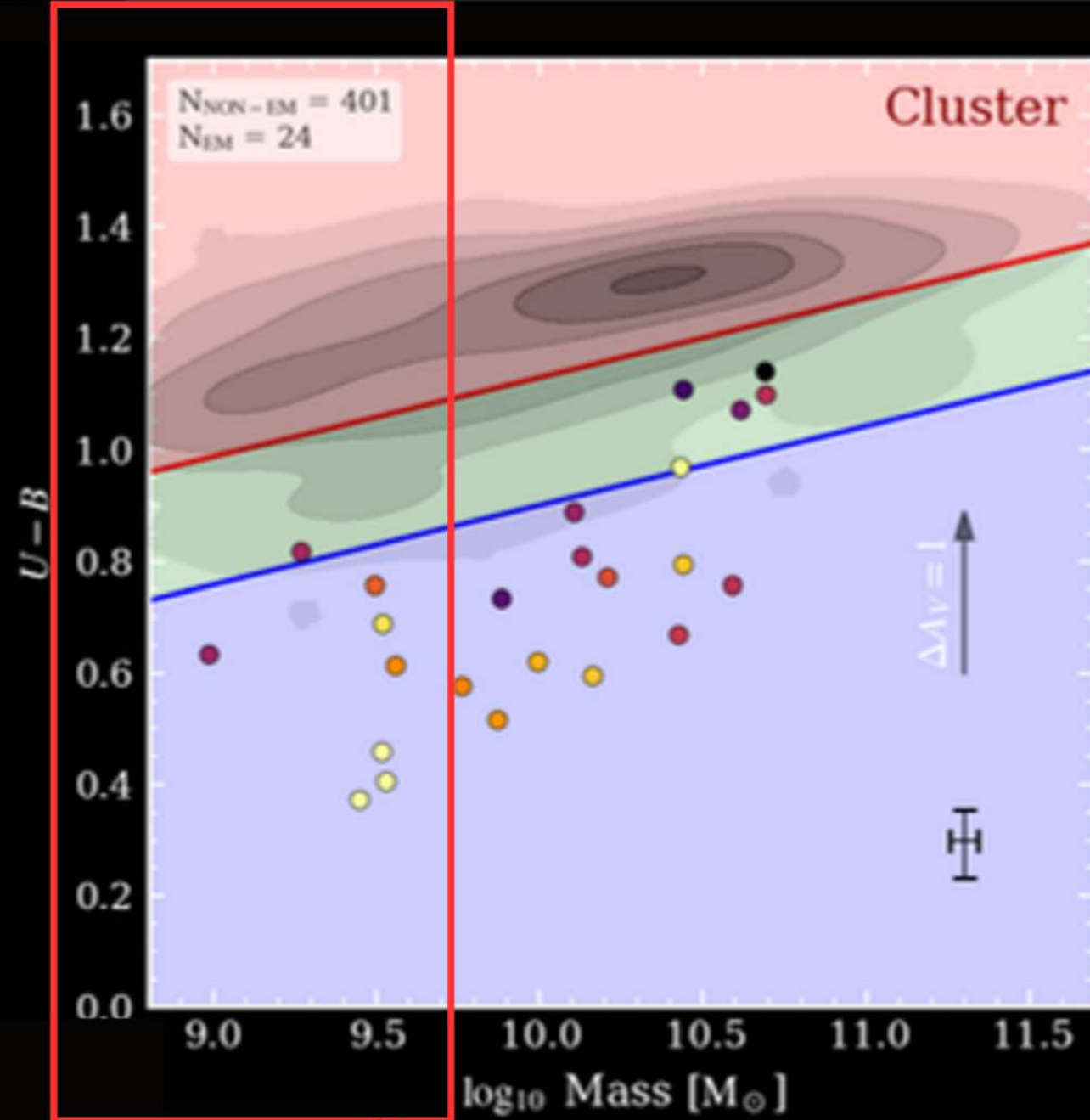
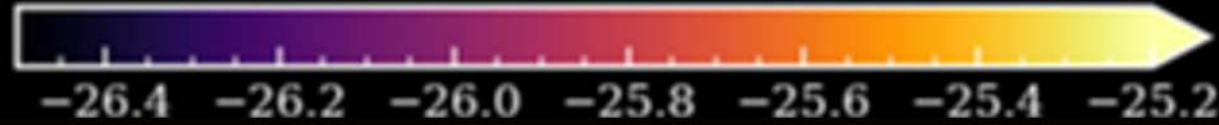


U-B vs Stellar Mass diagram

Fraction of non-emitters enclosed



$\log s\text{PAH}[\text{erg s}^{-1} \text{cm}^{-2} \text{M}_{\odot}^{-1}]$



Fraction of PAH emitters
in the green zone:
(47±6)% field vs (6±3)%
in the cluster

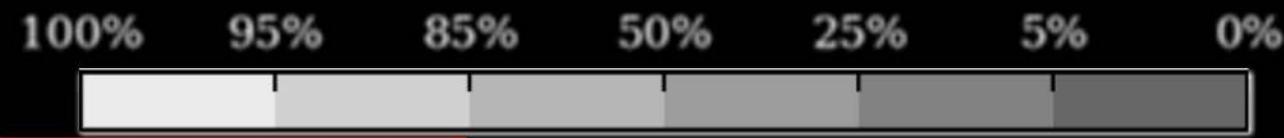
Pietro Benotto

Low-mass cluster galaxies
dominated by quenched spheroids

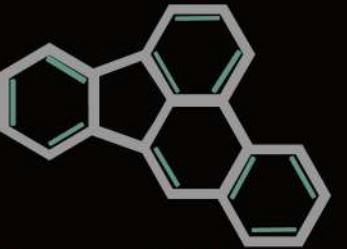
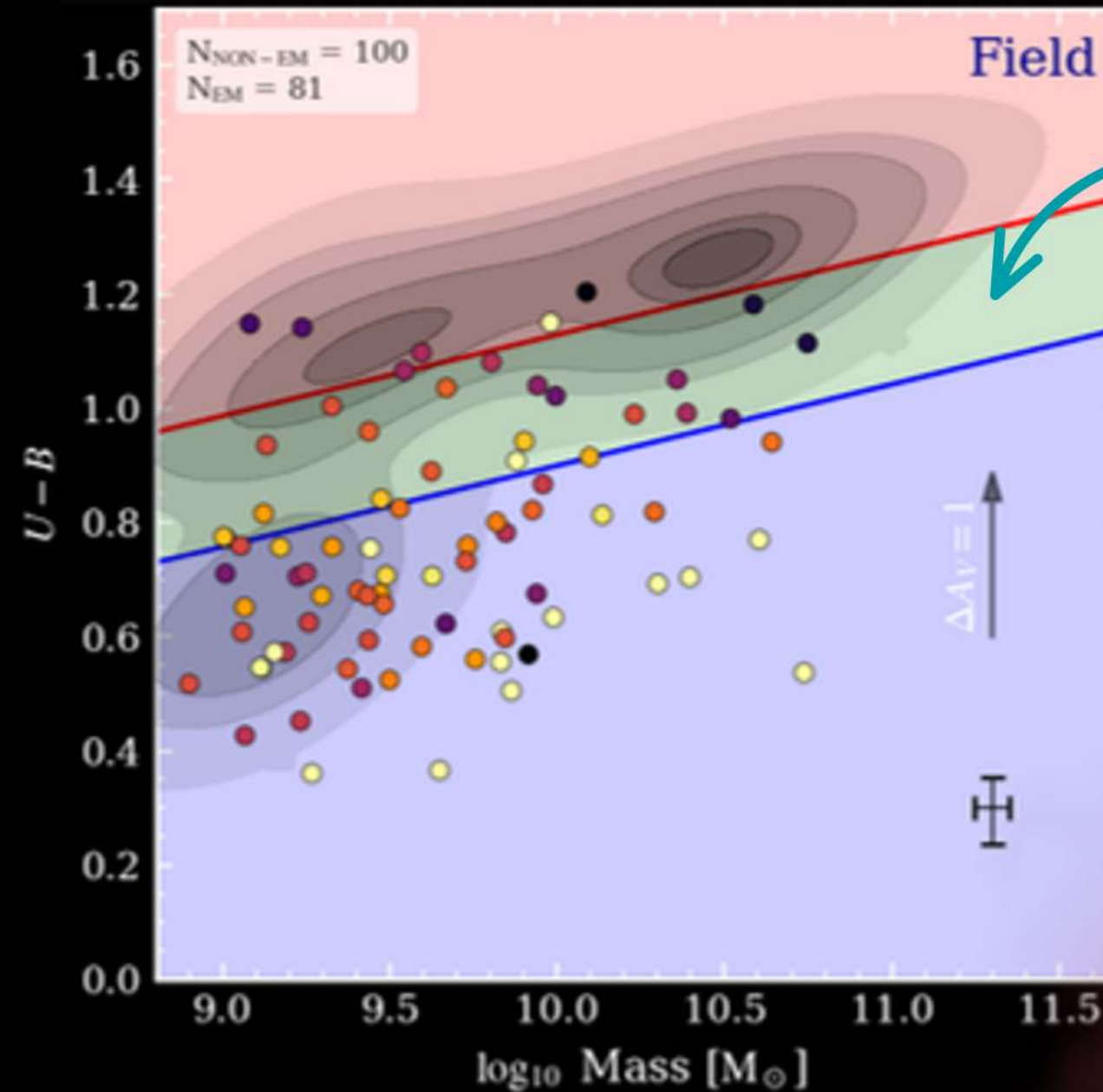
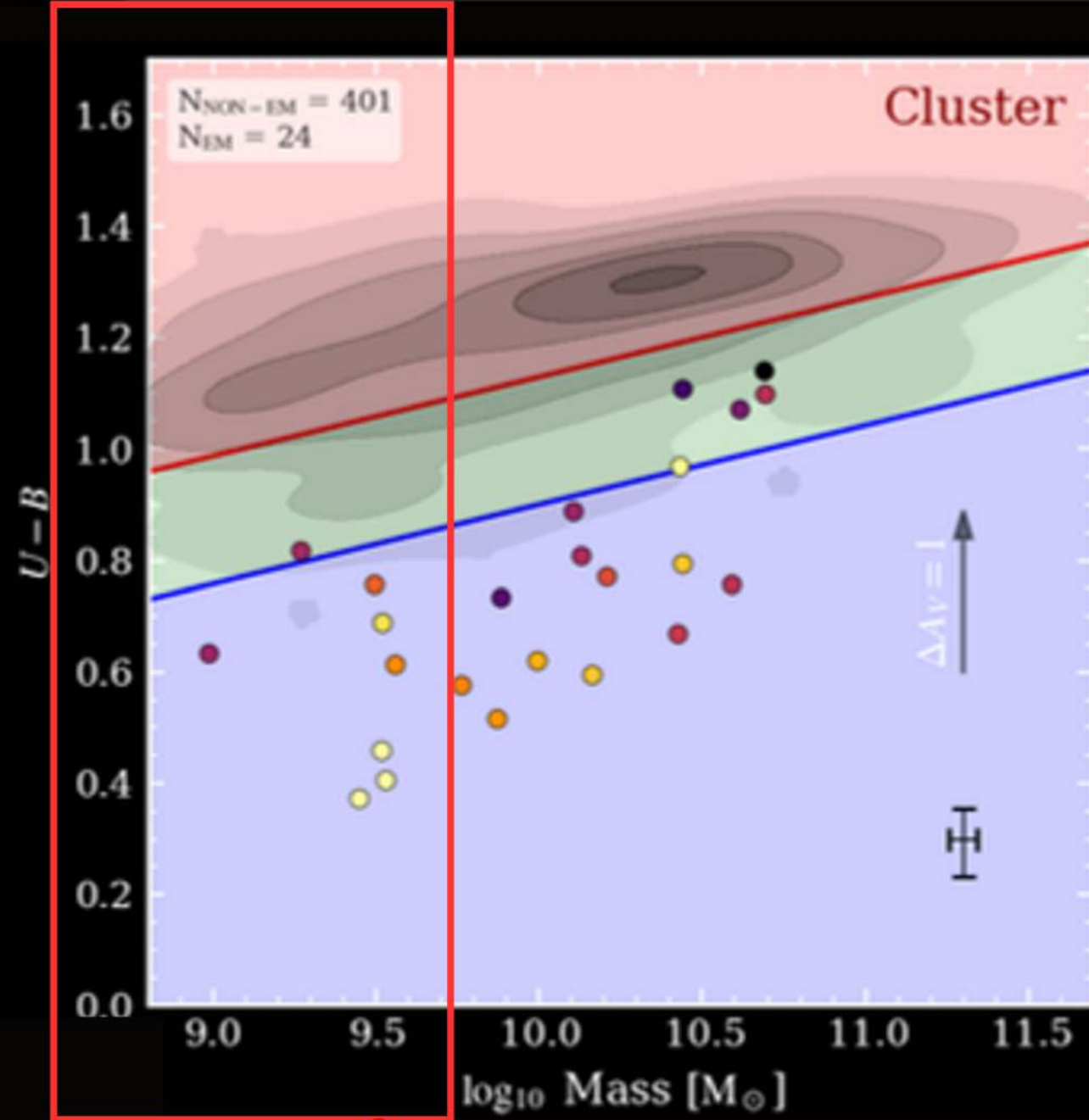
pietro.benotto@inaf.it

U-B vs Stellar Mass diagram

Fraction of non-emitters enclosed



$\log s\text{PAH}[\text{erg s}^{-1} \text{cm}^{-2} \text{M}_{\odot}^{-1}]$

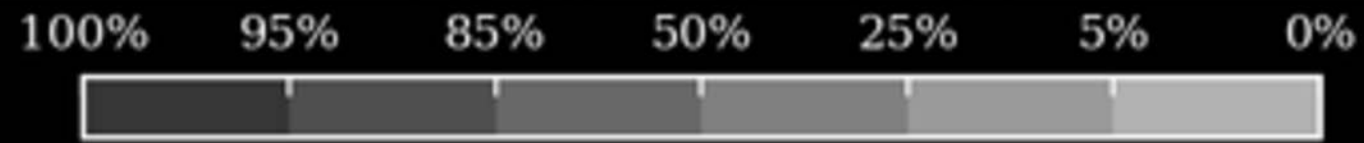


Fraction of PAH emitters
in the green zone:
(47±6)% field vs (6±3)%
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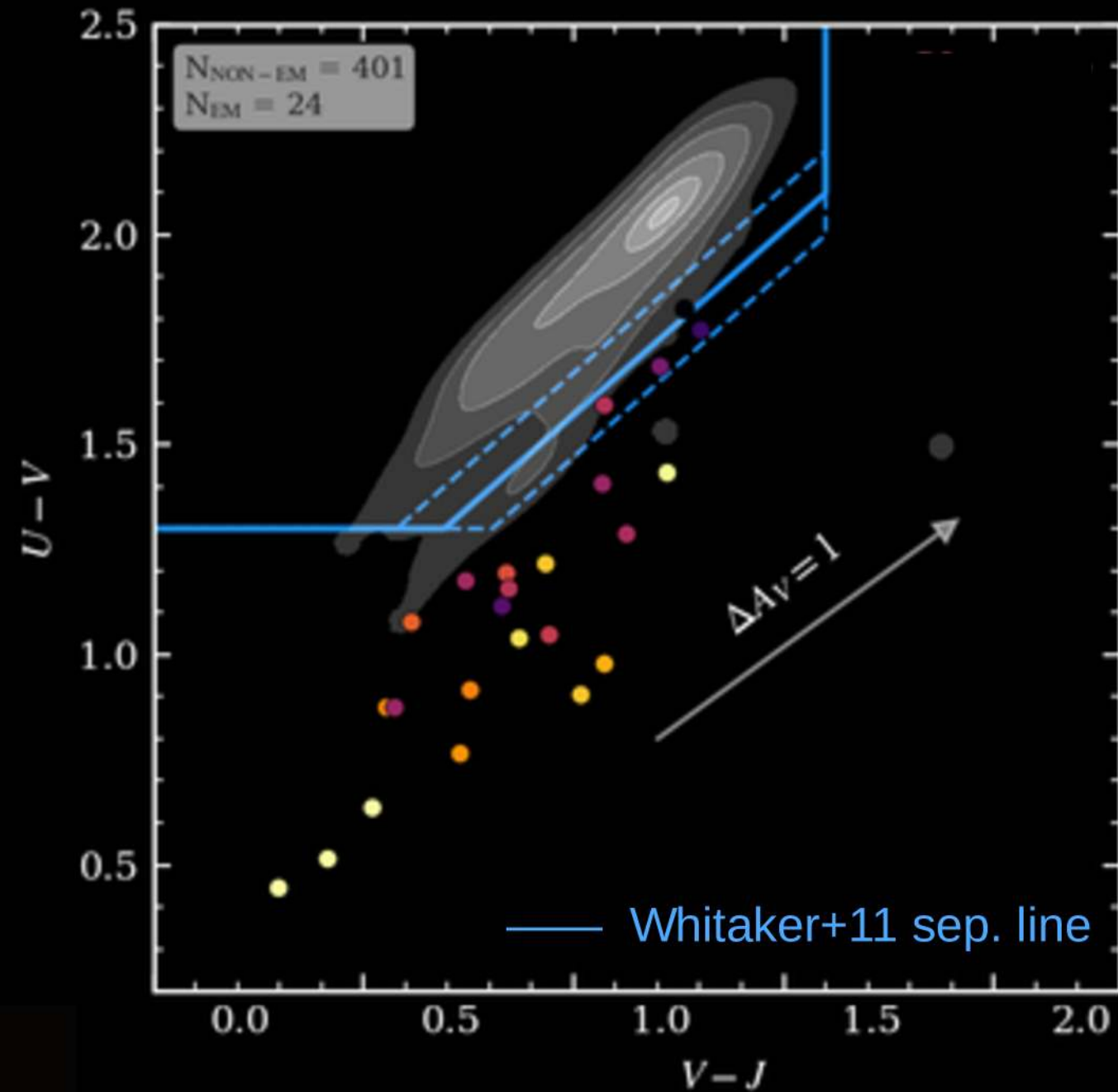
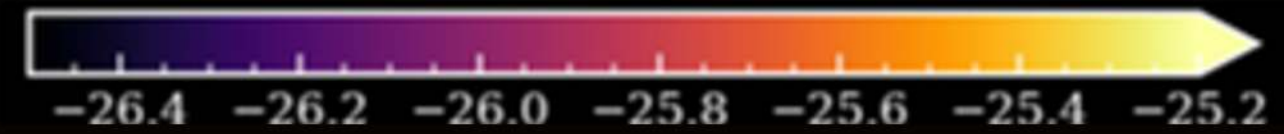
Dust attenuation may
affects the result!

UVJ diagram

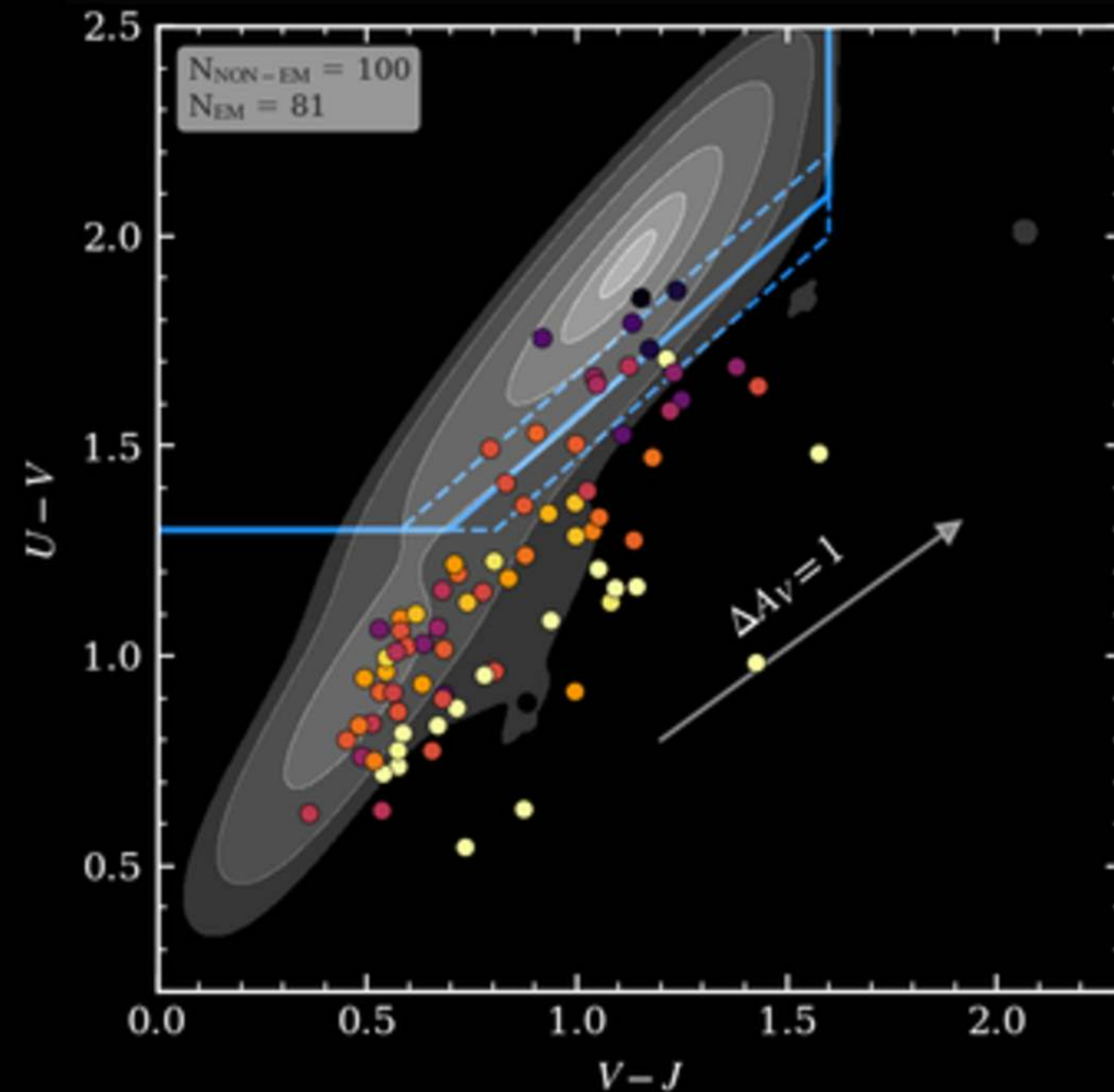
Fraction of non-emitters enclosed



$\log s\text{PAH}[\text{erg s}^{-1} \text{cm}^{-2} \text{M}_{\odot}^{-1}]$

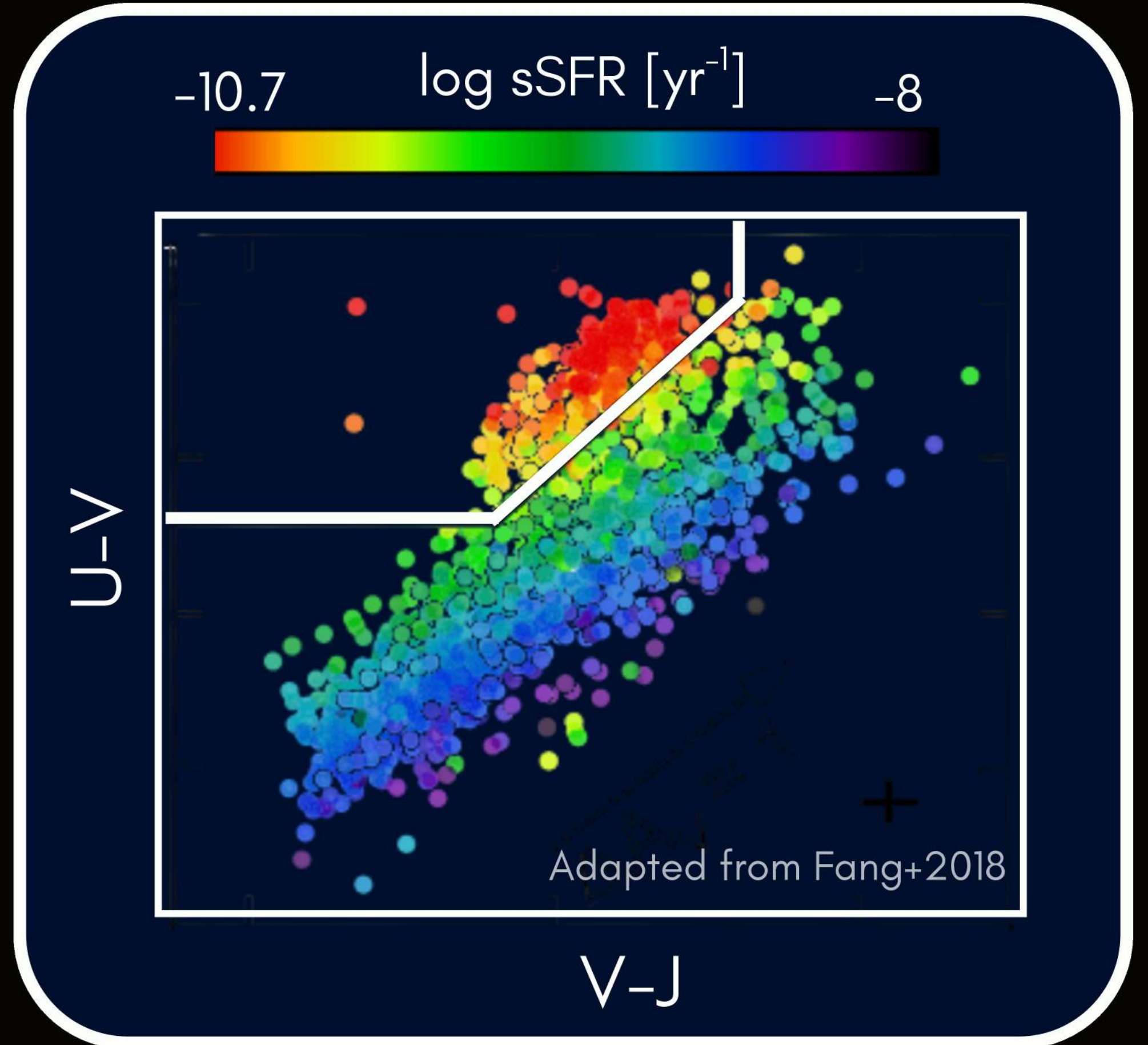
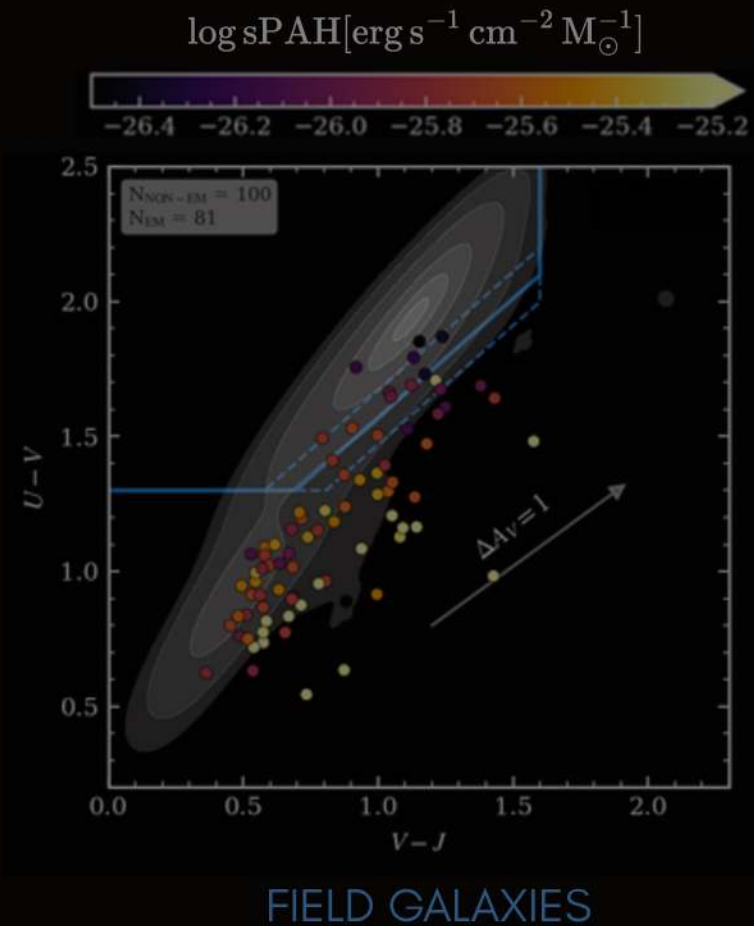
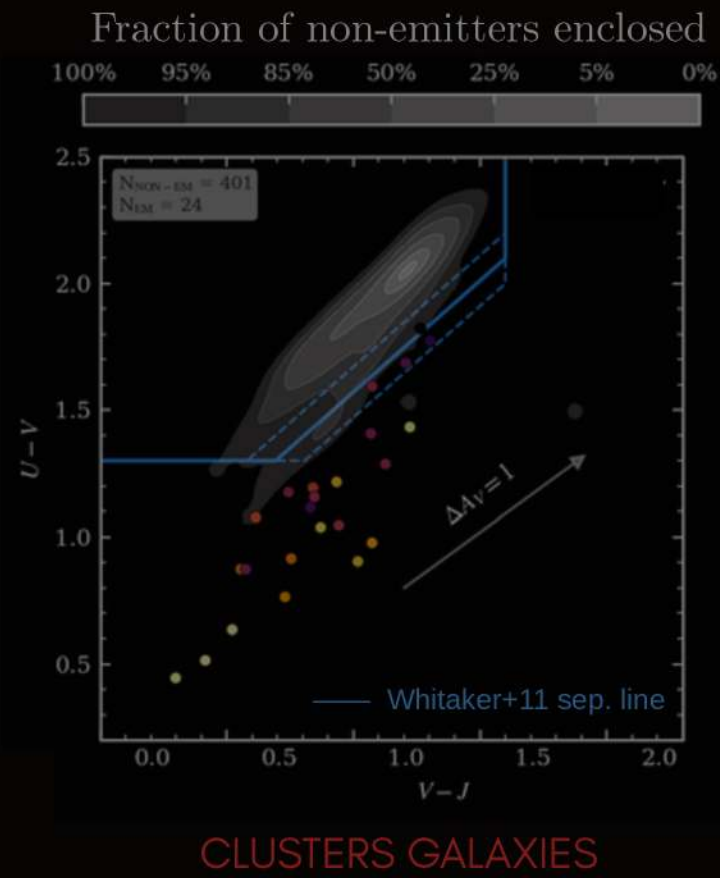


CLUSTERS GALAXIES

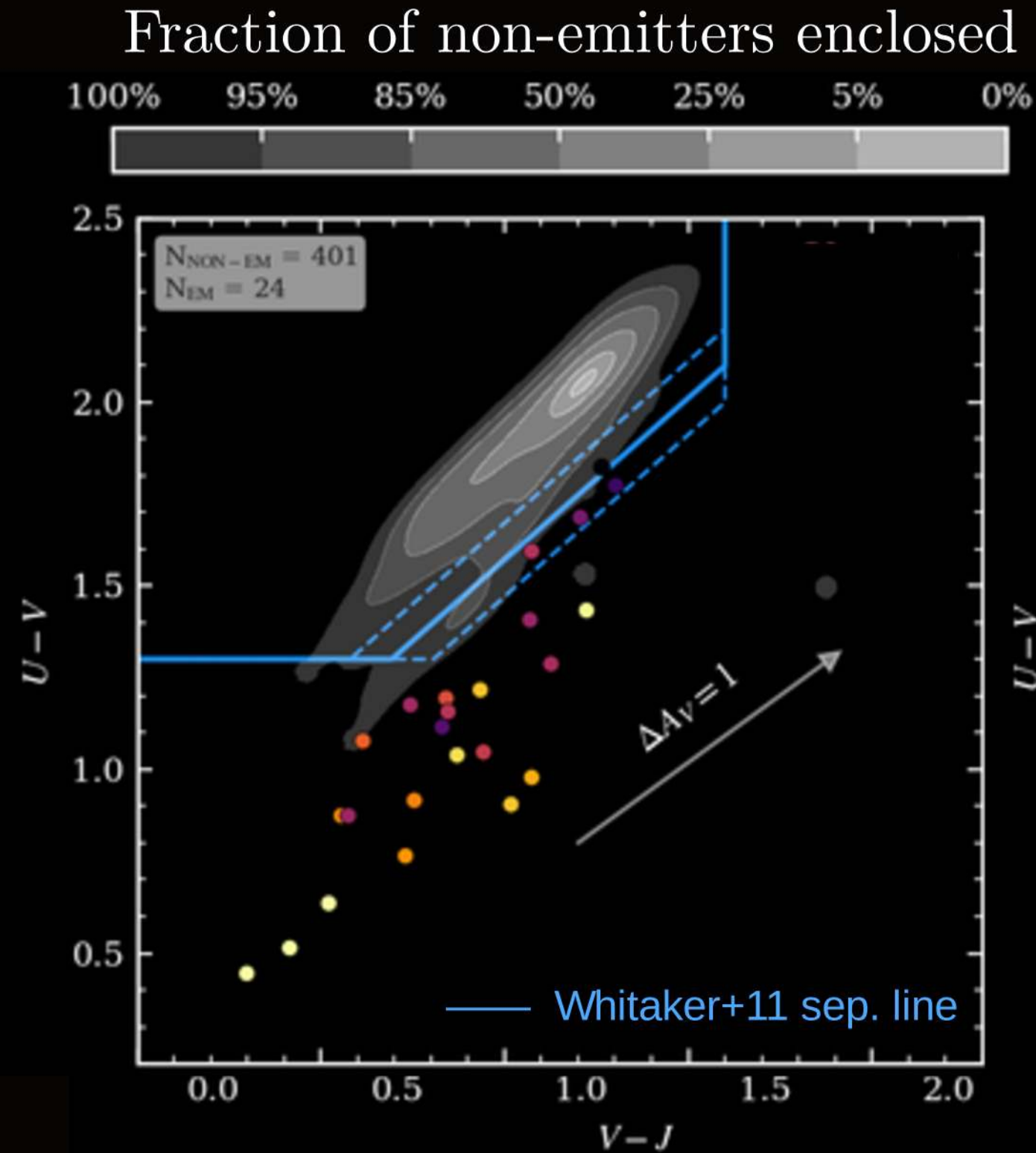


FIELD GALAXIES

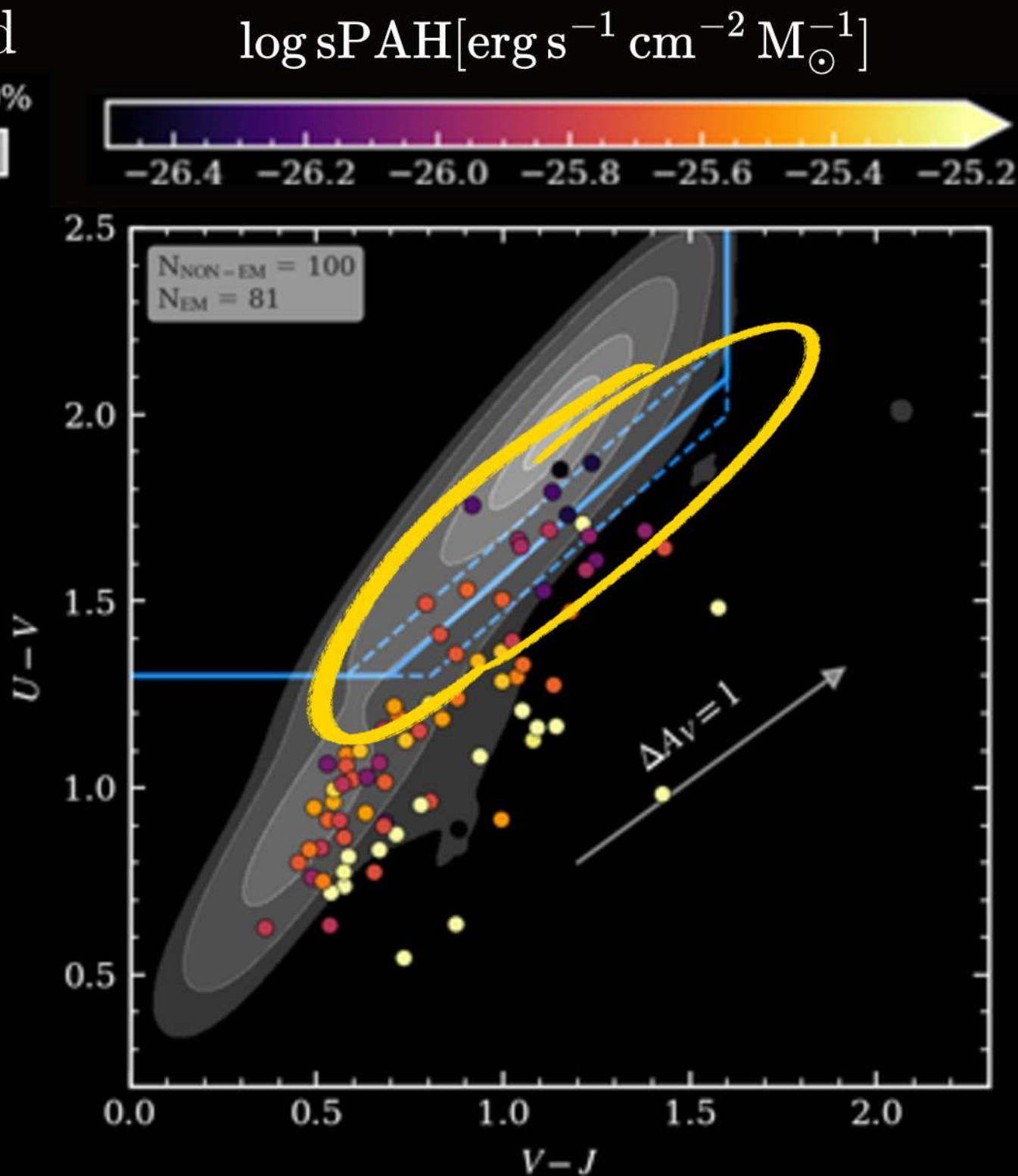
UVJ diagram



UVJ diagram



CLUSTERS GALAXIES

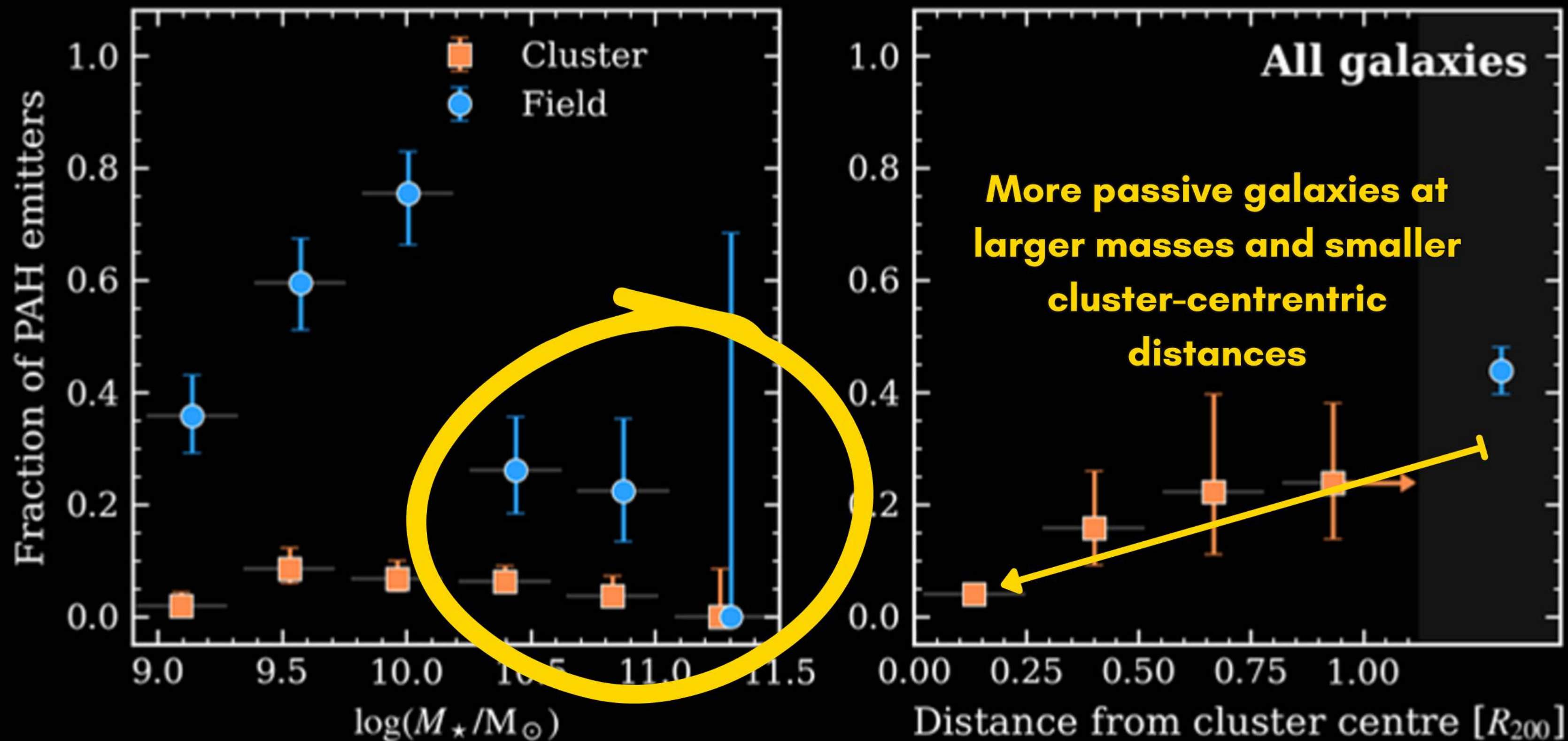


FIELD GALAXIES

Transition zone similar to McNab+22 and Mao+22

Defined as the region extending within ± 0.1 magnitudes on either side of the Whitaker+11 separation line

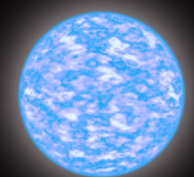
Fraction of PAH emitters



Fraction of PAH emitters

CLUSTER

FIELD



STAR-FORMING

59%

69%

($\pm 6\%$)



TRANSITION

9%

54%

($\pm 6\%$)



QUIESCENT

0%

5%

($\pm 2\%$)

	CLUSTER	FIELD	
<u>STAR-FORMING</u>	59%	69%	($\pm 6\%$)
<u>TRANSITION</u>	9%	54%	($\pm 6\%$)
<u>QUIESCENT</u>	0%	5%	($\pm 2\%$)



$$M_{\star} < 10^{10.2} M_{\odot}$$

CLUSTER	FIELD
$0^{+3}_{-0}\%$	$72^{+9}_{-12}\%$

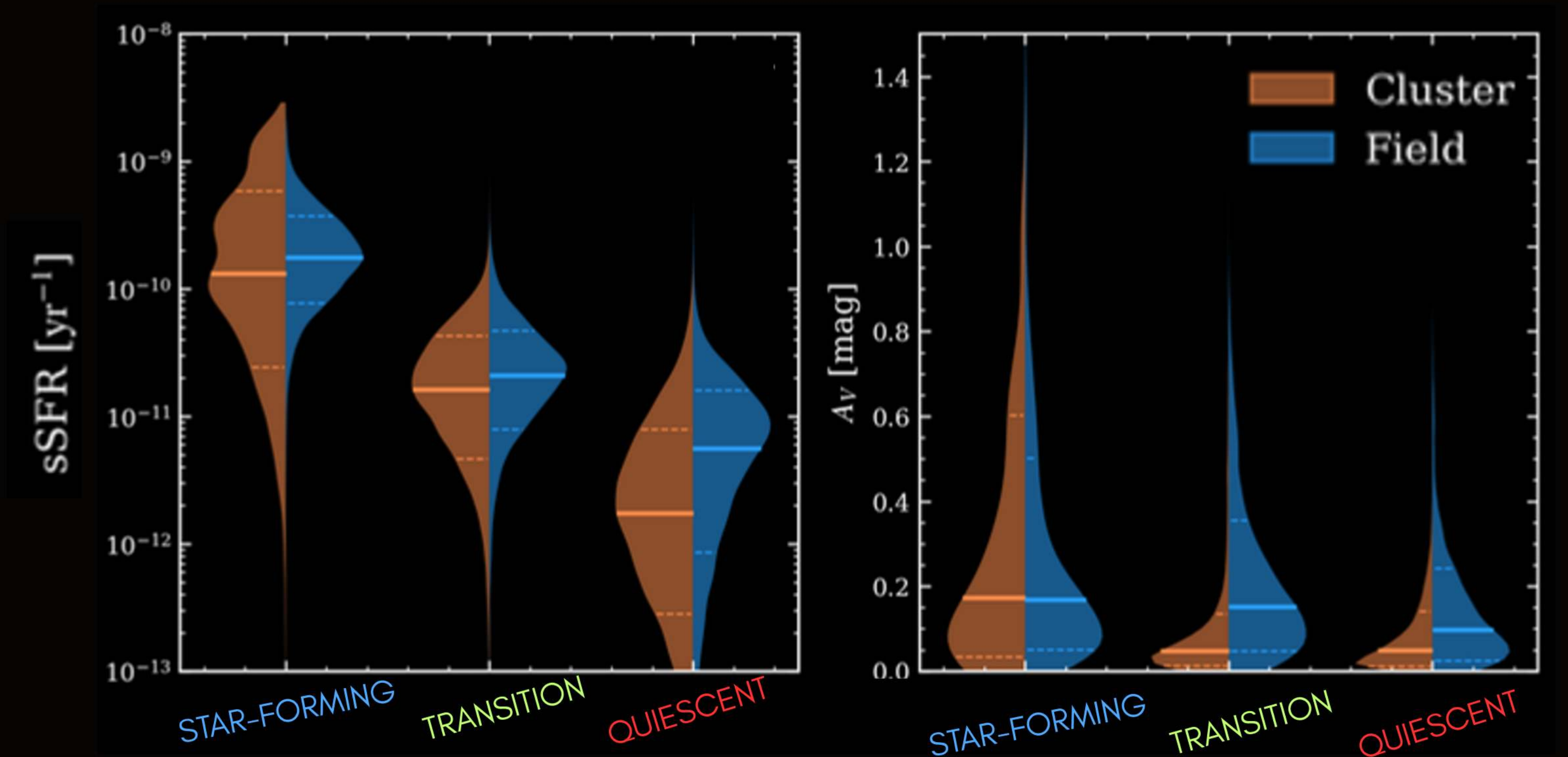
$> 5\sigma$
significance

Benotto+2026c
(in prep.)

	CLUSTER	FIELD	
<u>STAR-FORMING</u>	59%	69% ($\pm 6\%$)	
<u>TRANSITION</u>	9%	54% ($\pm 6\%$)	
<u>QUIESCENT</u>	0%	5% ($\pm 2\%$)	

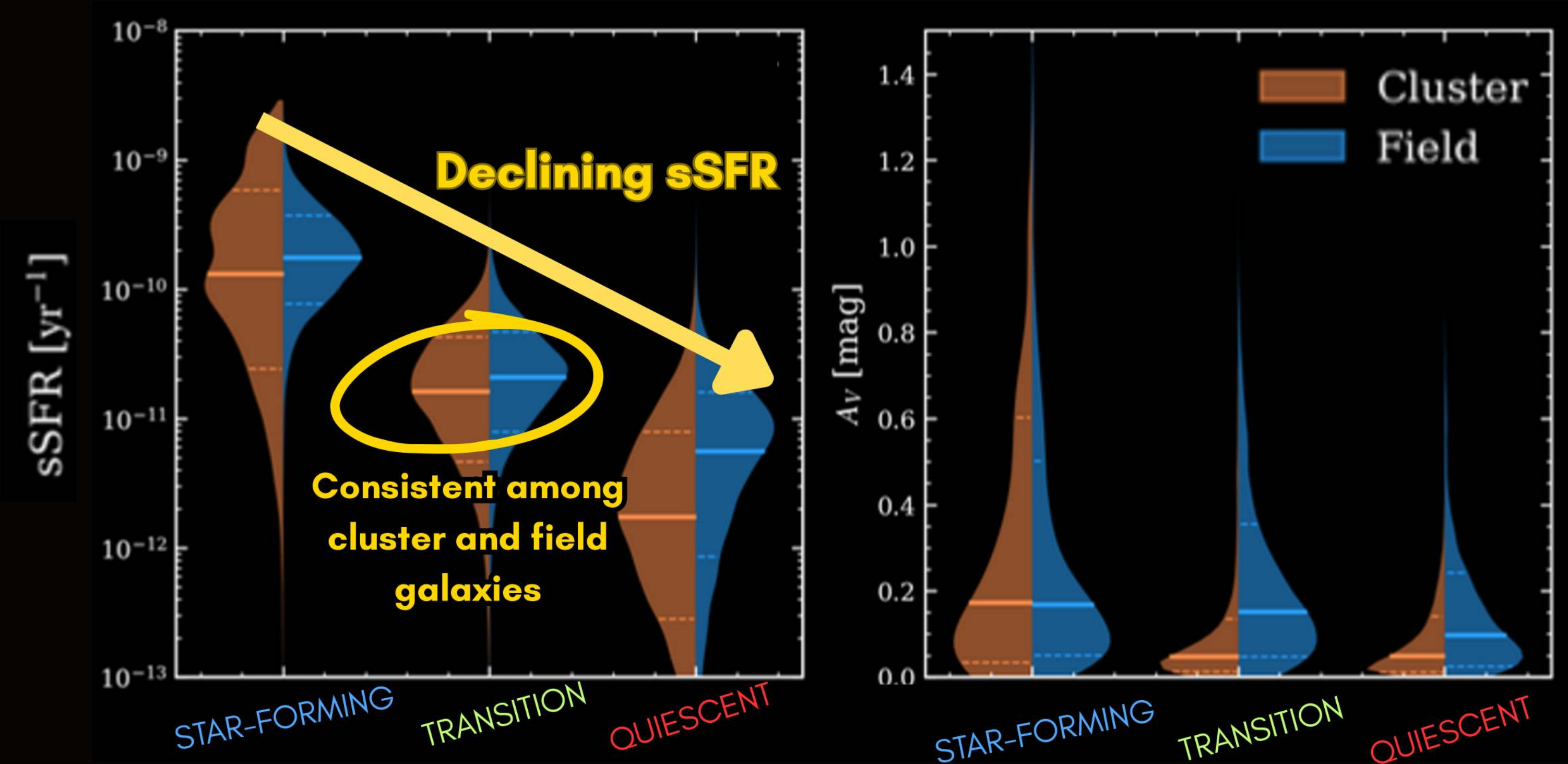
*Benotto+2026c
(in prep.)*

Pietro Benotto



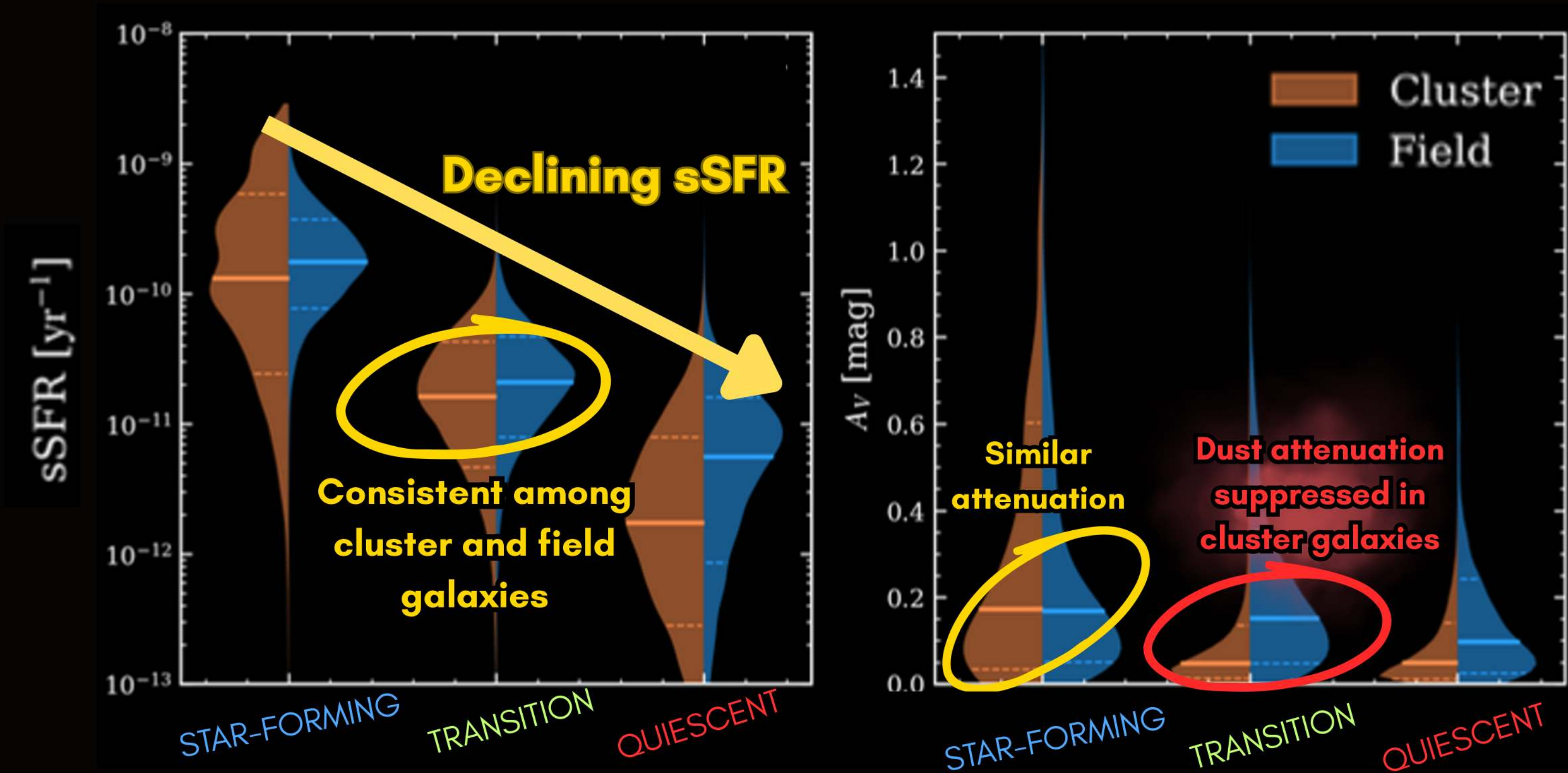
	CLUSTER	FIELD	
<u>STAR-FORMING</u>	59%	69% ($\pm 6\%$)	
<u>TRANSITION</u>	9%	54% ($\pm 6\%$)	
<u>QUIESCENT</u>	0%	5% ($\pm 2\%$)	

Benotto+2026c
(in prep.)

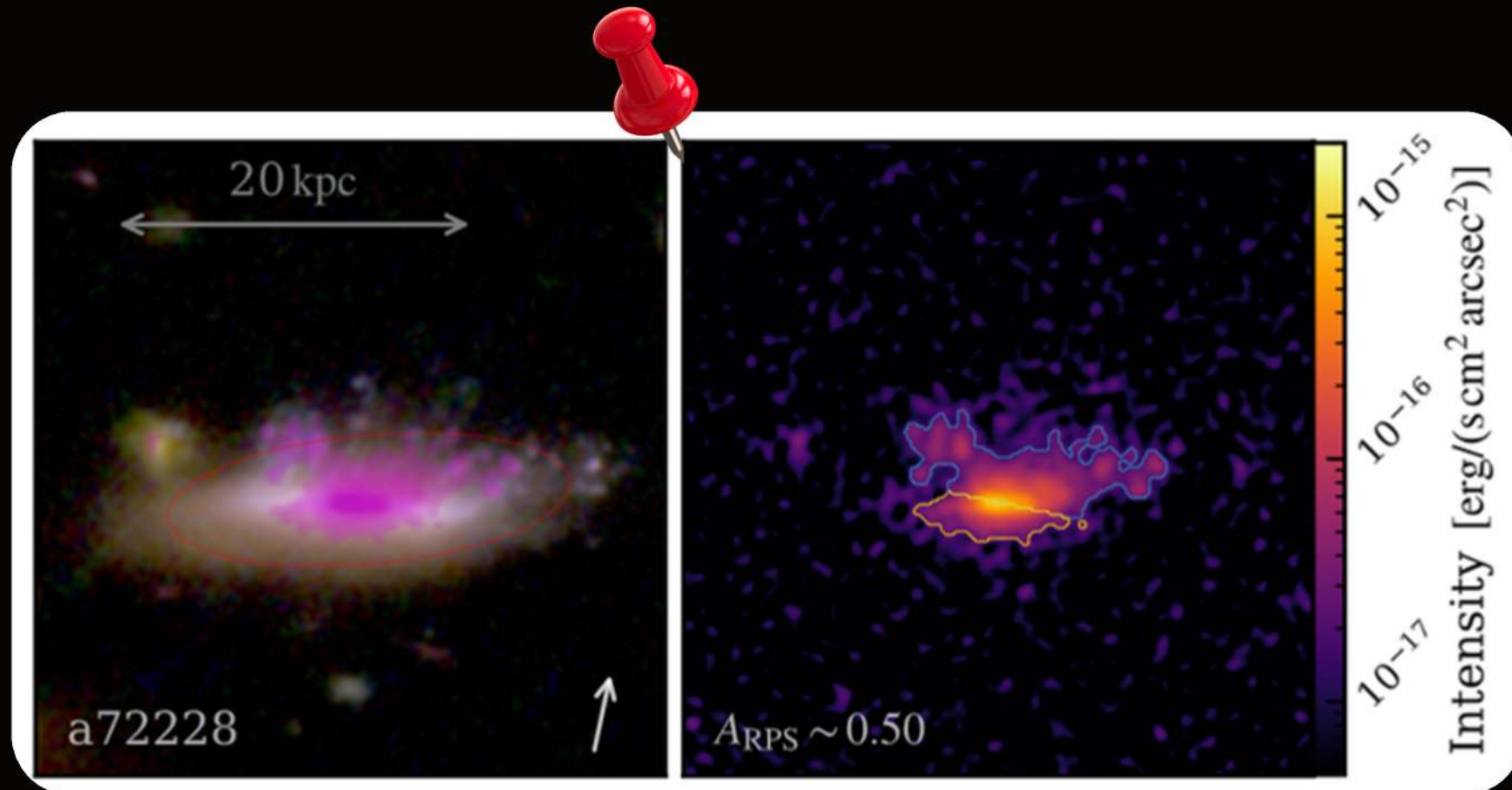
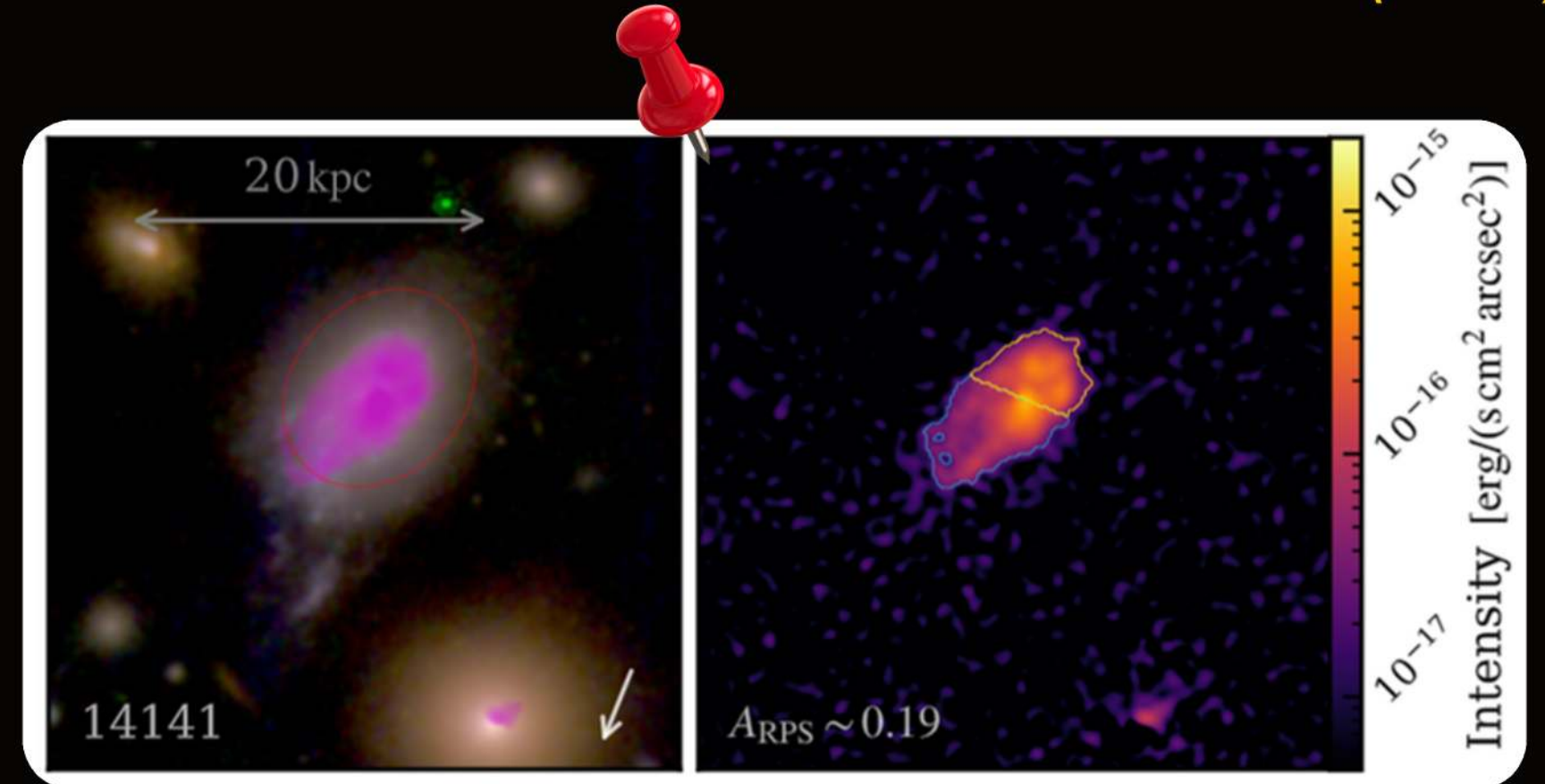
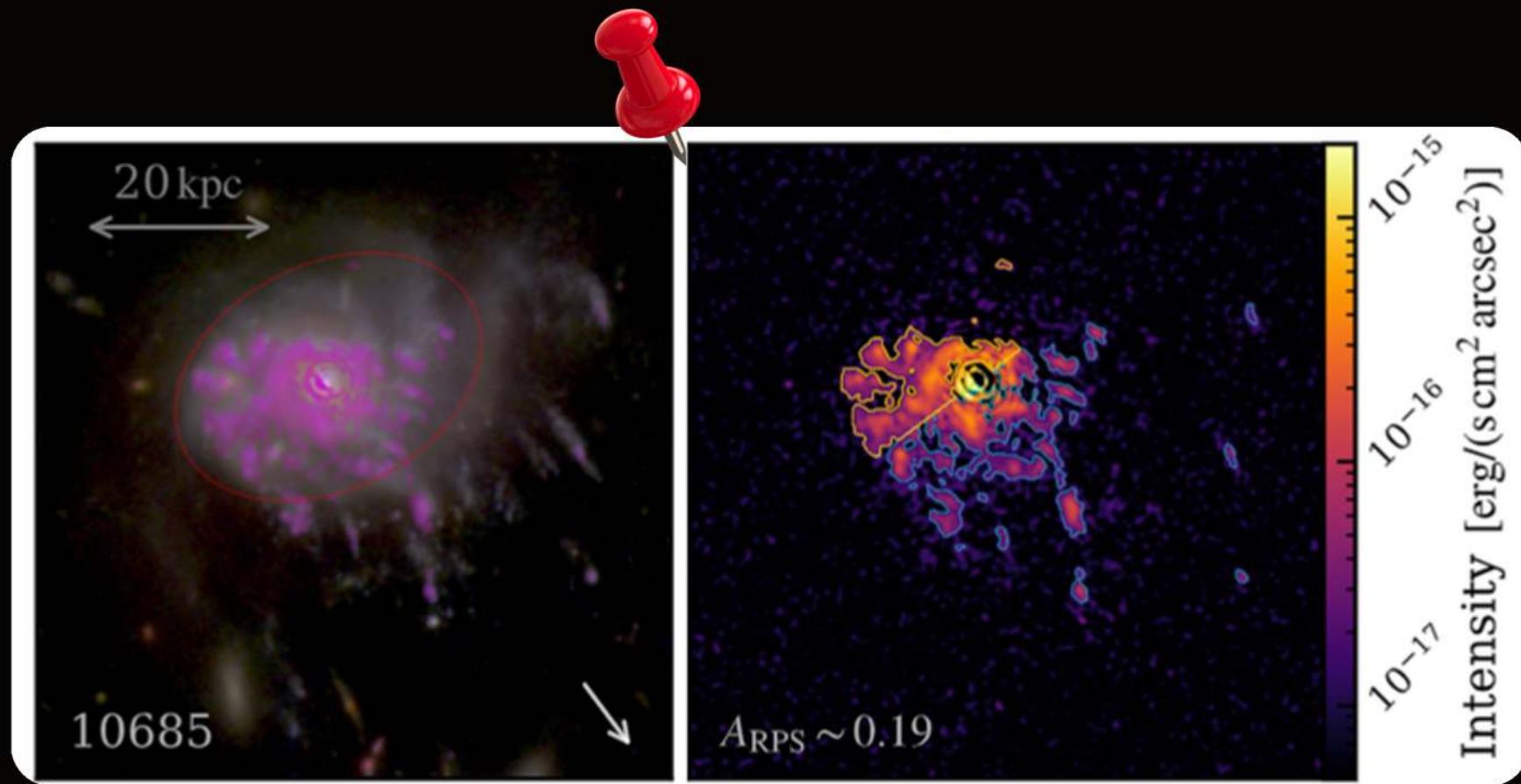


	CLUSTER	FIELD	
<u>STAR-FORMING</u>	59%	69% ($\pm 6\%$)	
<u>TRANSITION</u>	9%	54% ($\pm 6\%$)	
<u>QUIESCENT</u>	0%	5% ($\pm 2\%$)	

Benotto+2026c
(in prep.)



Benotto+2026a (A&A)



PAH emission in ram pressure stripped galaxies

PAH molecules can be stripped together with the atomic gas

Conclusions

PAH emitting galaxies

$5 \pm 1\%$ (clusters)
VS
 $44 \pm 4\%$ (fields)



The **PAH-emitting fraction** in clusters increases with cluster-centric distance and peak at $M = 10^{10} M_{\odot}$

Strong PAH suppression in cluster galaxies in the UVJ transition zone at 5σ

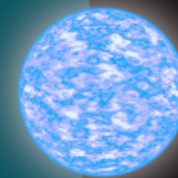


PAH EMISSION

found in stripped gas clumps

Environmental processing

has to be kept in mind to correctly calibrated the PAH emission as a SFR tracer



can have an impact on the PAH molecules directly



Conclusions

PAH emitting galaxies

$5 \pm 1\%$ (clusters)
VS
 $44 \pm 4\%$ (fields)



The **PAH-emitting fraction** in clusters increases with cluster-centric distance and peak at $M = 10^{10} M_{\odot}$

Strong PAH suppression in cluster galaxies in the UVJ transition zone at 5σ



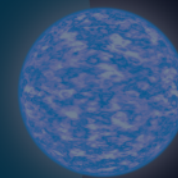
Thank you!

PAH EMISSION

found in stripped gas clumps

Environmental processing

has to be kept in mind to correctly calibrated the PAH emission as a SFR tracer



can have an impact on the PAH molecules directly



Table 4. Parameters used for SED fitting with BAGPIPES

Parameter	Fixed / Prior	Range
Redshift (z)	Fixed	Spectroscopic redshift
Metallicity (Z)	Logarithmic	$[0.05; 3] Z_{\odot}$
Ionisation parameter ($\log \mathcal{U}$)	Uniform	$[-4; -1]$
Birth clouds age (t_{bc})	Fixed	10 Myr
Dust		
Extinction curve model	Fixed	Calzetti et al. (2000)
Dust emission model	Fixed	Draine & Li (2007)
V-band extinction (A_V)	Uniform	$[0; 6]$
Continuity SFH		
Initial mass function	Fixed	Kroupa (2001)
Stellar population	Fixed	Bruzual & Charlot (2003)
Age bins	Fixed	$(0, 0.03, 0.10, 0.25, 0.6, 1.4, 3.2, 7.5, 10^{(a)})$ Gyr
$\log\left(\frac{\text{SFR}(t)}{\text{SFR}(t+\Delta t)}\right)$	Student-t	$\nu = 2, \sigma = 0.6$ Physical $\mu^{(b)}$
$\log(\text{Stellar mass } [M_{\odot}])$	Uniform	$[6; 15]$

Notes. ^(a)The last age bin ends at redshift $z = 20$ for each galaxy.

^(b)Prior parameters: ν is the wings weight, σ is the scale and μ is the distribution location. The latter is set to follow the [Madau & Dickinson \(2014\)](#) SFR across cosmic time.

Table 2. Spectroscopic catalogues used in each field.

Catalogue	Reference	Fields
ASTRODEEP	Merlin et al. (2024)	A2744, COSMOS, UDS, GOODS-S
All the Little Things	Naidu et al. (2024)	A2744
UNCOVER	Price et al. (2025)	A2744
GLASS-JWST NIRISS	Watson et al. (2025a)	A2744
MUSE (archival)	Owers et al. (2011) ; Braglia et al. (2009) ; Bergamini et al. (2023a)	A2744
MUSE A2744	ESO 114.27M6 (PI: Matthee)	A2744
GLASS	Schmidt et al. (2014) ; Treu et al. (2015)	A2744, MACS0416, A370
CANUCS	Sarrouh et al. (2026)	MACS0416, A370
MUSE clusters	Richard et al. (2021)	MACS0416, A370
VIMOS	Caminha et al. (2017)	MACS0416
HST/MUSE A370	Lagattuta et al. (2022)	A370
COSMOS compilation	Khostovan et al. (2026)	COSMOS
DESI-XMM	Ratajczak et al. (2026)	UDS
X-ray to FIR	Zou et al. (2022)	UDS
HLF	Whitaker et al. (2019)	GOODS-S
MUSE HUDF	Bacon et al. (2023)	GOODS-S
MuseWide	Urrutia et al. (2019)	GOODS-S
3D-HST	Momcheva et al. (2016)	All fields
DJA	Brammer & Valentino (2025)	All fields

Table A.1. Photometric observations used in this analysis.

Instrument	Filters	Use
HST/WFC3/UVIS	F225W, F275W, F336W, F350LP, F606W, F814W, F850LP	SED fitting
HST/ACS	F435W, F475W, F606W, F625W, F775W, F814W, F850LP	SED fitting
HST/WFC3/IR	F105W, F110W, F125W, F140W, F160W	SED fitting
JWST/NIRCAM	F070W, F090W, F115W, F140M, F150W, F162M, F164N, F182M, F187N, F200W, F210M, F250M, F277W, F300M, F335M, F356W, F360M, F410M, F430M, F460M, F480M	SED fitting
JWST/NIRCAM	F335M, F360M, F410M, F430M, F460M, F480M	PAH maps

COMPLETENESS

Galaxies with N_{ph} in the range of interest = $q N_{\text{tr}}$ + $f N_{\text{tr}}$ Galaxies NOT in the redshift range of interest

Galaxies in the redshift range of interest

Detection efficiency

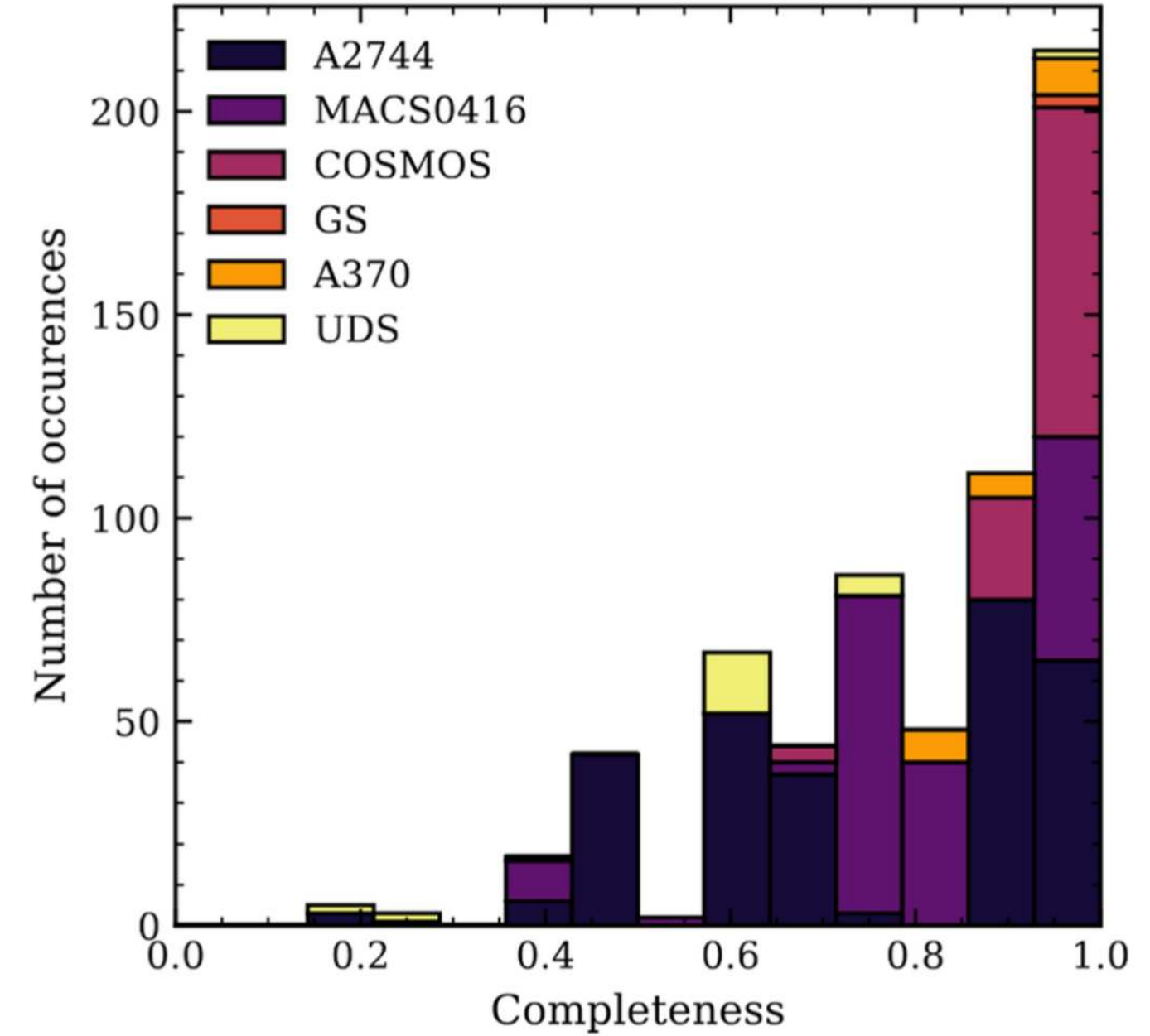
$$q = \frac{N_{\text{ph,sp}}}{N_{\text{sp}}}$$

False positives fraction

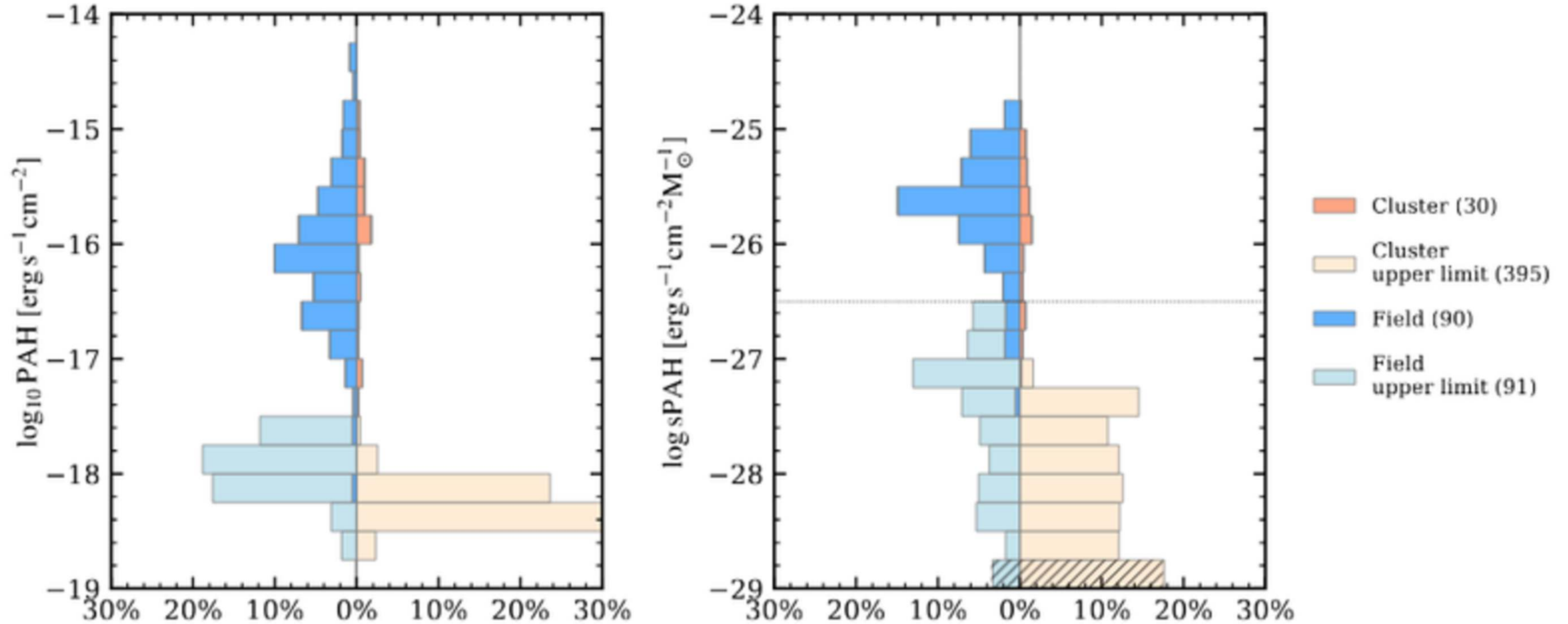
$$f = \frac{N_{\text{ph},\overline{\text{sp}}}}{N_{\overline{\text{sp}}}}$$

Table 1. Fields used in this work and number of selected galaxies per field.

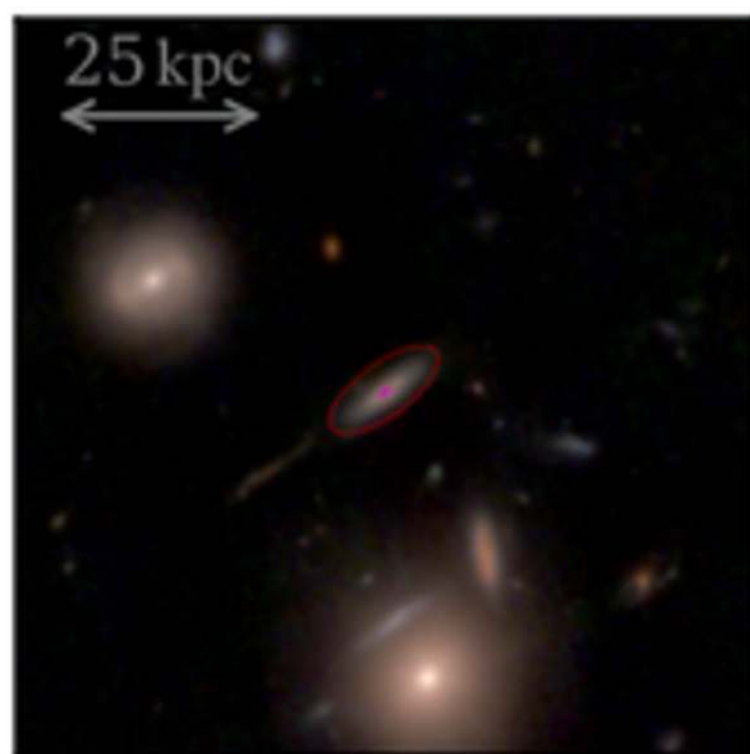
Field name	Cluster/field	RA [hms]	DEC [dms]	z_{cl}	N_{spec}	N_{good}
A2744	cluster	00:14:20.2	-30:22:26	[0.284, 0.325]	365	268
A2744	field			-	82	20
A370	cluster	02:39:54.8	-01:35:45	[0.350, 0.400]	8	2
A370	field			-	78	22
MACS0416	cluster	04:16:13.0	-24:04:59	[0.382, 0.412]	293	170
MACS0416	field			-	74	19
COSMOS	field	10:00:32.6	+02:21:18	-	276	110
GS	field	03:32:27.4	-27:49:37	-	68	3
UDS	field	02:17:28.3	-05:08:38	-	75	26



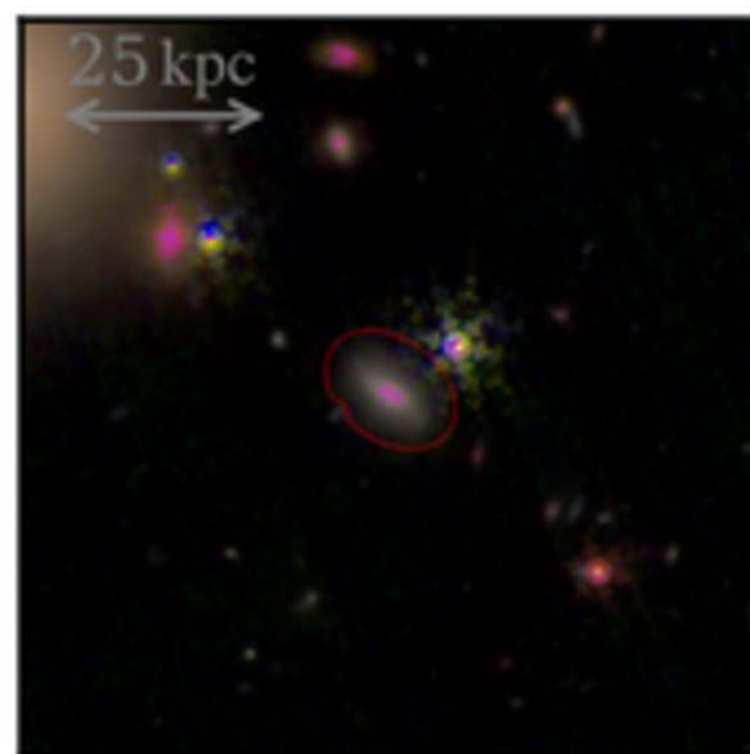
PAH distribution



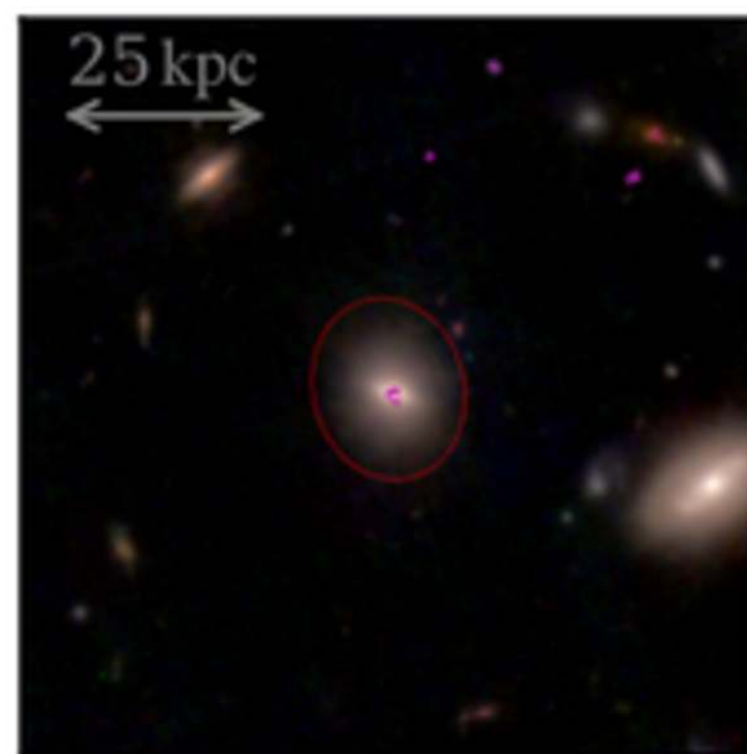
COSMOS_81592



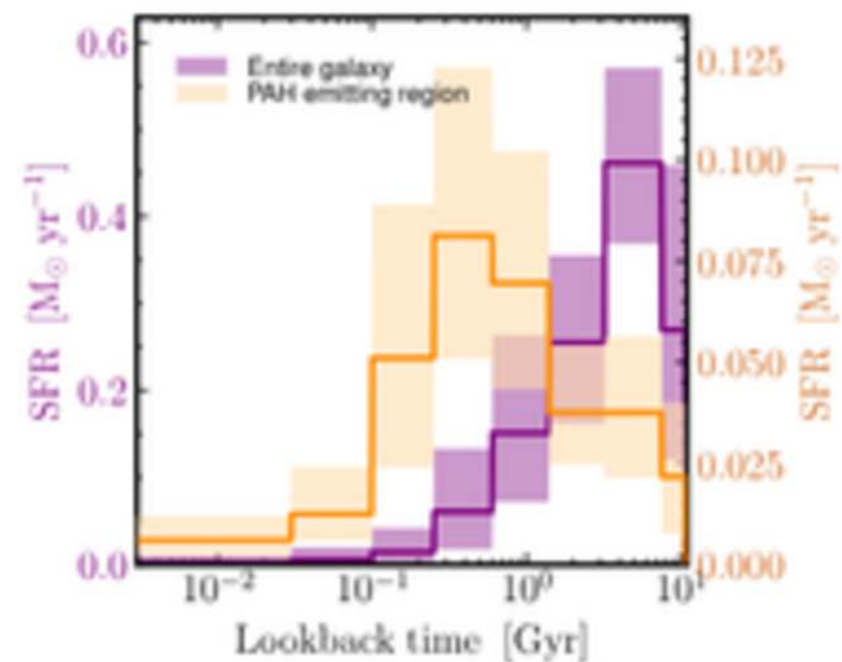
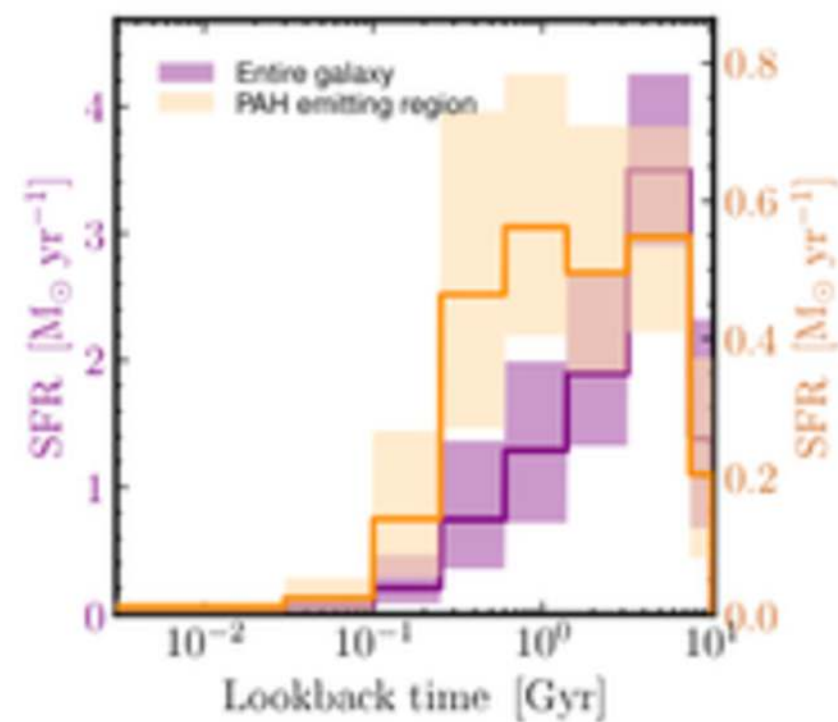
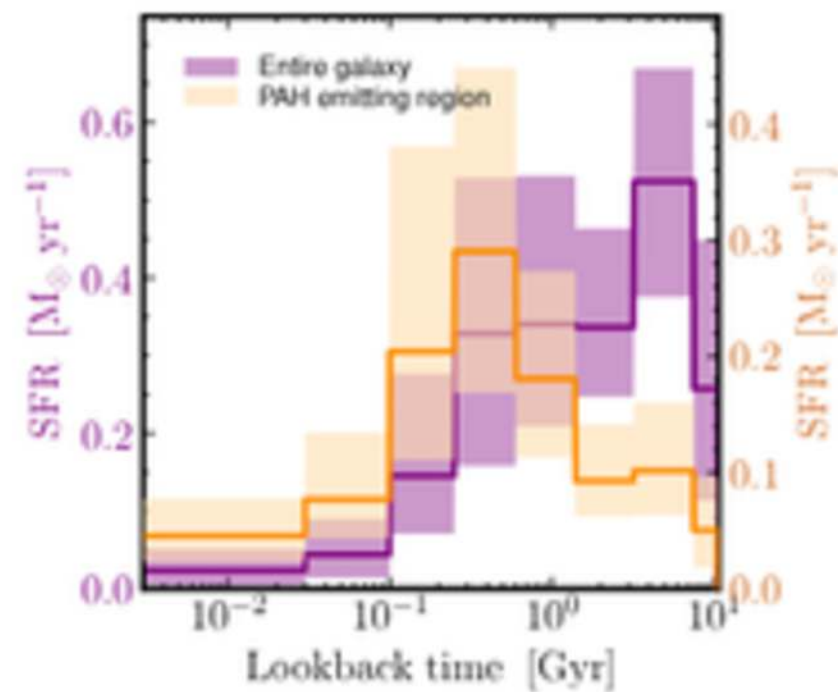
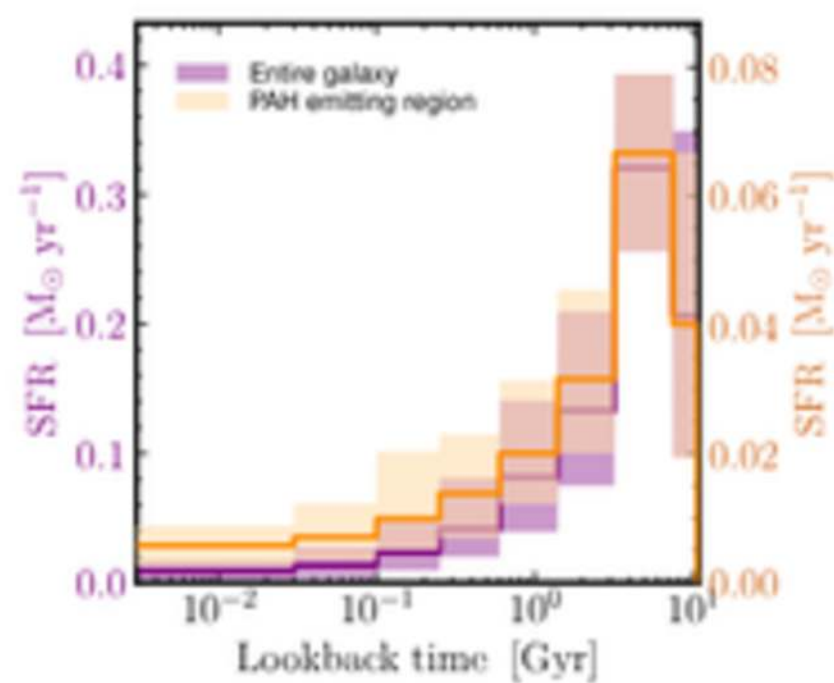
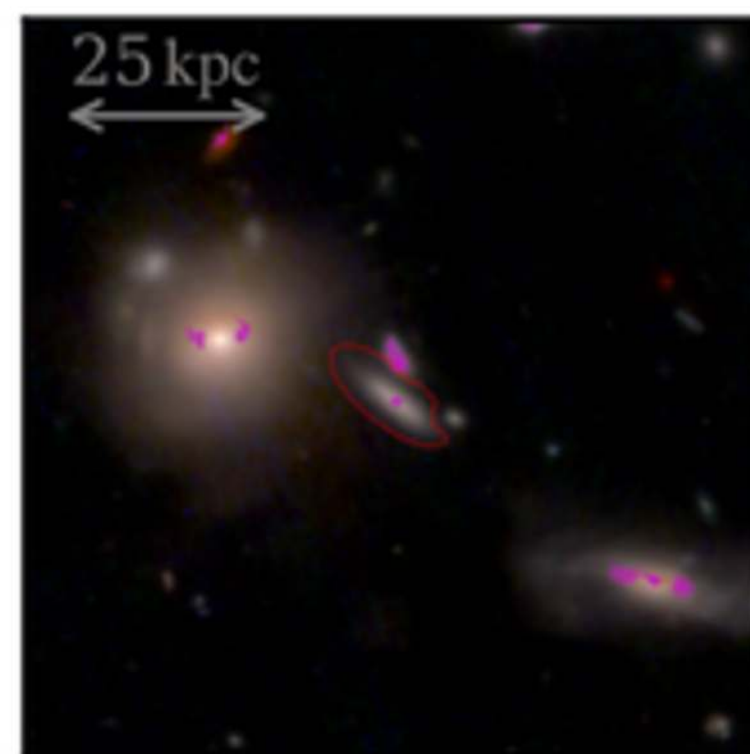
COSMOS_104342



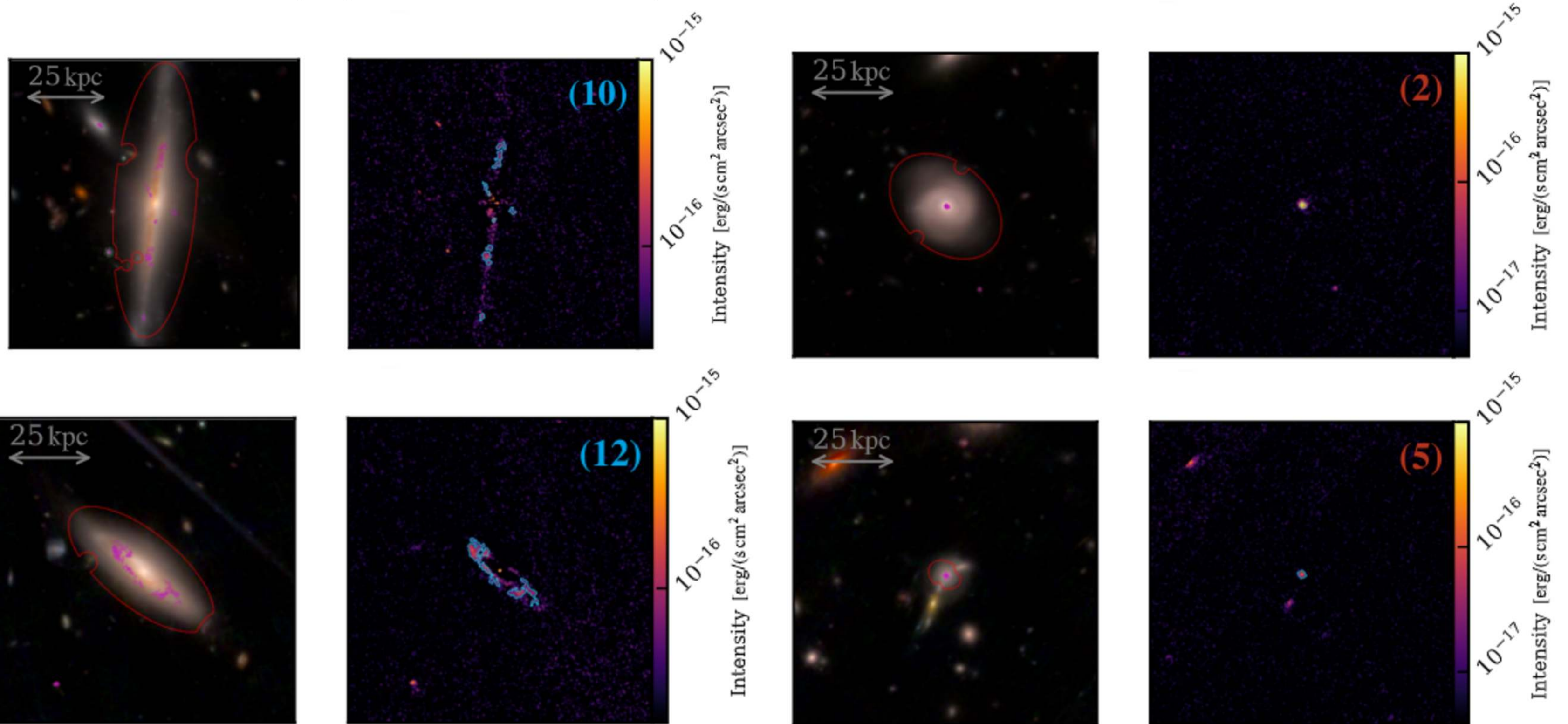
COSMOS_72201



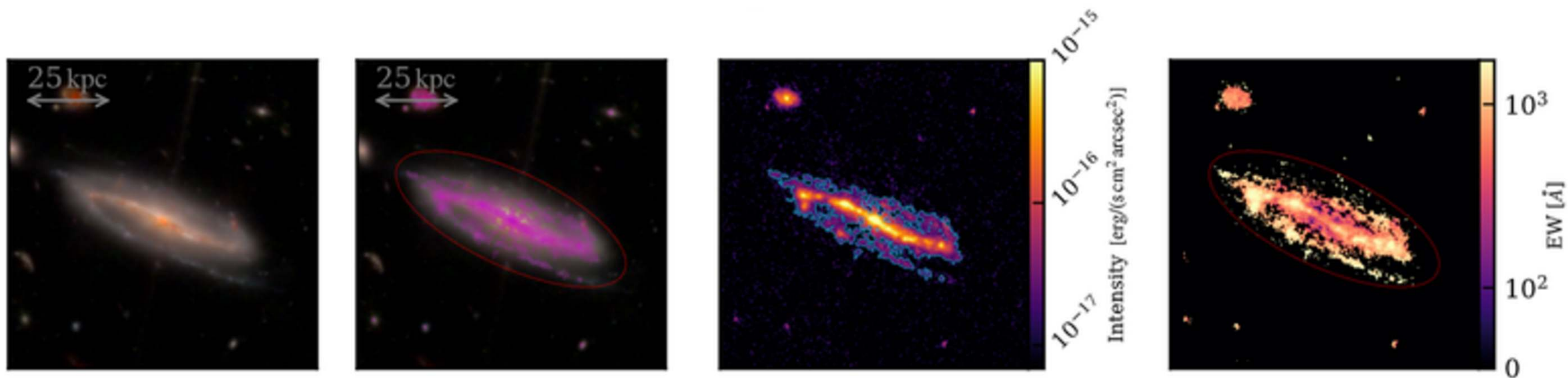
COSMOS_110273



low sPAH galaxies

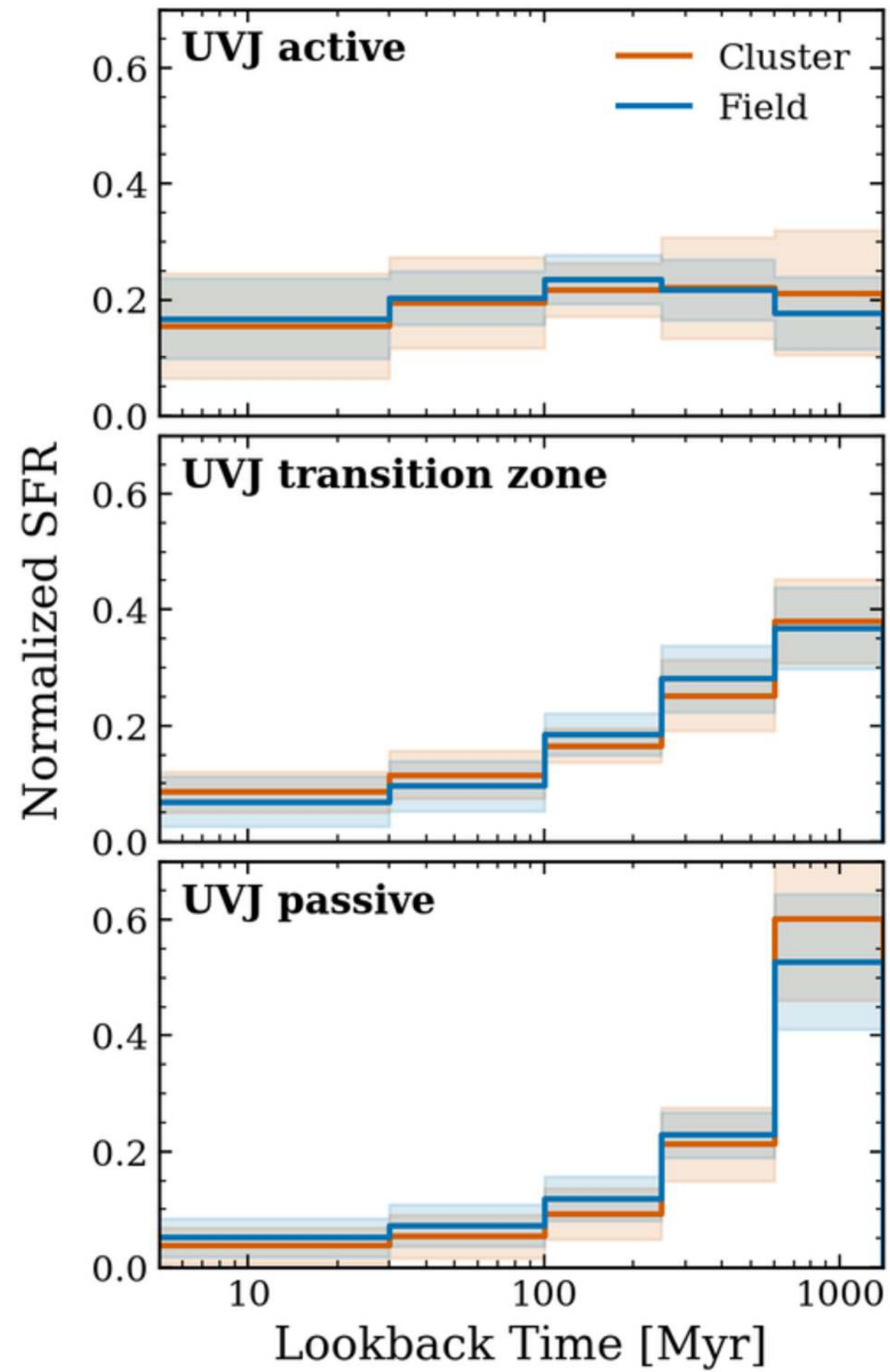


PAH maps

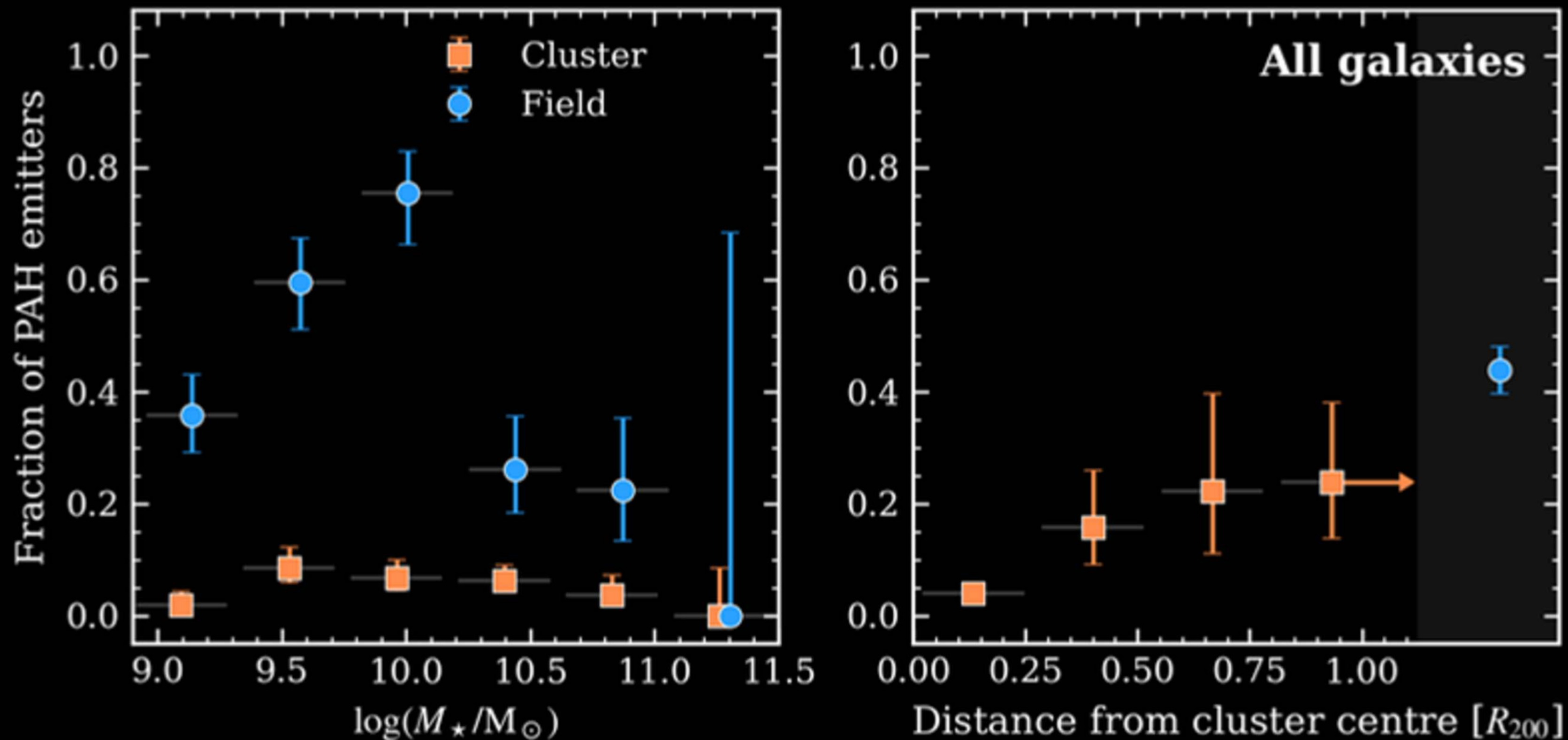


Transition zone

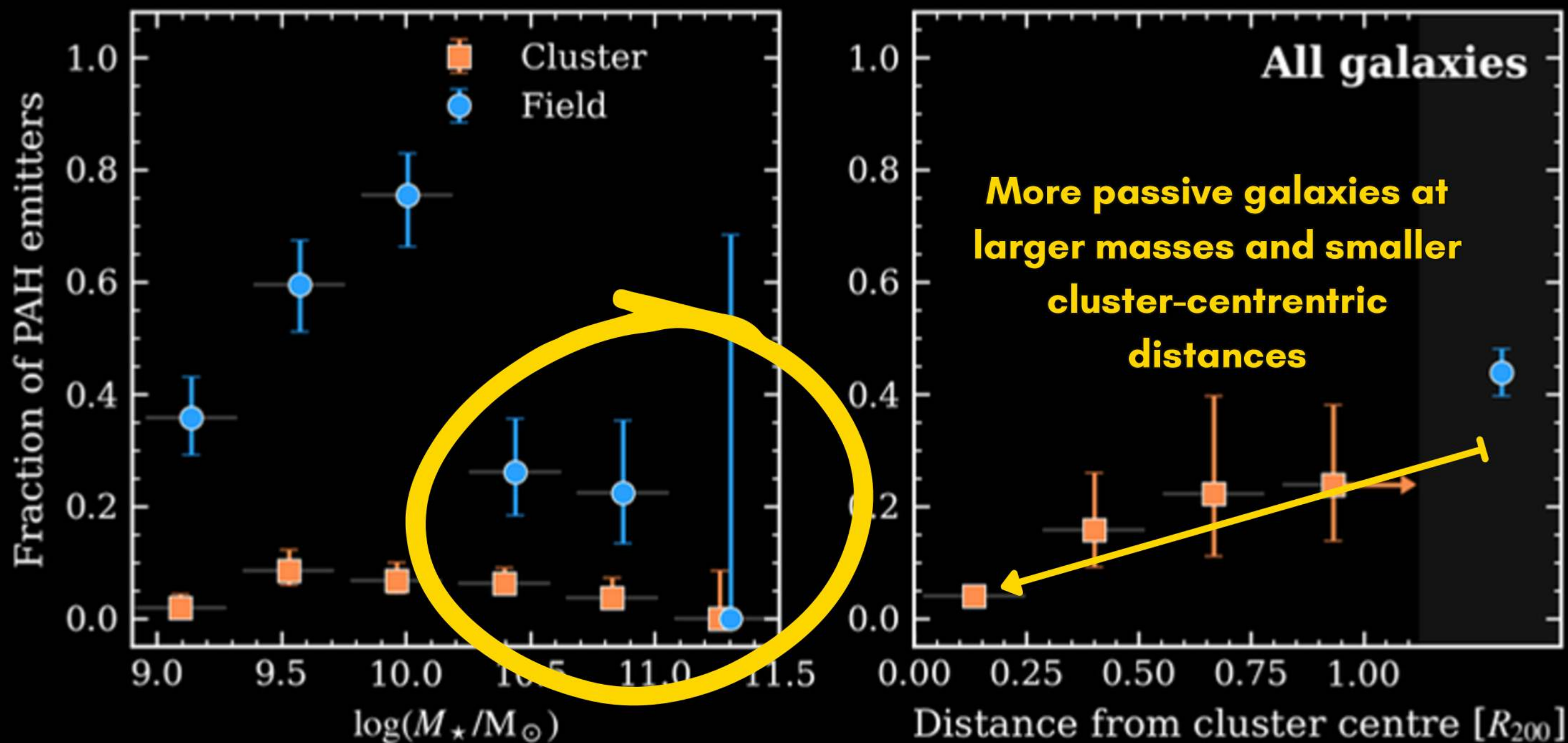
$$0.88(V - J) + 0.59 < (U - V)_{\text{trans}} < 0.88(V - J) + 0.79$$



Fraction of PAH emitters

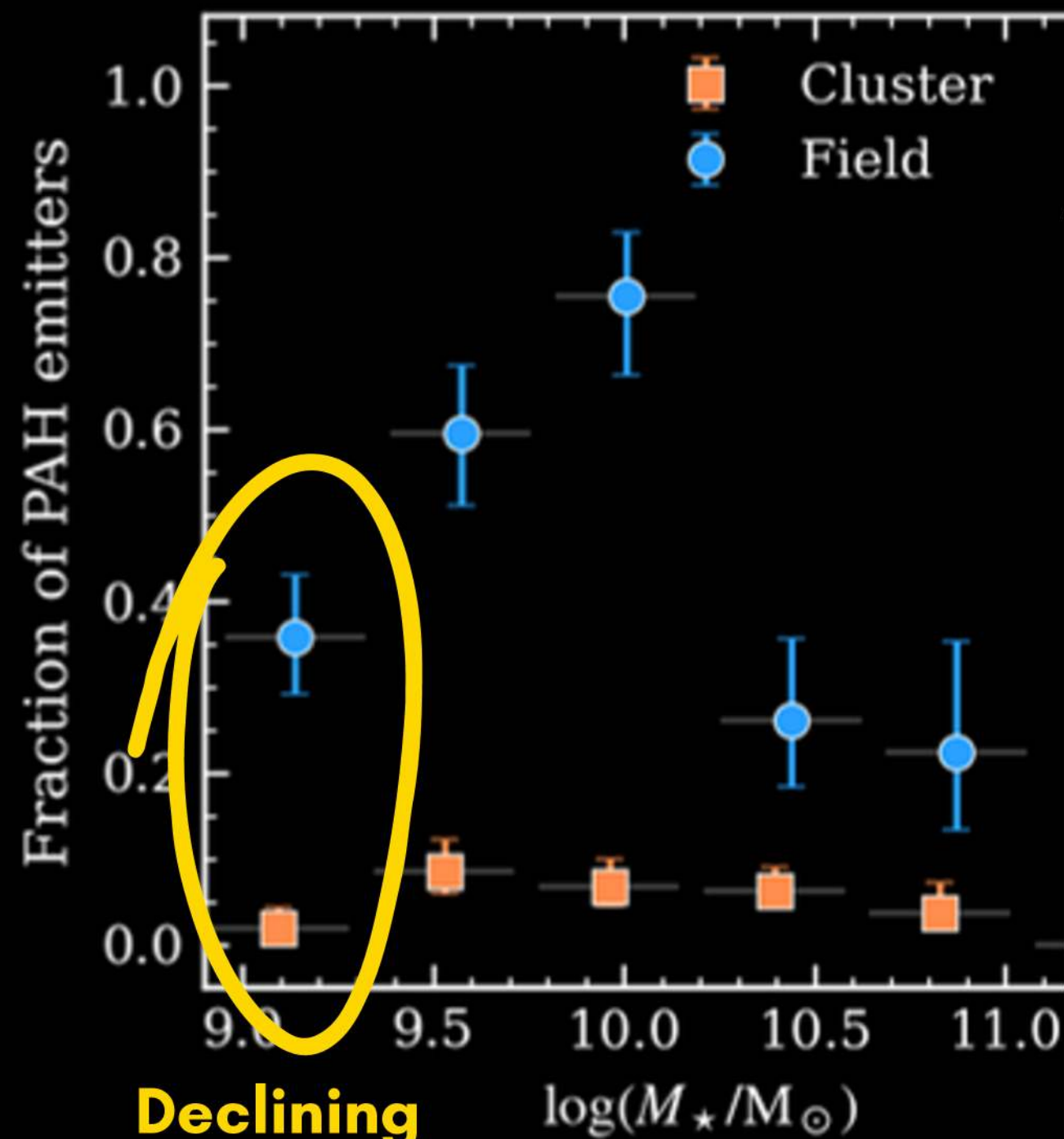


Fraction of PAH emitters



Qualitatively, this is consistent with what we already know about SF in clusters

Fraction of PAH emitters



**Declining
emitter fraction
at low masses**

